

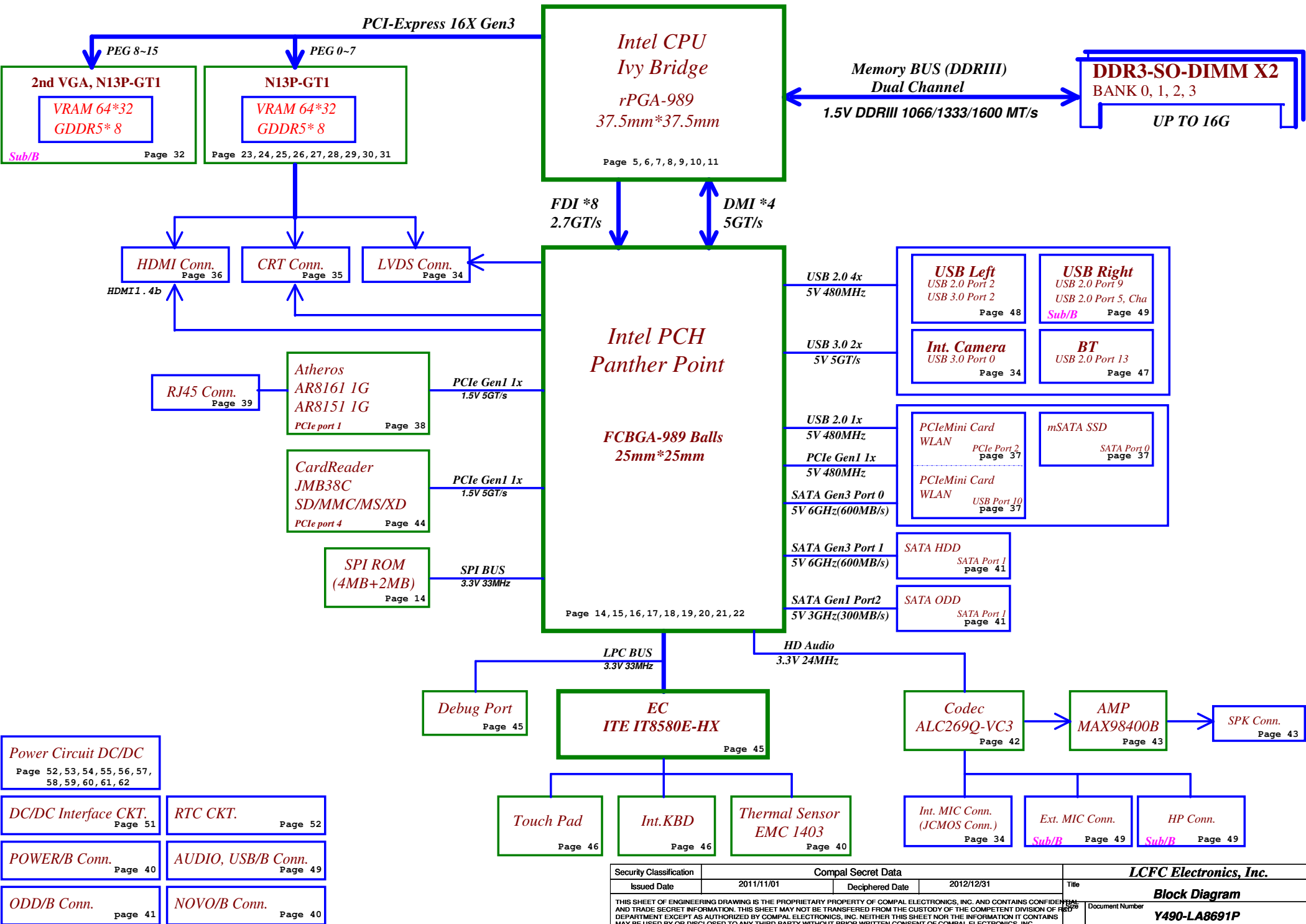
QIQY5

Whisky3.0 (Y400)

LA-8691P Rev0.3 Schematic

*Intel IVY Bridge Processor with DDRIII + Panther Point PCH
nVIDIA N13P GT1-A2 + 2nd VGA N13P GT1-A2
2012-05-10 Rev0.3*

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Voltage Rails (O --> Means ON , X --> Means OFF)

Power Plane / State	B+	+3VALW +5VALW	+1.5V	+5VS +3VS +1.5VS +VCCSA +V1.5S_VCCP +CPU_CORE +VGA_CORE +GFX_CORE +1.8VS +1.05VS +0.75VS +3.3VS_VGA +1.5VS_VGA +1.05VS_VGA
S0	O	O	O	O
S3	O	O	O	X
S5 S4/AC Only	O	O	X	X
S5 S4 Battery only	O	X	X	X
S5 S4 AC & Battery don't exist	X	X	X	X

STATE \ SIGNAL	SLP_S1#	SLP_S3#	SLP_S4#	SLP_S5#	+VALW	+V	+VS	Clock
Full ON	HIGH	HIGH	HIGH	HIGH	ON	ON	ON	ON
S1 (Power On Suspend)	LOW	HIGH	HIGH	HIGH	ON	ON	ON	LOW
S3 (Suspend to RAM)	LOW	LOW	HIGH	HIGH	ON	ON	OFF	OFF
S4 (Suspend to Disk)	LOW	LOW	LOW	HIGH	ON	OFF	OFF	OFF
S5 (Soft OFF)	LOW	LOW	LOW	LOW	ON	OFF	OFF	OFF

USB Port Table

USB 2.0	USB 3.0	Port	4 External USB Port
	XHCI 1	0	Camera
		1	
		2	USB Port (Left Side)
		3	
		4	
		5	USB Port (Right Side)
		6	
		7	
		8	
		9	USB Port (Right Side)
		10	Mini Card(WLAN)
		11	
		12	
		13	Blue Tooth

BOM Structure Table

BOM Structure	BTO Item
HDMI@	HDMI part
CHG@	USB charger part
NOCHG@	No USB charger part
CMOS@	CMOS Camera part
8161@	AR8161 LAN part
8151@	AR8151 LAN part
8161S@	AR8161 LAN surge part
8151S@	AR8151 LAN surge part
SURGE@	AR8151&8161 LAN surge part
61@	X76 P/N for AR8161
51@	X76 P/N for AR8151
X76@	X76 Level part for VRAM
GC6@	NV CG6 support part
NOGC6@	NV no CG6 support part
AOAC@	AOAC support part
KBL@	K/B Light part
ME@	ME part
OPT@	For optimus function part
SLI@	For SLI function part
DS3@	Deep S3 support part
S3@	For S3 function part
GT@	NV chip part
@	Unpop

SMBUS Control Table

	SOURCE	Main VGA	2nd VGA	BATT	IT8580E	SODIMM	WLAN WiMAX	Thermal Sensor	PCH	TP Module
EC_SMB_CK1 EC_SMB_DA1	IT8580E +3VALW	X	X	V +3VALW	X	X	X	X	X	X
EC_SMB_CK2 EC_SMB_DA2	IT8580E +3VS	V +3VS	V +3VS	X	X	X	X	V +3VS	V +3V_PCH	X
SMB_CLK_S3 SMB_DATA_S3	PCH +3VS	X	X	X	X	V +3VS	V +3VS	X	V +3V_PCH	V +3VS

PCIE PORT LIST

Port	Device
1	LAN
2	WLAN
3	
4	Card Reader
5	
6	
7	
8	

EC SM Bus1 address

Device	Address
Smart Battery	0001 011X b

EC SM Bus2 address

Device	Address
Thermal Sensor EMC1403-2	1001_101xb
Master VGA	0x9E
Slave VGA	0x9C

PCH SM Bus address

Device	Address
DDR DIMM0	1001 000Xb
DDR DIMM2	1001 010Xb



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Hot plug detect for IFP link E

VGA and GDDR5 Voltage Rails (N13Px GPIO)

GPIO	I/O	ACTIVE	Function Description
GPIO0	OUT	-	GPU VID4
GPIO1	OUT	-	GPU VID3
GPIO2	OUT	-	VGA_BL_PWM
GPIO3	OUT	-	VGA_ENVDD
GPIO4	OUT	-	VGA_ENBKL
GPIO5	OUT	-	GPU VID1
GPIO6	OUT	-	GPU VID2
GPIO7	OUT	-	DPRSLPVR_VGA
GPIO8	I/O	-	Thermal Catastrophic Over Temperature
GPIO9	OUT	-	GPIO9
GPIO10	OUT	-	Memory VREF Control
GPIO11	OUT	-	GPU VID0
GPIO12	IN		AC Power Detect Input (10K pull High)
GPIO13	OUT	-	GPU VID5
GPIO14	OUT	-	FB_CLAMP_TOGGLE_REQ#
GPIO15	IN	N/A	(100K pull low)
GPIO16	OUT	-	FRMLCK#
GPIO17	IN	N/A	
GPIO18	IN	-	dGPU_HDMI_HPD
GPIO19	IN	-	HPD_IRQ

Performance Mode P0 TDP at Tj = 102 C* (GDDR5)

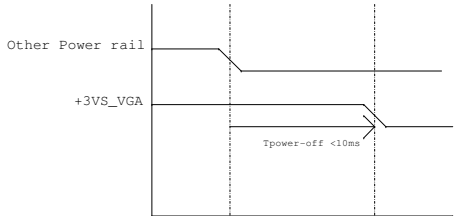
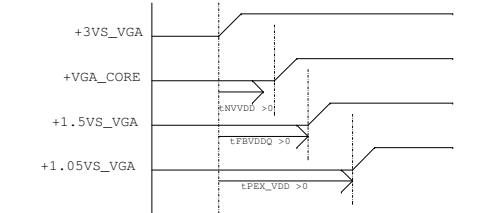
	GPU (4)	Mem (1,5)	NVCLK /MCLK	NVVDD			FBVDD (1.35V)		FBVDDQ (GPU+Mem) (1.35V)		PCI Express (1.05V) (6)		I/O and PLLVDD (1.8V)		I/O and PLLVDD (1.05V)		Other (3.3V)	
Products	(W)	(W)	(MHz)	(V)	(A)	(W)	(A)	(W)	(A)	(W)	(mA)	(W)	(mA)	(W)	(mA)	(W)	(mA)	(W)
N13X 128bit 1GB GDDR5	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Physical Strapping pin	Power Rail	Logical Strapping Bit3	Logical Strapping Bit2	Logical Strapping Bit1	Logical Strapping Bit0
ROM_SCLK	+3VS_VGA	PCI_DEVID[4]	SUB_VENDOR	SLOT_CLK_CFG	PEX_PLL_EN_TERM
ROM_SI	+3VS_VGA	RAM_CFG[3]	RAM_CFG[2]	RAM_CFG[1]	RAM_CFG[0]
ROM_SO	+3VS_VGA	FB[1]	FB[0]	SMB_ALT_ADDR	VGA_DEVICE
STRAP0	+3VS_VGA	USER[3]	USER[2]	USER[1]	USER[0]
STRAP1	+3VS_VGA	3GIO_PAD_CFG_ADR[3]	3GIO_PAD_CFG_ADR[2]	3GIO_PAD_CFG_ADR[1]	3GIO_PAD_CFG_ADR[0]
STRAP2	+3VS_VGA	PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
STRAP3	+3VS_VGA	SOR3_EXPOSED	SOR2_EXPOSED	SOR1_EXPOSED	SOR0_EXPOSED
STRAP4	+3VS_VGA	RESERVED	PCIE_SPEED_CHANGE_GEN3	PCIE_MAX_SPEED	DP_PLL_VDD33V

Device ID		setting		I2C Slave addresses ID	
N13P-GT (28nm)	0x0FDB	SMB_ALT_ADDR (ROM_SO Bit 1)	0	0x9E	
			1	0x9C	

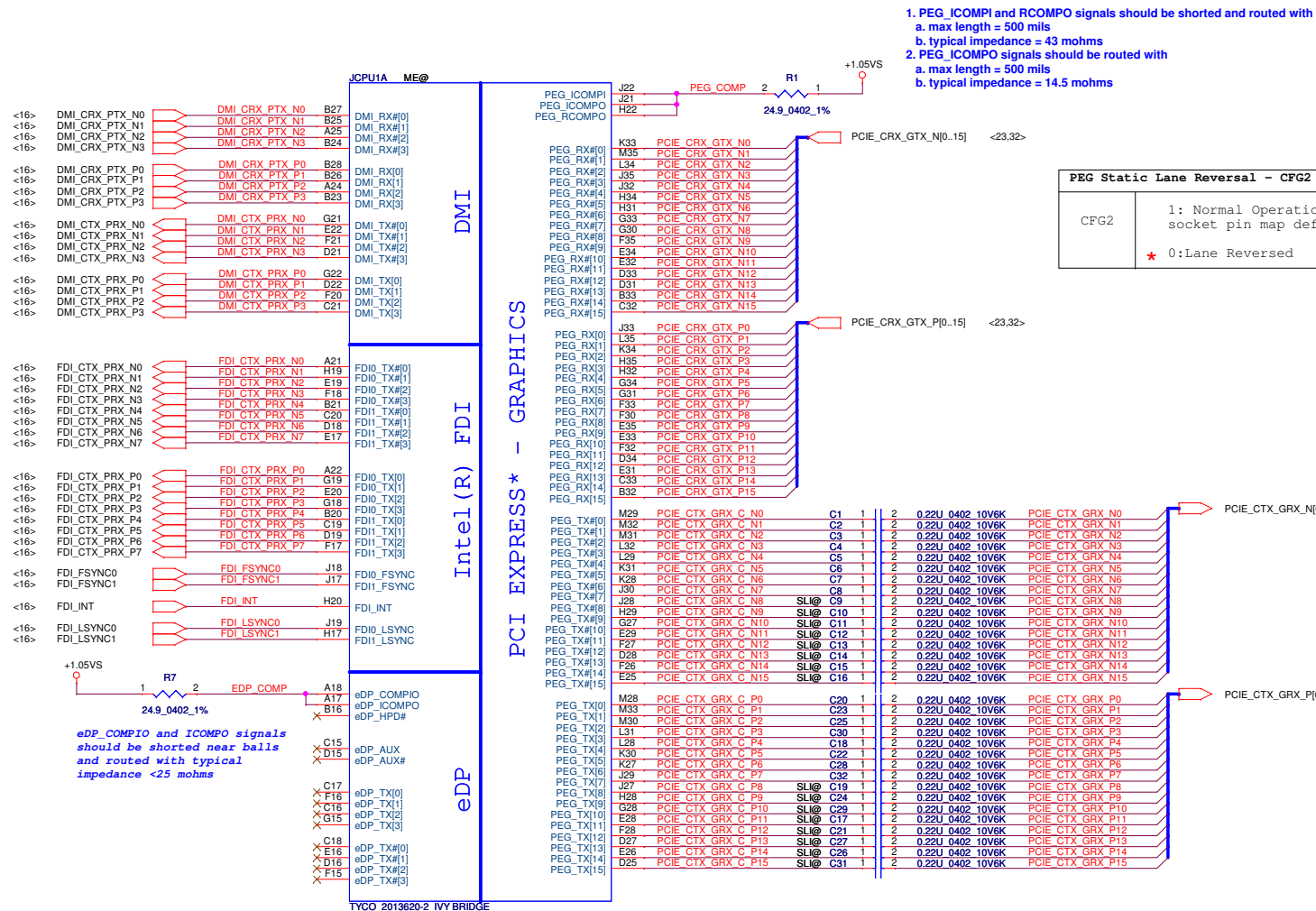
GPU	ROM_SO	ROM_SCLK	STRAP0	STRAP1	STRAP2	STRAP3	STRAP4	
N13P-GT1 28nm	PU 10K	PU 25K	PU 45K	PD 35K	PD 10K	PU 5K	PD 10K	Master
	PU 20K	PU 25K	PU 45K	PD 35K	PD 10K	PD 5K	PD 10K	Slave

GPU		N13P-GT		
FB Memory (GDDR5)		ROM_SI		
Samsung 2500MHz	K4G10325FG-HC04			
	32Mx32	PD 45K		
Hynix 2500MHz	H5GQ1H24BFR-T2C			
	32Mx32	PD 35K		
Samsung 2500MHz	K4G20325FD-FC04			
	64Mx32	PD 30K		
Hynix 2500MHz	H5GQ2H24MFR-T2C			
	64Mx32	PD 25K		

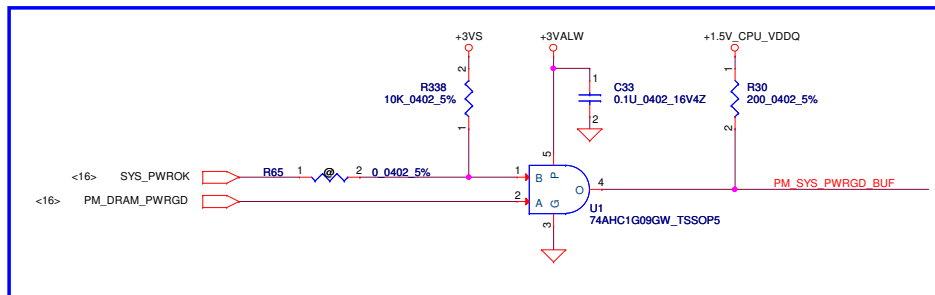
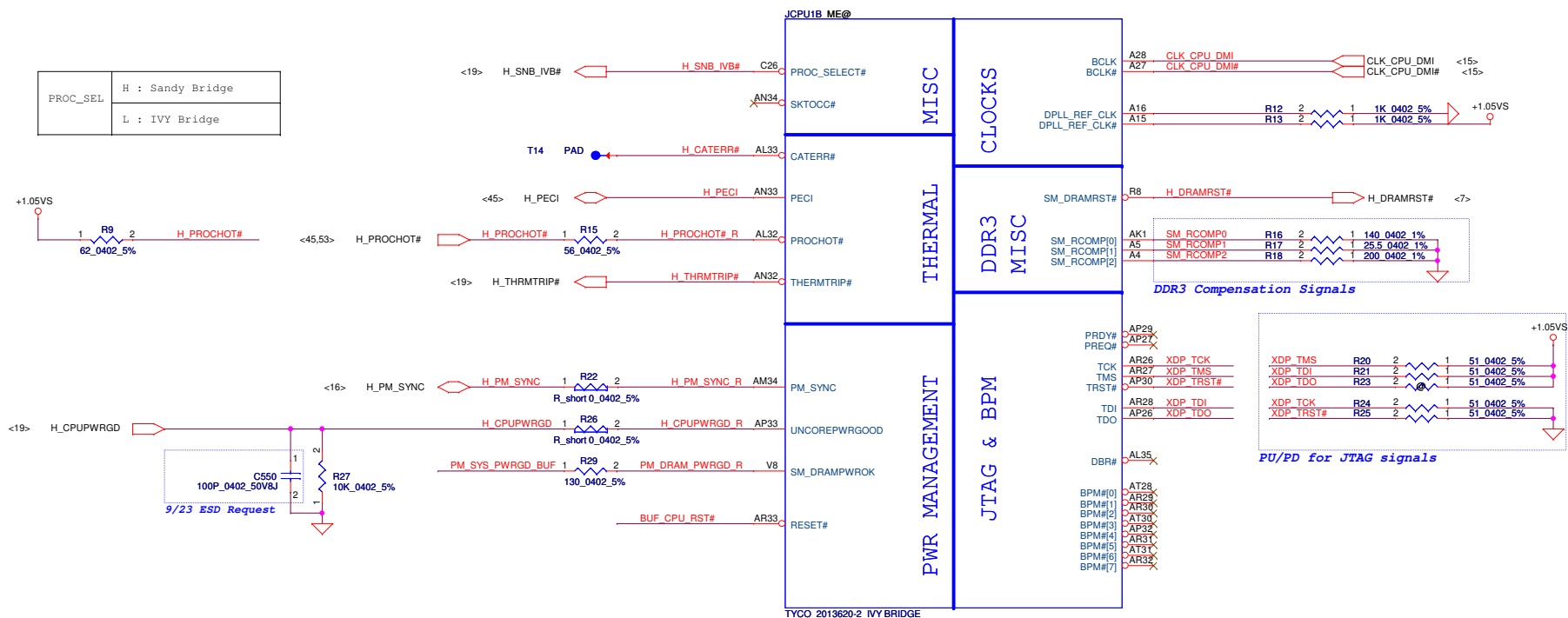


- 1.all GPU power rails should be turned off within 10ms
2. Optimus system VDD33 avoids drop down earlier than NVDD and FBVDDQ

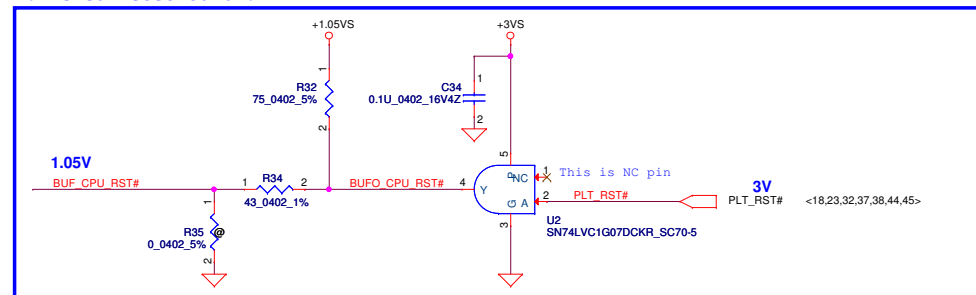
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PROC_SEL	H : Sandy Bridge
	L : IVY Bridge



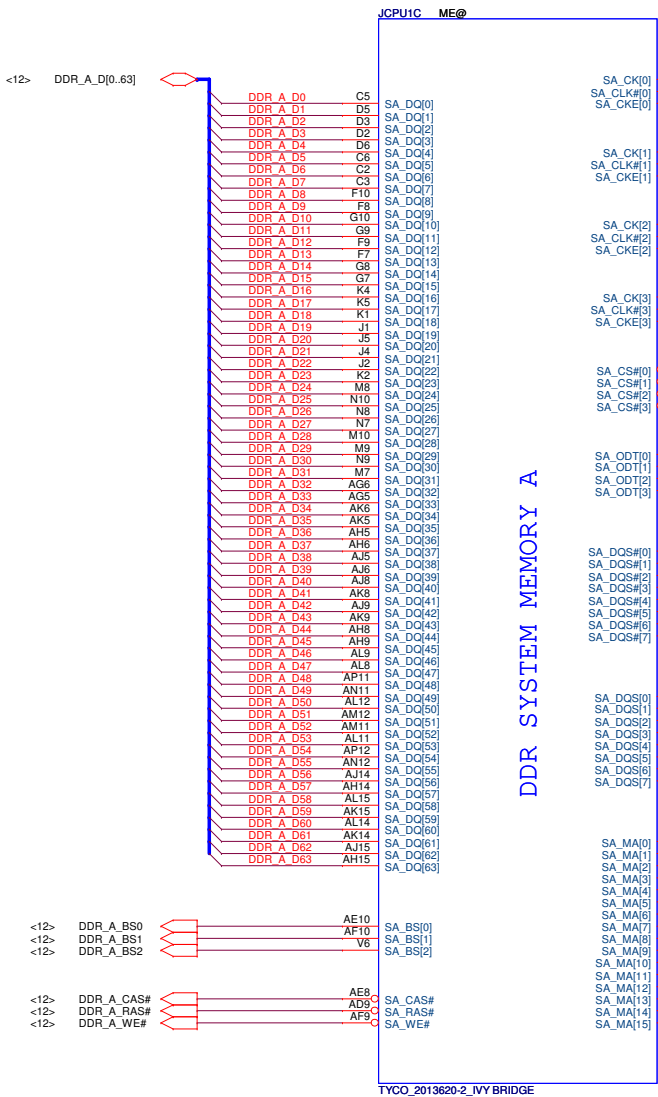
Buffered Reset to CPU



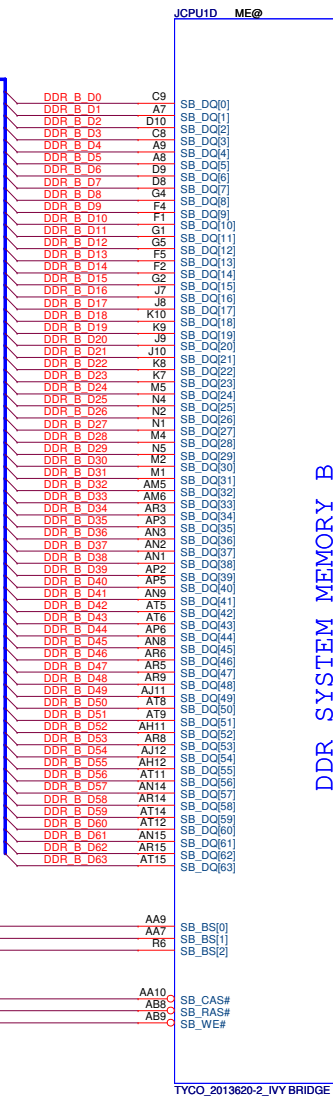
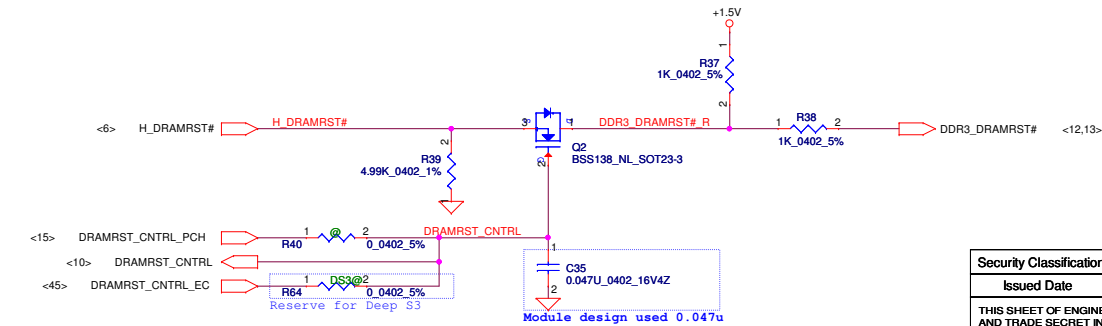
Compal Electronics, Inc.

PROCESSOR(2/7) PM,XDP,CLK

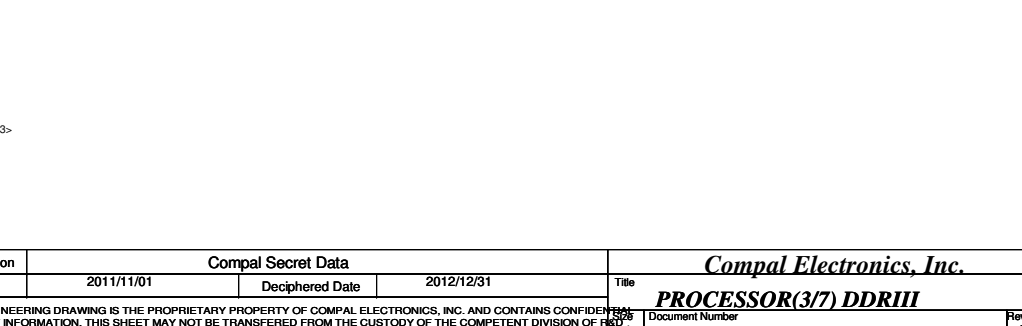
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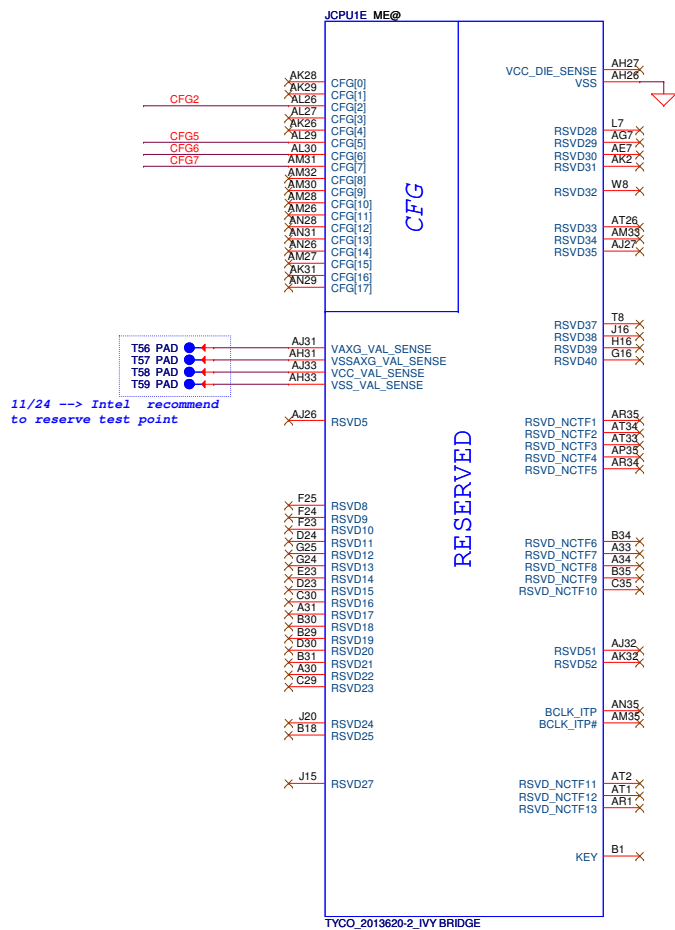
DDR SYSTEM MEMORY A



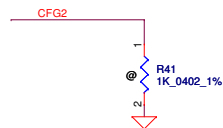
DDR SYSTEM MEMORY B



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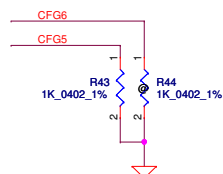


CFG Straps for Processor

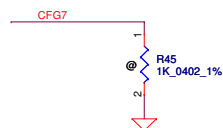


PEG Static Lane Reversal - CFG2 is for the 16x	
CFG2	★ 1: Normal Operation; Lane # definition matches socket pin map definition 0: Lane Reversed

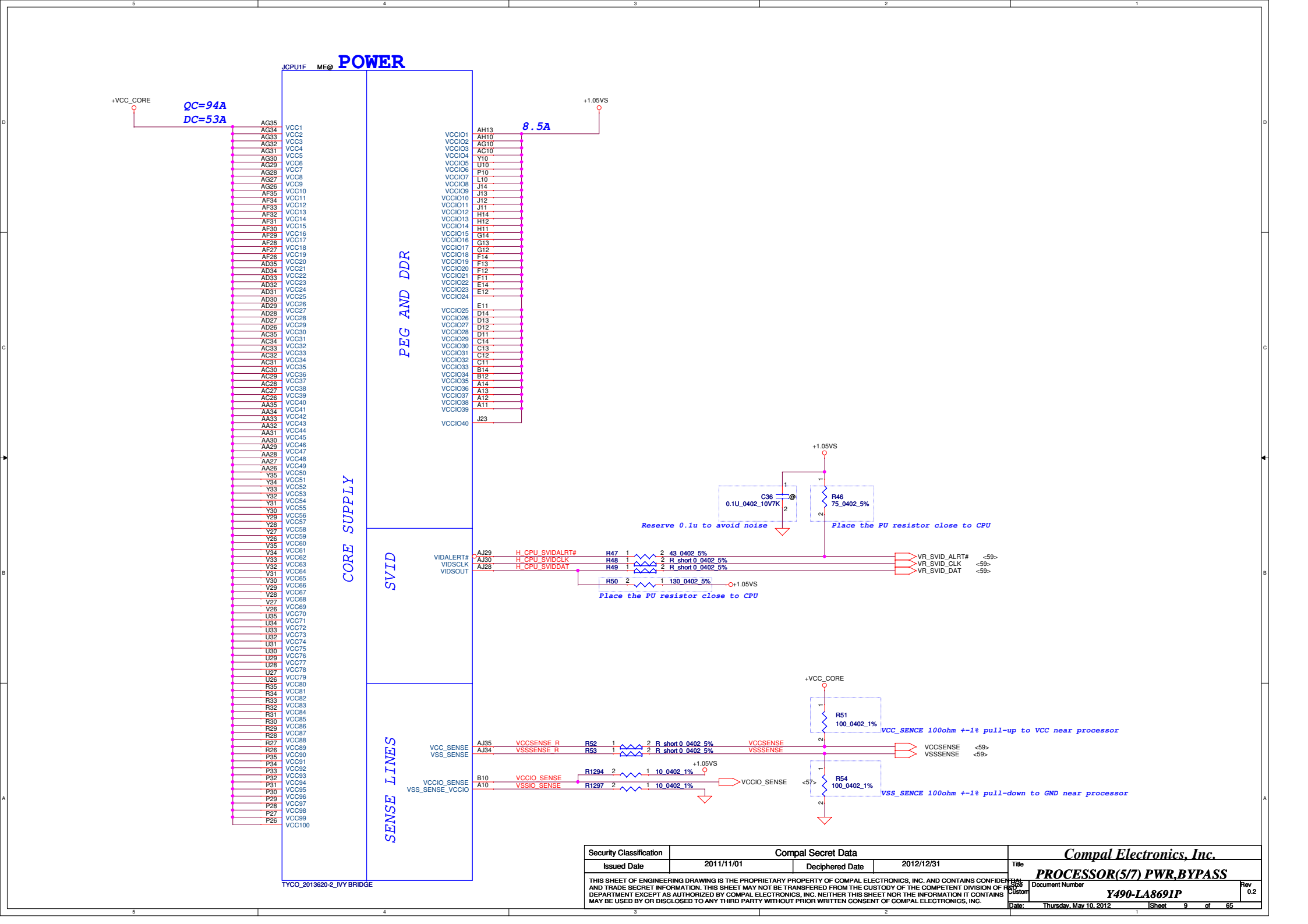
Display Port Presence Strap	
CFG4	★ 1 : Disabled; No Physical Display Port attached to Embedded Display Port 0 : Enabled; An external Display Port device is connected to the Embedded Display Port



PCIe Port Bifurcation Straps	
CFG[6:5]	11: (Default) x16 - Device 1 functions 1 and 2 disabled ★10: x8, x8 - Device 1 function 1 enabled ; function 2 disabled 01: Reserved - (Device 1 function 1 disabled ; function 2 enabled) 00: x8,x4,x4 - Device 1 functions 1 and 2 enabled

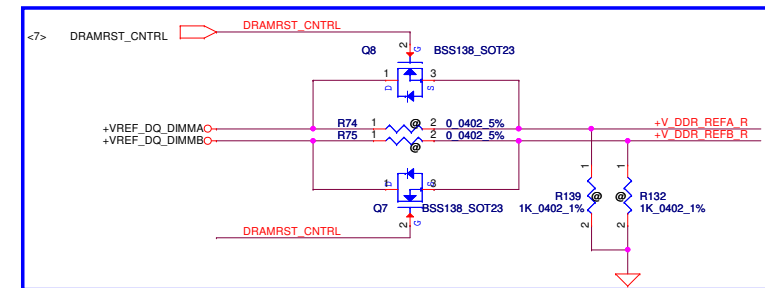
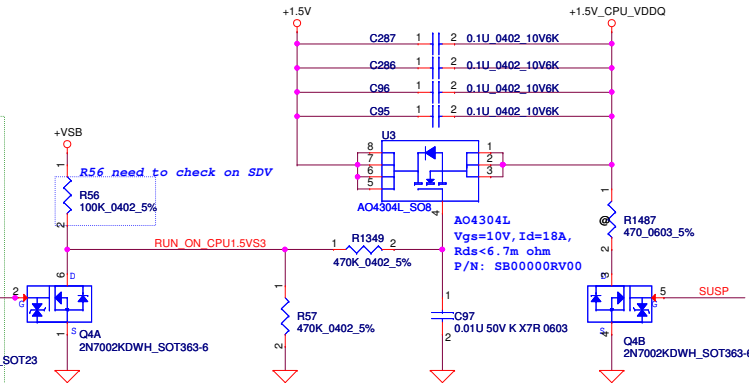
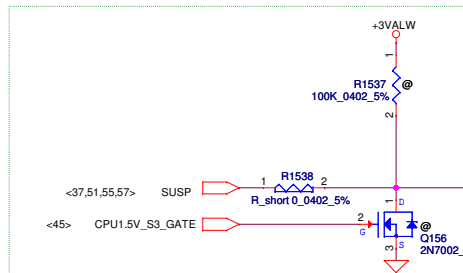


PEG DEFER TRAINING	
CFG7	1: (Default) PEG Train immediately following xxRESETB de assertion 0: PEG Wait for BIOS for training



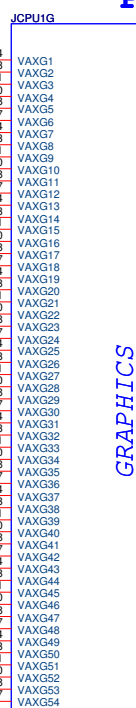
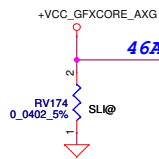
+1.5V_CPU_VDDQ

For Deep S3



6/8: Add M3 Circuit (Processor Generated SO-DIMM VREF_DQ)

POWER



SENSE LINES

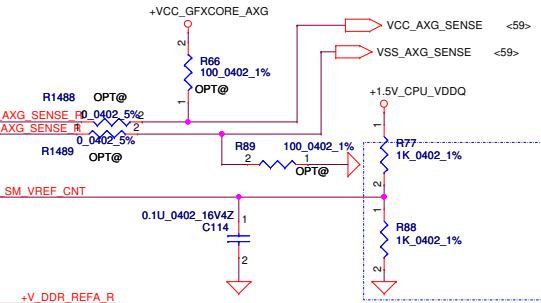
VREF

DDR3 - 1.5V RAILS

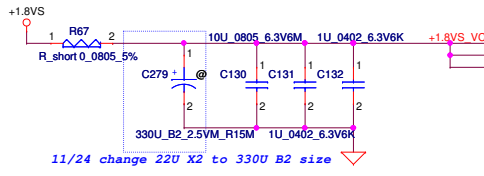
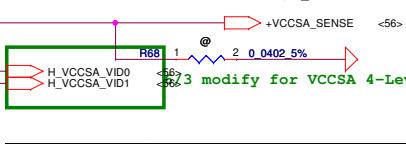
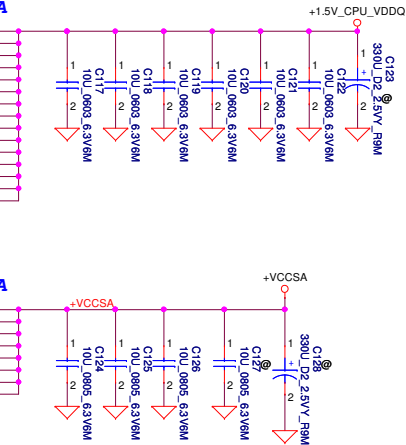
SA RAIL

MISC

1.8V RAIL



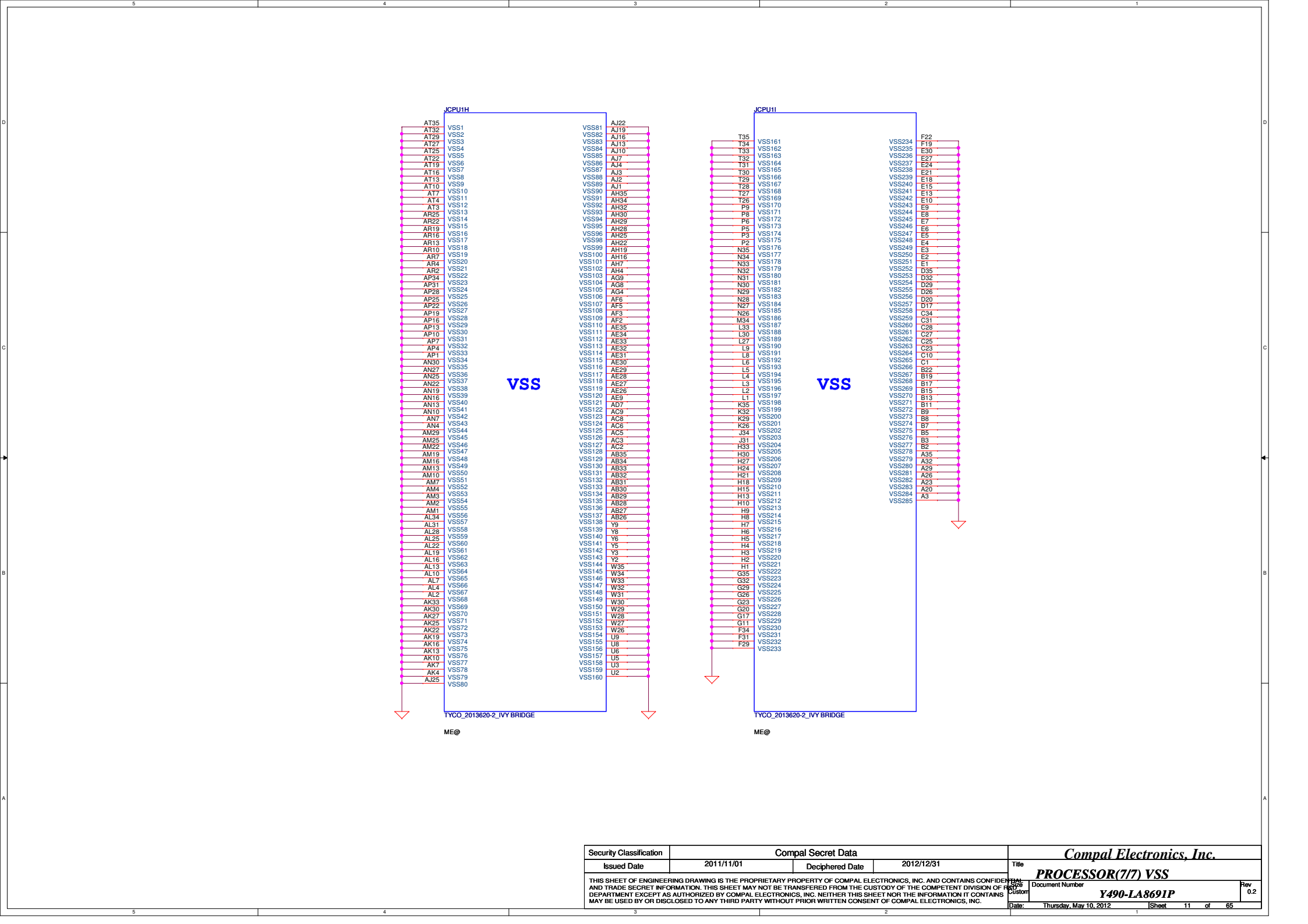
Place the PU/PD resistor close to CPU within 2 inch (Reserve power side)



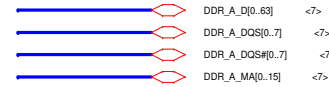
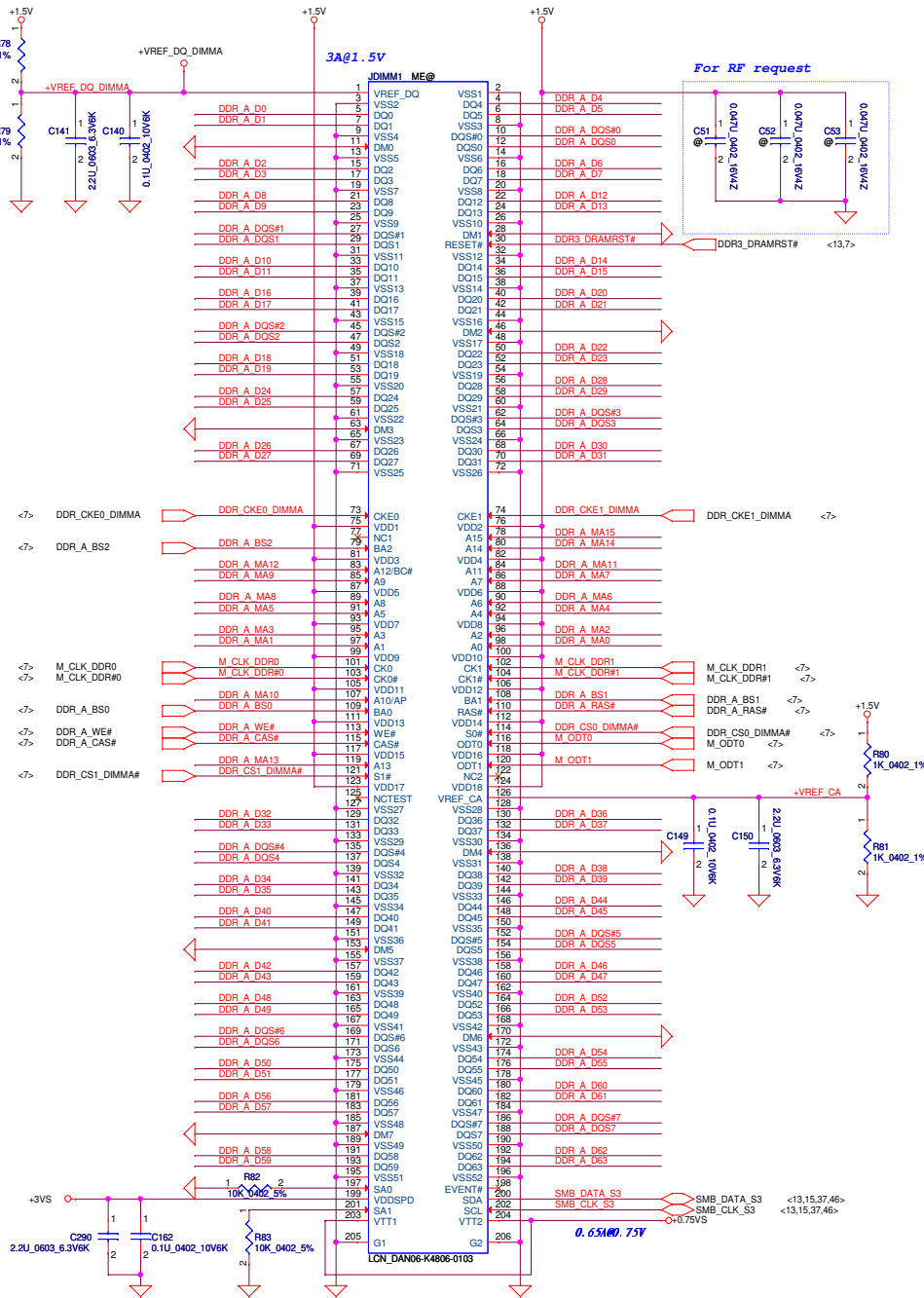
TYCO 2013620-2_VYBRIDGE

ME@

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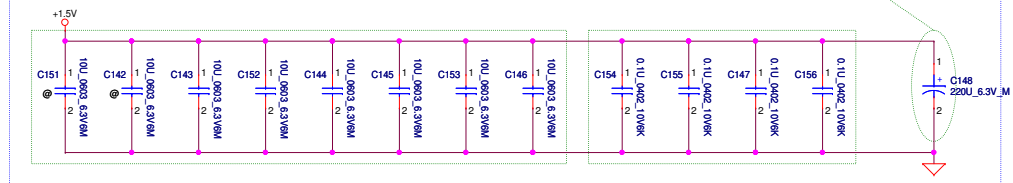


DDR3 SO-DIMM A

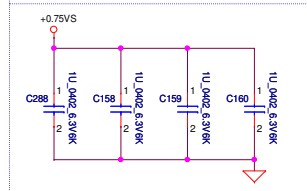


Layout Note:
Place near DIMM

OSCON (220uF 6.3V 4.2L ESR17m) *1=(SF000002Y00)
(10uF_0603_6.3V)*8
(0.1uF_402_10V)*4



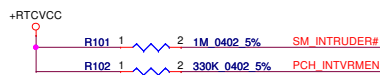
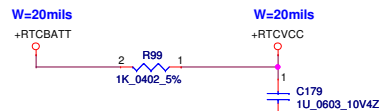
Layout Note:
Place near DIMM



Layout Note:
Place near DIMM

DDR_A_DM[0:7] connect to GND

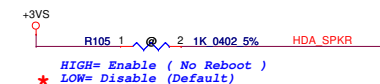
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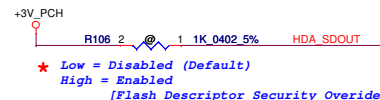
INTVRMEN

* H : Integrated VRM enable (Default)
L : Integrated VRM disable

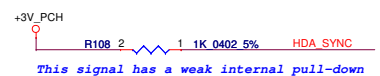
(INTVRMEN should always be pull high.)



HIGH= Enable (No Reboot)
* LOW= Disable (Default)



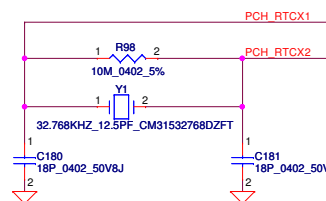
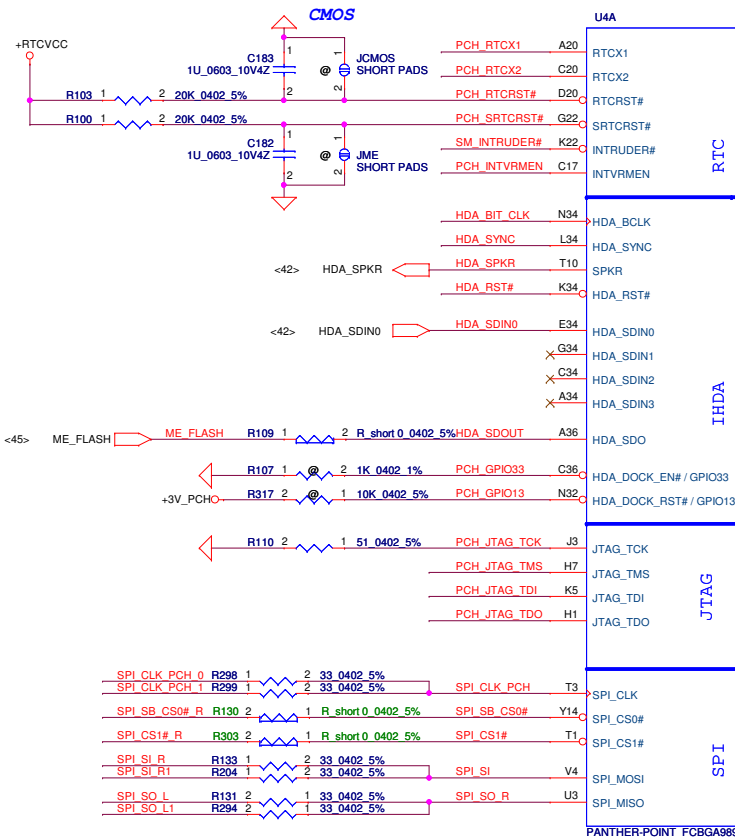
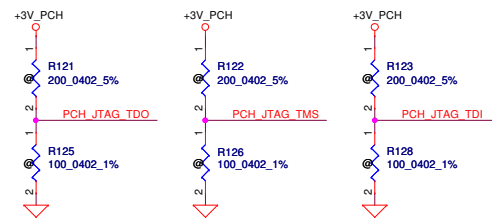
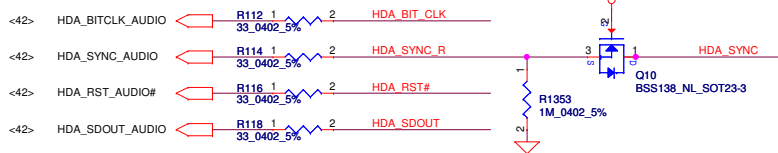
* Low = Disabled (Default)
High = Enabled
[Flash Descriptor Security Override]



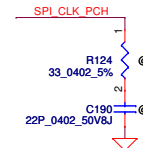
This signal has a weak internal pull-down

On Die PLL VR Select is supplied by
1.5V when sampled high (Default)
1.8V when sampled low
Needs to be pulled High for Chief River platform

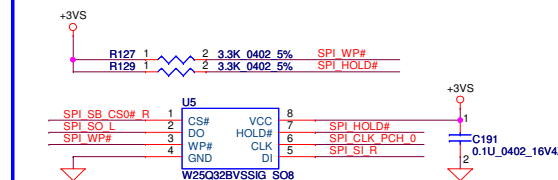
HDA AUDIO



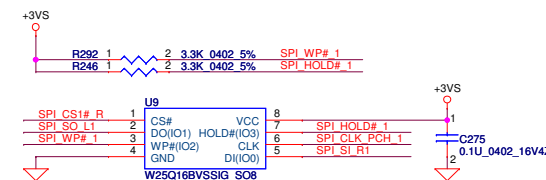
For EMI



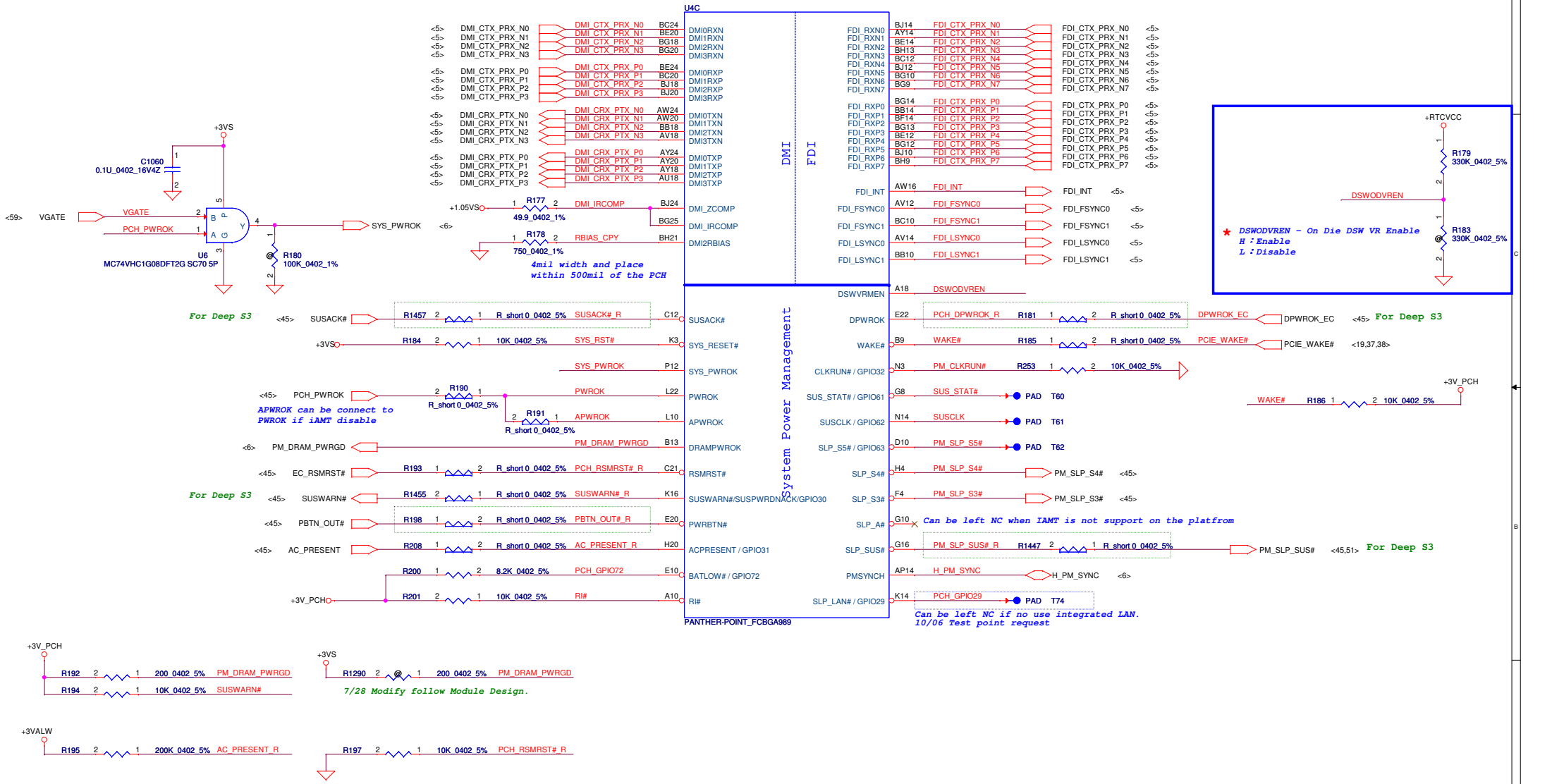
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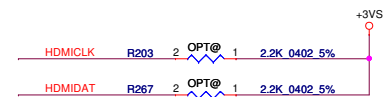
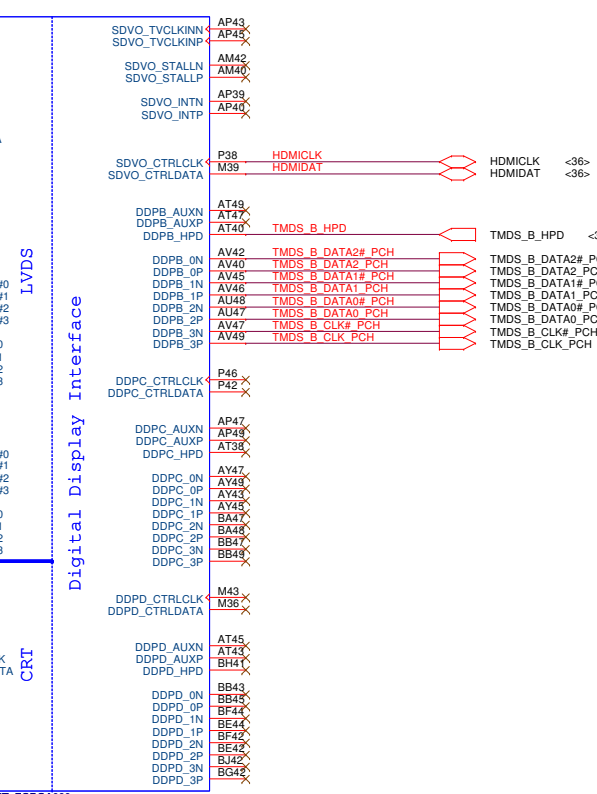
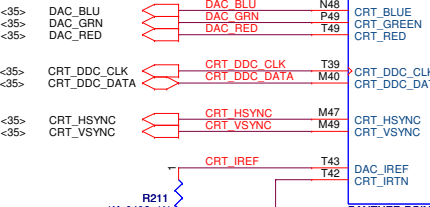
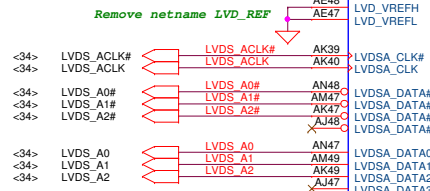
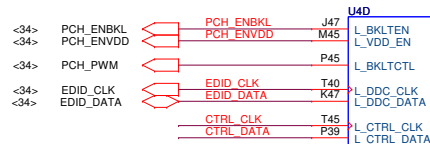
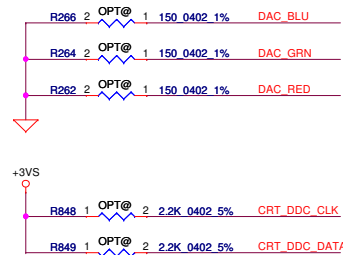
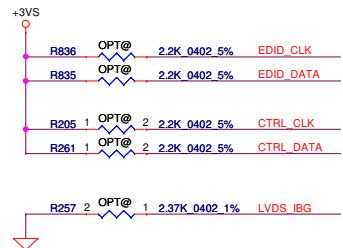


2MB P/N : SA00003FO10

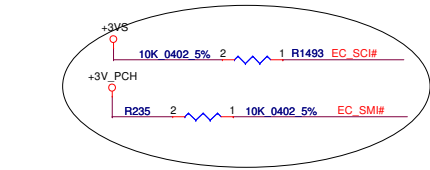


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Issued Date	2011/11/01	Deciphered Date	2012/12/31	PCH (U9) SATA,HDA,SPI, LPC, XDP
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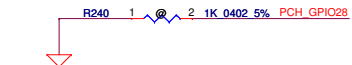


Function	PCH_GPIO38	PCH_GPIO67	PCH_GPIO70
Optimus	0	0	X
Reserve	0	1	X
DIS (SLI)	1	0	X
Reserve	1	1	X
14"	X	X	0
15"	X	X	1

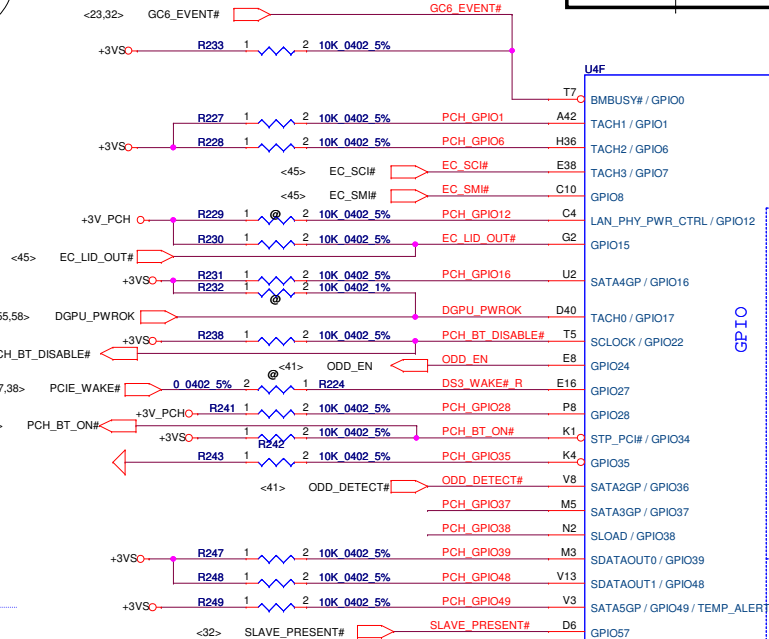
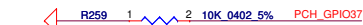
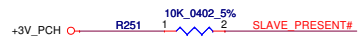
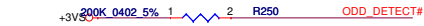
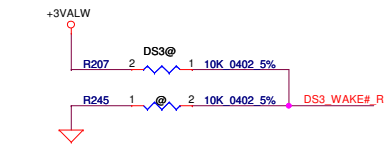


GPIO28
On-Die PLL Voltage Regulator
This signal has a weak internal pull up

* H : On-Die voltage regulator enable
L : On-Die PLL Voltage Regulator disable

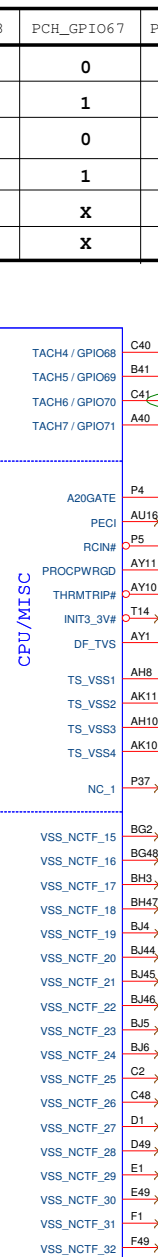


* PCH_GPIO27 (Have internal Pull-High)
High: VCCVRM VR Enable
Low: VCCVRM VR Disable



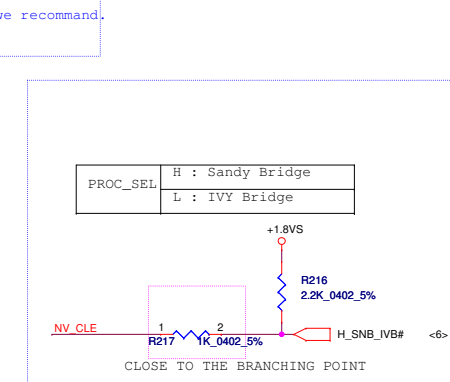
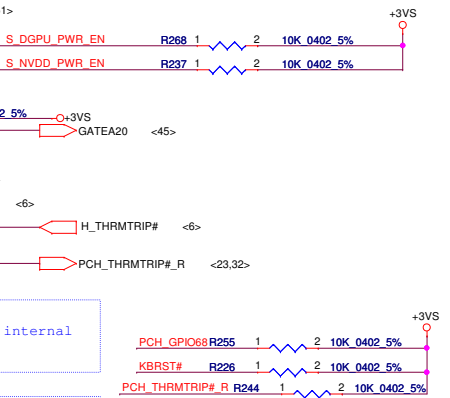
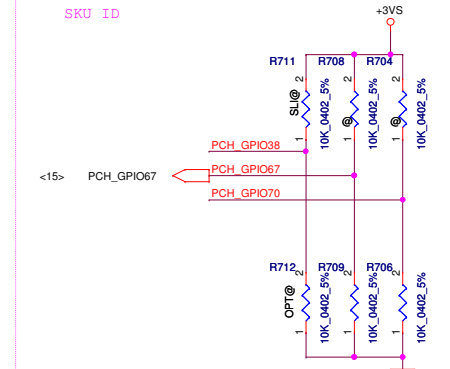
GPIO

NCTF



CPU/MISC

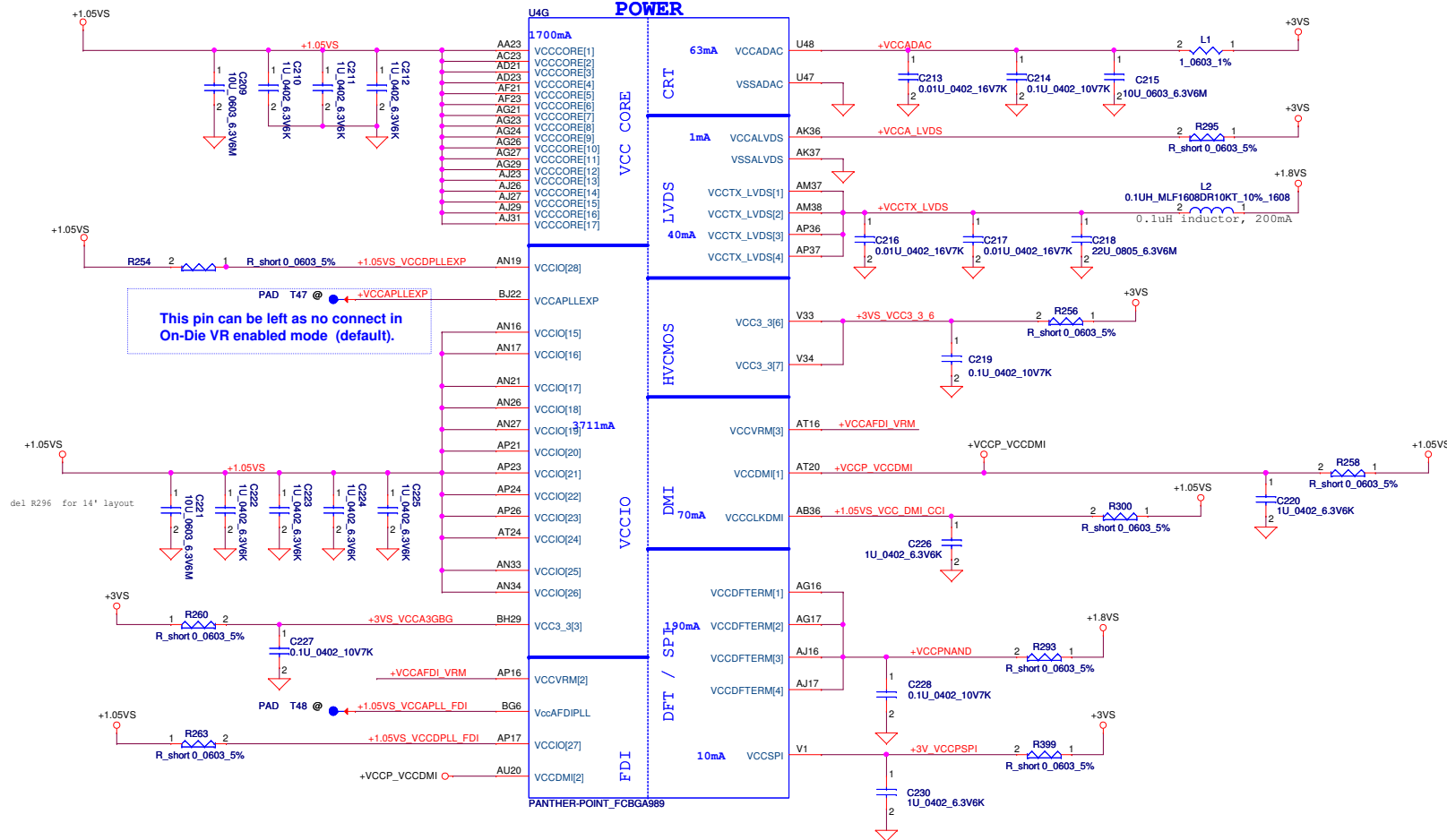
NCTF



PROC_SEL	H : Sandy Bridge L : IVY Bridge
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Title	PCH (6/9) GPIO, CPU, MISC
Rev	0.2

L1 change to 1 ohm P/N
S RES 1/10W 1 +-1% 0603



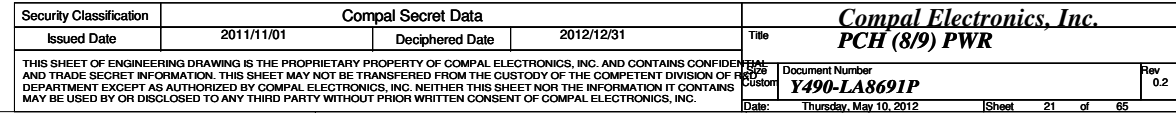
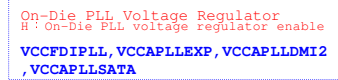
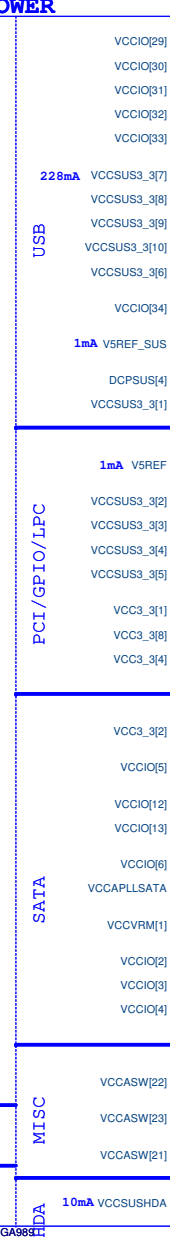
This pin can be left as no connect in
On-Die VR enabled mode (default).

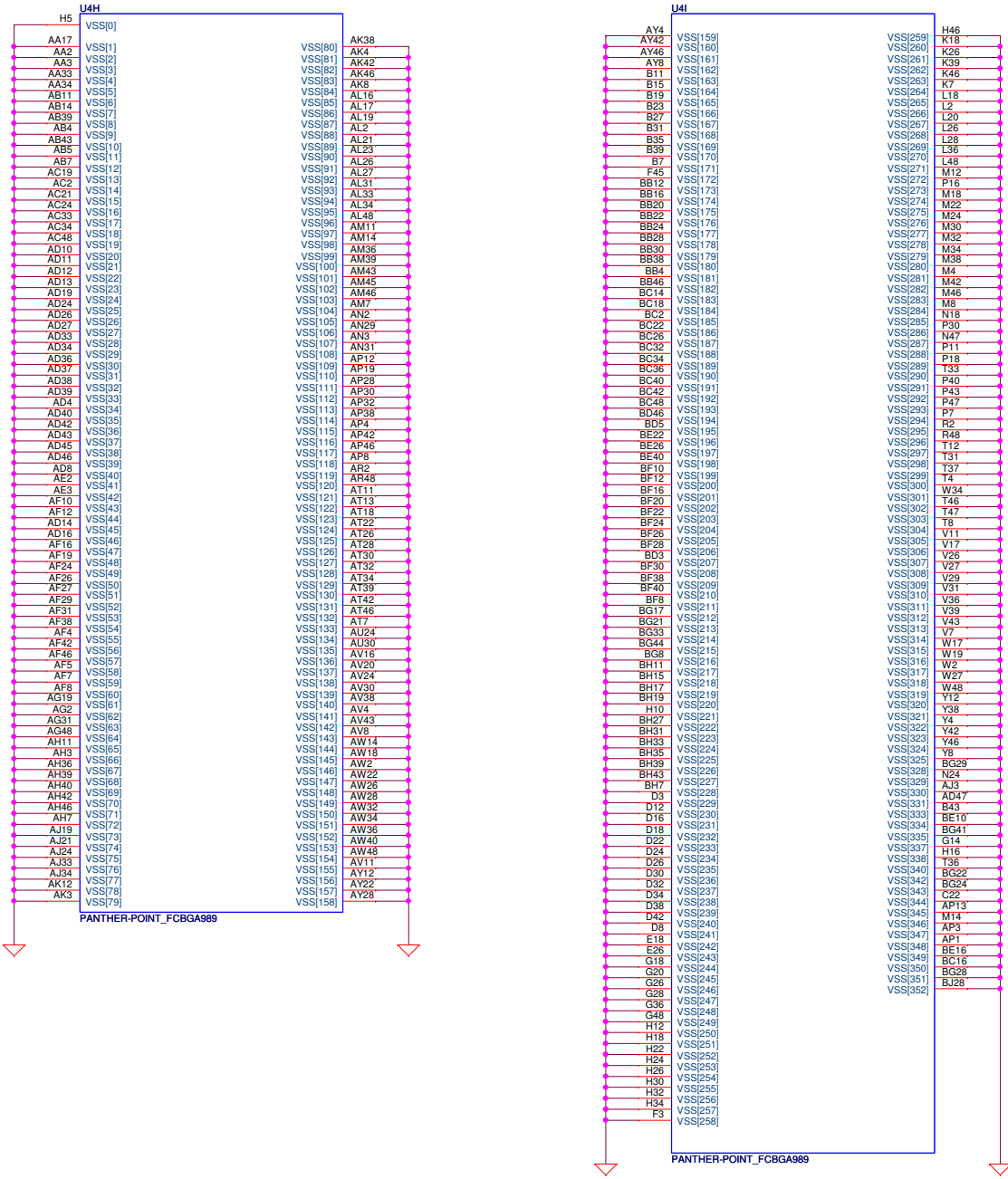
Intel recommend VCCVRM==>1.5V FOR MOBILE
stuff R265 and unstuff R266 VCCVRM==>1.8V FOR DESKTOP
VCCVRM = 160mA detal waiting for newest spec

PCH Power Rail Table
Refer to CPU EDS R1.5

Voltage Rail	Voltage	S0 Iccmax Current (A)
V_PROC_IO	1.05	0.001
V5REF	5	0.001
V5REF_Sus	5	0.001
Vcc3_3	3.3	0.228
VccADAC	3.3	0.063
VccADPLLA	1.05	0.08
VccADPLLB	1.05	0.08
VccCore	1.05	1.7
VccDMI	1.05	0.047
VccIO	1.05	3.711
VccASW	1.05	0.903
VccSPI	3.3	0.01
VccDSW	3.3	0.001
VccDFTERM	1.8	0.002
VccRTC	3.3	6 uA
VccSus3_3	3.3	0.095
VccSusHDA	3.3 / 1.5	0.01
VccVRM	1.8 / 1.5	0.167
VccCLKDMI	1.05	0.07
VccSSC	1.05	0.095
VccDIFFCLKN	1.05	0.055
VccALVDS	3.3	0.001
VccTX_LVDS	1.8	0.04

OT
T_FCB

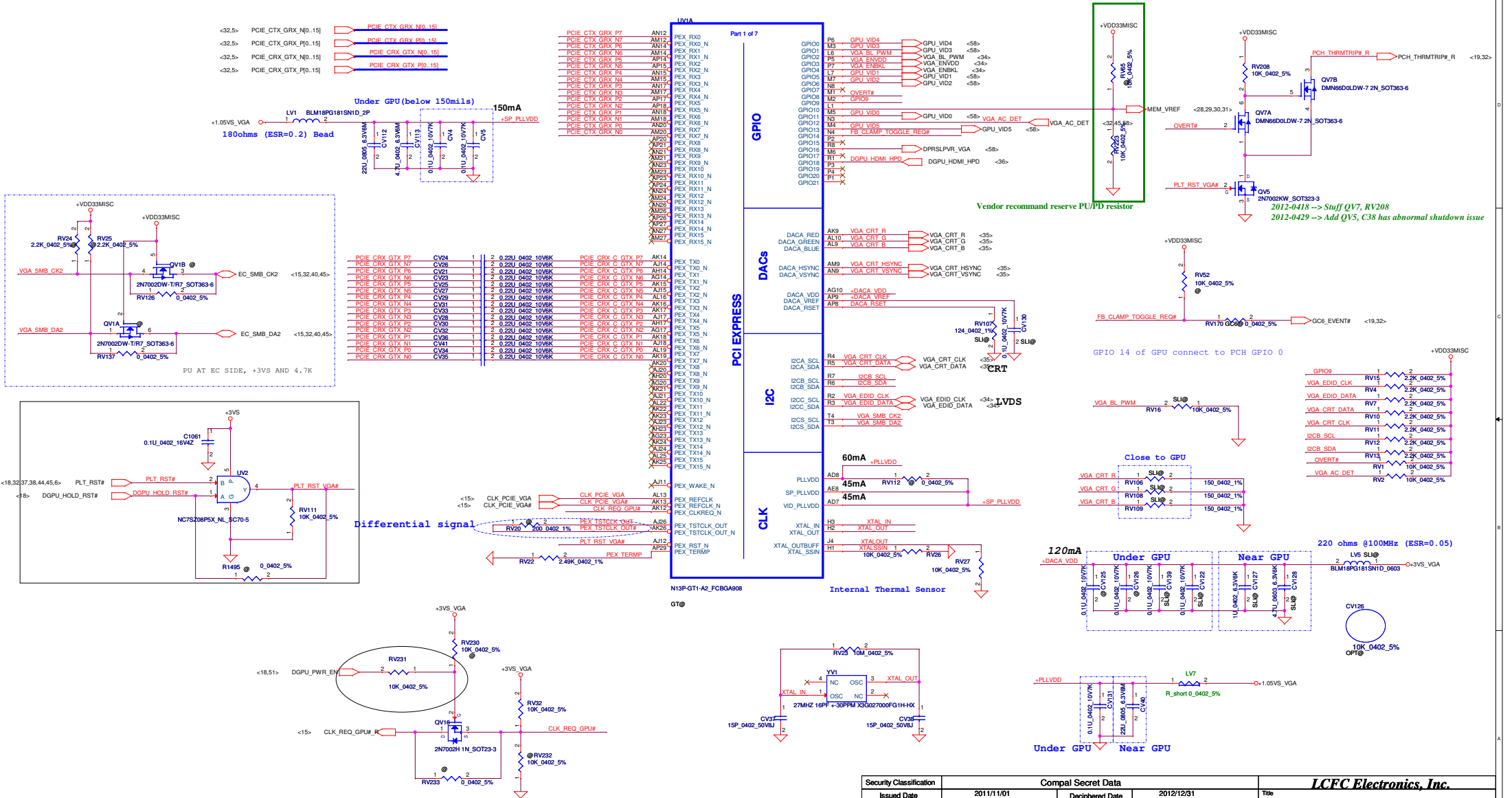




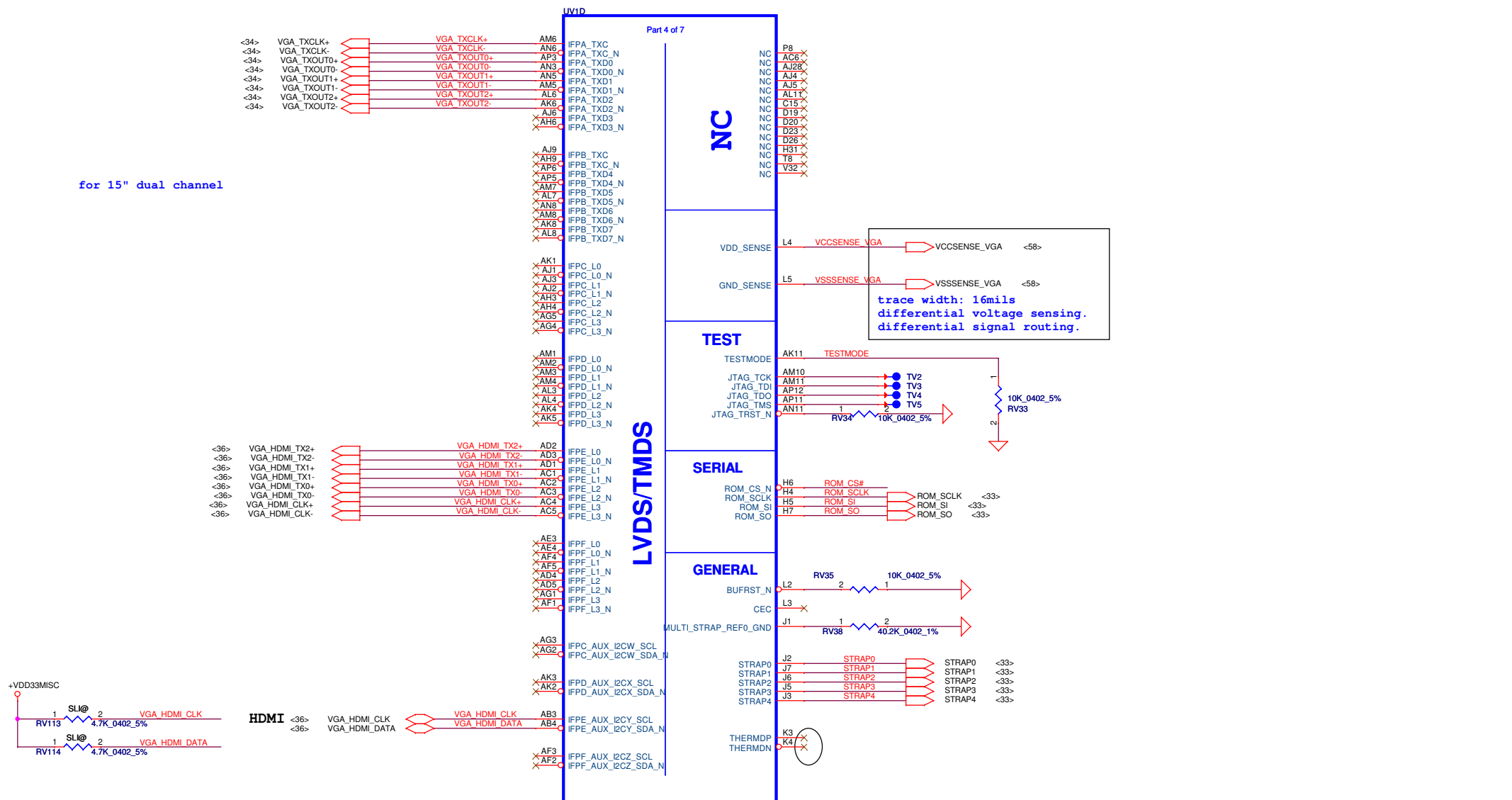
PANTHER-POINT_FCBGA989

PANTHER-POINT_FCBGA989

Security Classification	Compal Secret Data			Compal Electronics, Inc.	
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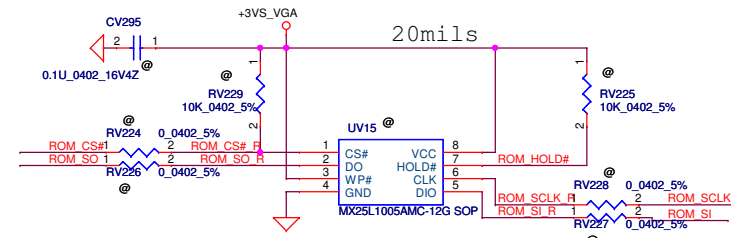
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N13P-PCIE/DAC/GPIO				Document Number
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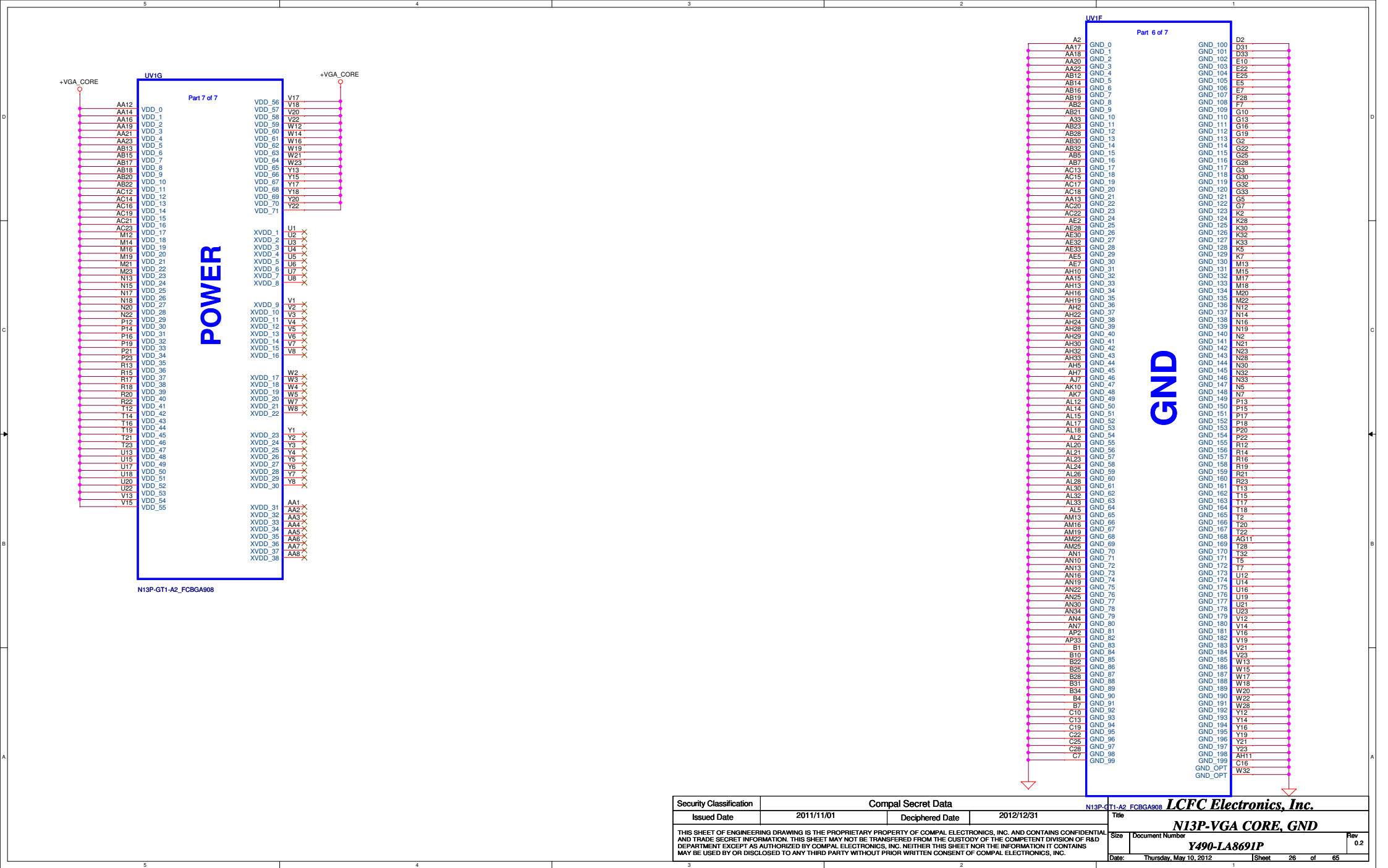
for 15" dual channel

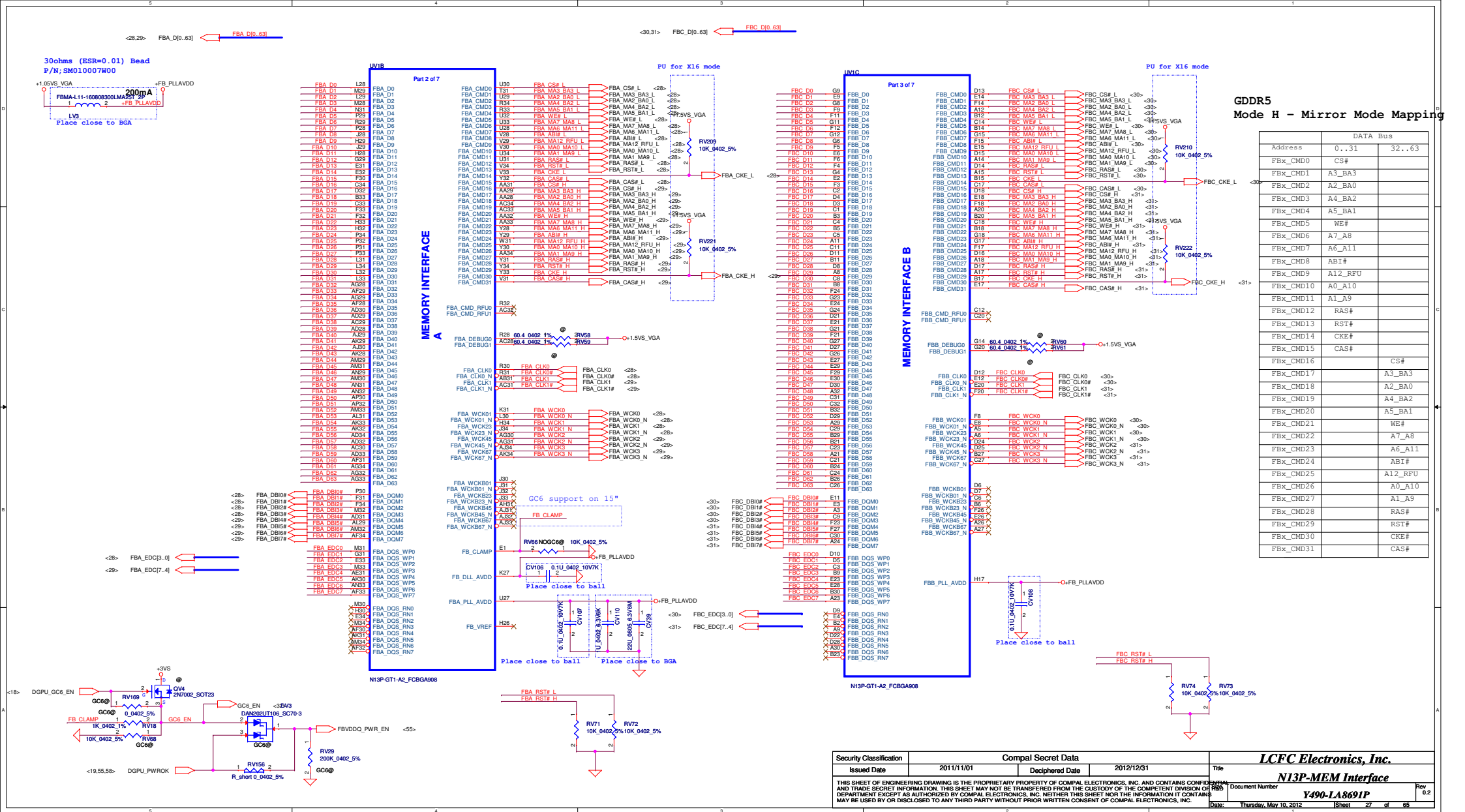
VCCSENSE_VGA <58>
VSSSENSE_VGA <58>
trace width: 16mils
differential voltage sensing.
differential signal routing.

1MB SPI ROM FOR VBIOS ROM (SLI)

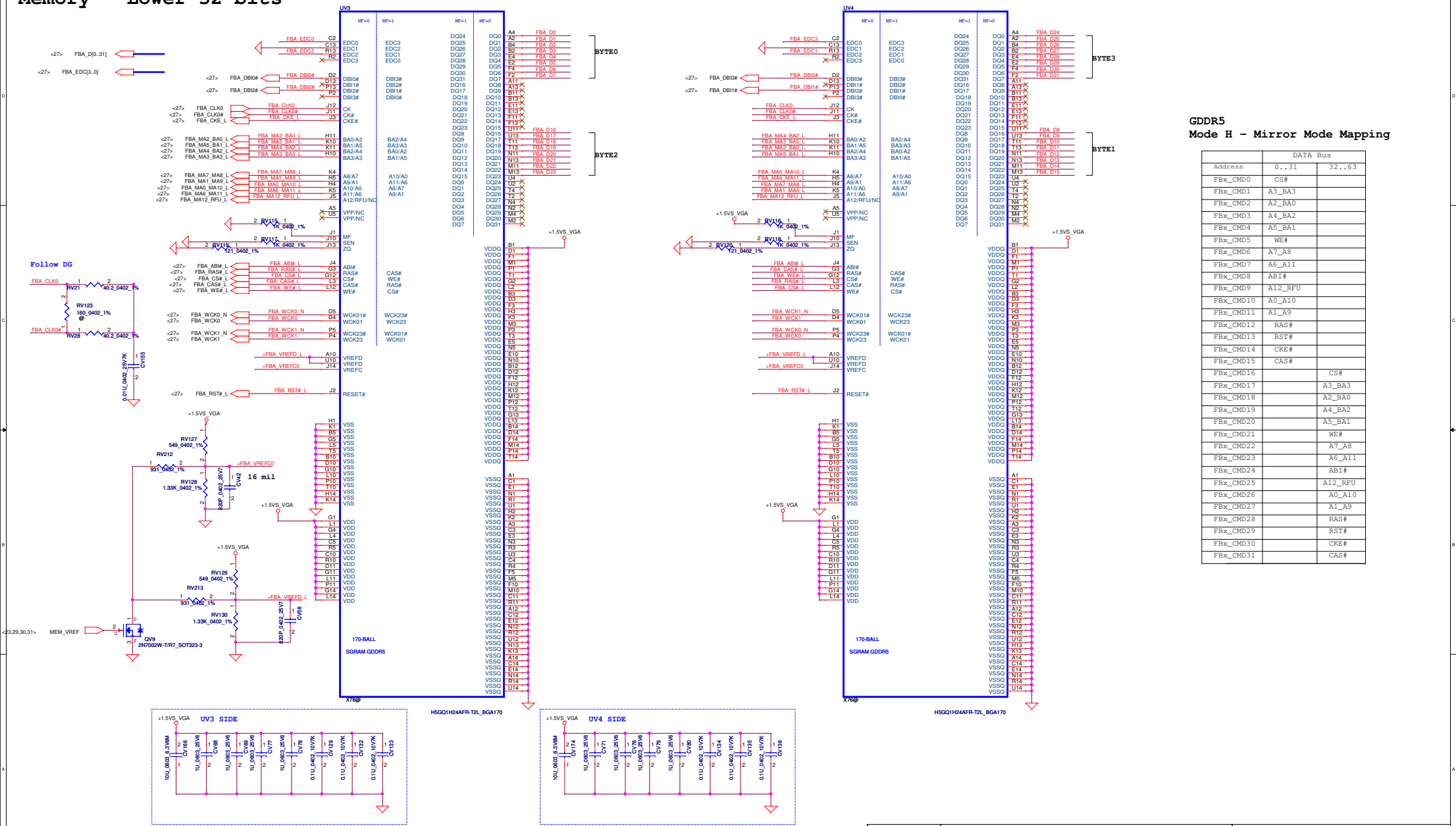


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				Size	Document Number
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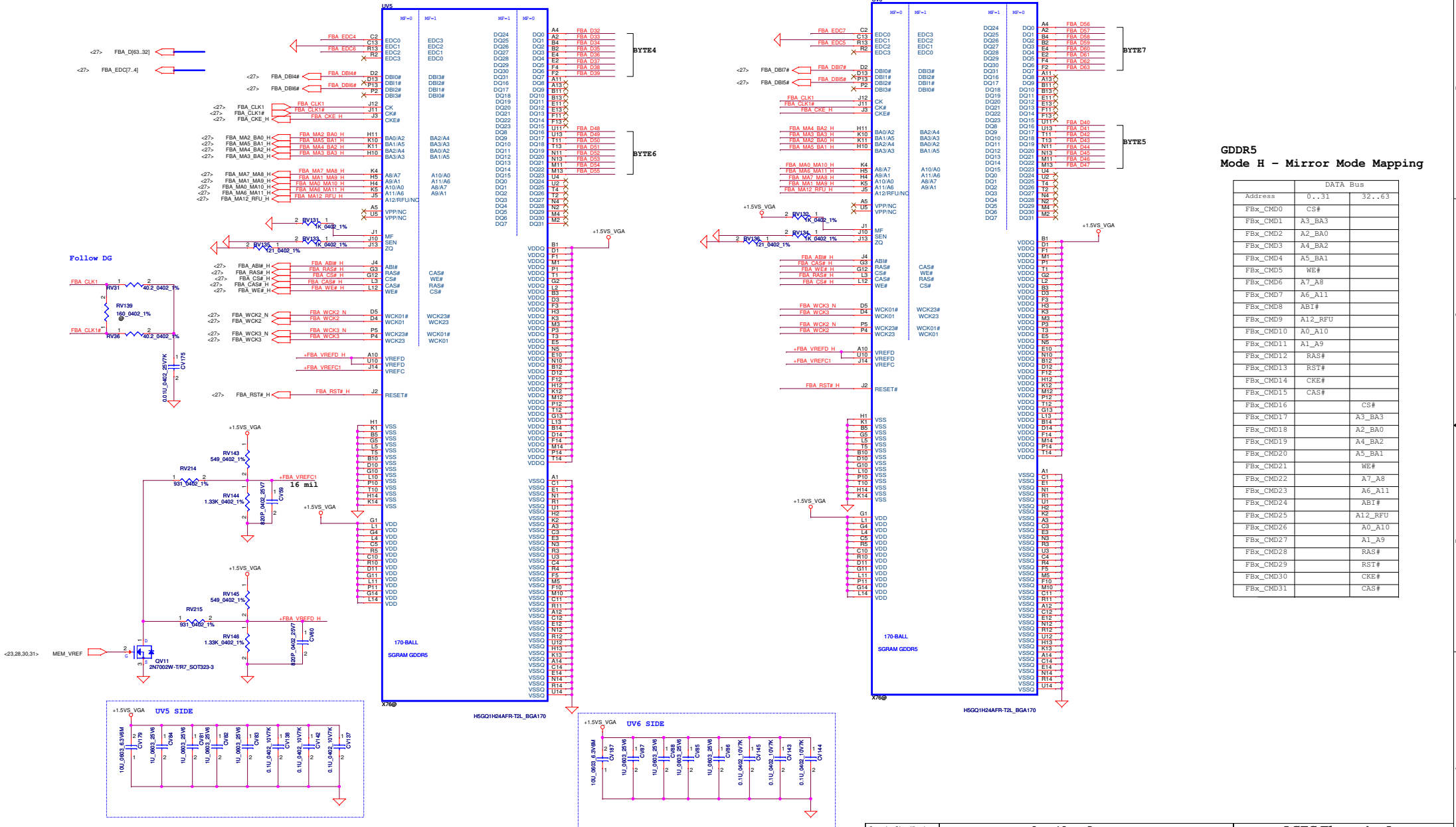
Memory - Lower 32 bits



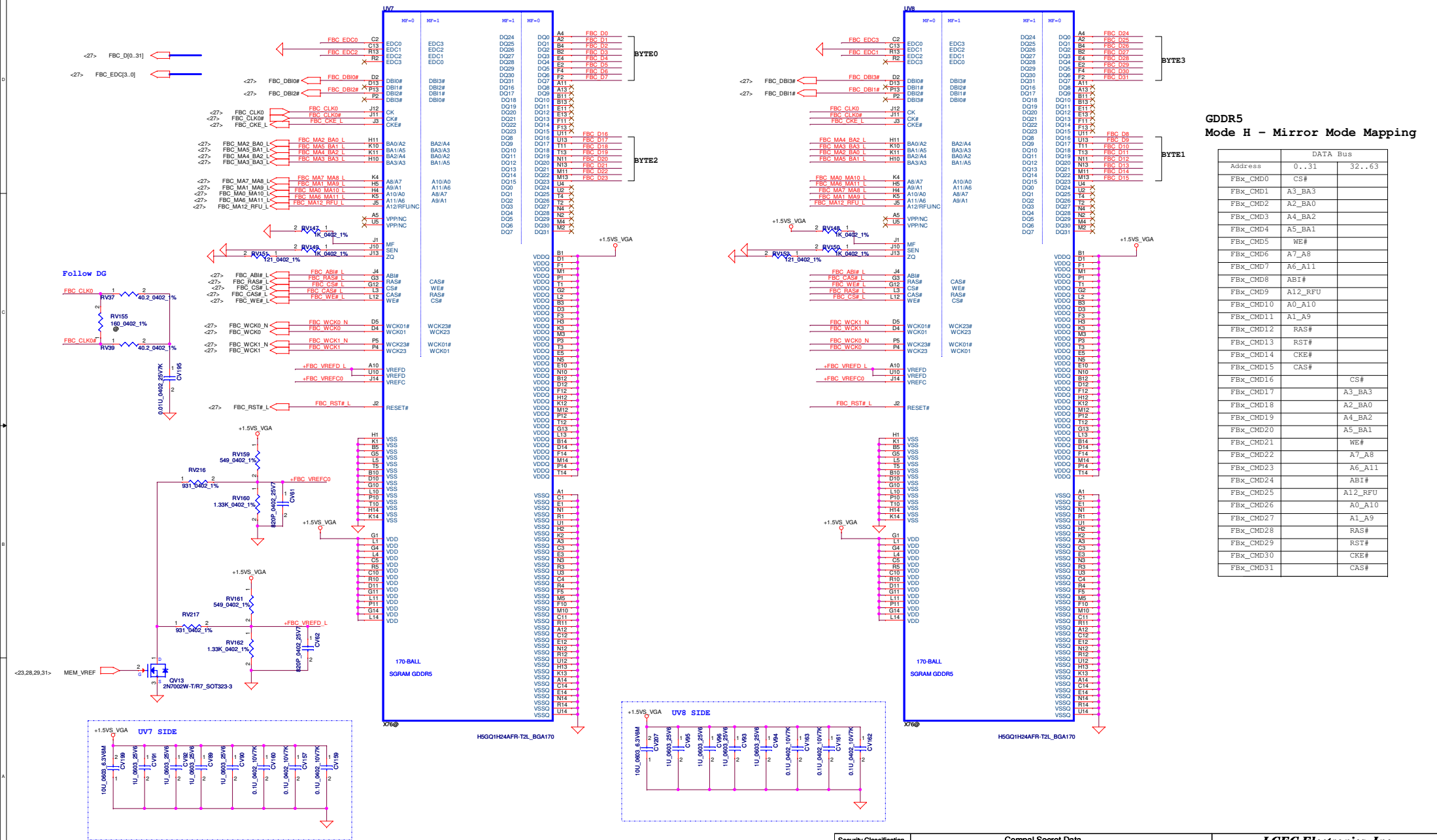
GDDR5
Mode H - Mirror Mode Mapping

Address	DATA Bus	
	0..31	32..63
FBx_CMD0	CS#	
FBx_CMD1	A3_BA3	
FBx_CMD2	A2_BA0	
FBx_CMD3	A4_BA2	
FBx_CMD4	A5_BA1	
FBx_CMD5	WE#	
FBx_CMD6	A7_A8	
FBx_CMD7	A6_A11	
FBx_CMD8	AB1#	
FBx_CMD9	A12_RFU	
FBx_CMD10	A0_A10	
FBx_CMD11	A1_A9	
FBx_CMD12	RAS#	
FBx_CMD13	RST#	
FBx_CMD14	CKE#	
FBx_CMD15	CAS#	
FBx_CMD16		CS#
FBx_CMD17		A3_BA3
FBx_CMD18		A2_BA0
FBx_CMD19		A4_BA2
FBx_CMD20		A5_BA1
FBx_CMD21		WE#
FBx_CMD22		A7_A8
FBx_CMD23		A6_A11
FBx_CMD24		AB1#
FBx_CMD25		A12_RFU
FBx_CMD26		A0_A10
FBx_CMD27		A1_A9
FBx_CMD28		RAS#
FBx_CMD29		RST#
FBx_CMD30		CKE#
FBx_CMD31		CAS#

Memory - Upper 32 bits



Memory Partition C - Lower 32 bits



GDDR5

Mode H - Mirror Mode Mapping

	DATA Bus	
Address	0..31	32..63
FBX_CMD0	CS#	
FBX_CMD1	A3_BA3	
FBX_CMD2	A2_BA0	
FBX_CMD3	A4_BA2	
FBX_CMD4	A5_BA1	
FBX_CMD5	WE#	
FBX_CMD6	A7_A8	
FBX_CMD7	A6_A11	
FBX_CMD8	ABI#	
FBX_CMD9	A12_RFU	
FBX_CMD10	A0_A10	
FBX_CMD11	A1_A9	
FBX_CMD12	RAS#	
FBX_CMD13	RST#	
FBX_CMD14	CKE#	
FBX_CMD15	CAS#	
FBX_CMD16		CS#
FBX_CMD17		A3_BA3
FBX_CMD18		A2_BA0
FBX_CMD19		A4_BA2
FBX_CMD20		A5_BA1
FBX_CMD21		WE#
FBX_CMD22		A7_A8
FBX_CMD23		A6_A11
FBX_CMD24		ABI#
FBX_CMD25		A12_RFU
FBX_CMD26		A0_A10
FBX_CMD27		A1_A9
FBX_CMD28		RAS#
FBX_CMD29		RST#
FBX_CMD30		CKE#
FBX_CMD31		CAS#

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[illegible]

Memory Partition C - Upper 32 bits

FBK_CMD Values Table:

FBK_CMD	Value
FBK_CMD0	CS#
FBK_CMD1	A3_BA3
FBK_CMD2	A2_BA0
FBK_CMD3	A4_BA2
FBK_CMD4	A5_BA1
FBK_CMD5	WE#
FBK_CMD6	A7_A8
FBK_CMD7	A6_A11
FBK_CMD8	AB1#
FBK_CMD9	A12_RFU
FBK_CMD10	A0_A10
FBK_CMD11	A1_A9
FBK_CMD12	RA5#
FBK_CMD13	RST#
FBK_CMD14	CKE#
FBK_CMD15	CAS#
FBK_CMD16	
FBK_CMD17	A3_BA3
FBK_CMD18	A2_BA0
FBK_CMD19	A4_BA2
FBK_CMD20	A5_BA1
FBK_CMD21	WE#
FBK_CMD22	A7_A8
FBK_CMD23	A6_A11
FBK_CMD24	AB1#
FBK_CMD25	A12_RFU
FBK_CMD26	A0_A10
FBK_CMD27	A1_A9
FBK_CMD28	RA5#
FBK_CMD29	RST#
FBK_CMD30	CKE#
FBK_CMD31	CAS#

FBK_CMD Values Table:

FBK_CMD	Value
FBK_CMD32	A2_BA0
FBK_CMD33	A4_BA2
FBK_CMD34	A5_BA1
FBK_CMD35	WE#
FBK_CMD36	A7_A8
FBK_CMD37	A6_A11
FBK_CMD38	AB1#
FBK_CMD39	A12_RFU
FBK_CMD40	A0_A10
FBK_CMD41	A1_A9
FBK_CMD42	RA5#
FBK_CMD43	RST#
FBK_CMD44	CKE#
FBK_CMD45	CAS#
FBK_CMD46	
FBK_CMD47	A3_BA3
FBK_CMD48	A2_BA0
FBK_CMD49	A4_BA2
FBK_CMD50	A5_BA1
FBK_CMD51	WE#
FBK_CMD52	A7_A8
FBK_CMD53	A6_A11
FBK_CMD54	AB1#
FBK_CMD55	A12_RFU
FBK_CMD56	A0_A10
FBK_CMD57	A1_A9
FBK_CMD58	RA5#
FBK_CMD59	RST#
FBK_CMD60	CKE#
FBK_CMD61	CAS#
FBK_CMD62	
FBK_CMD63	A3_BA3
FBK_CMD64	A2_BA0
FBK_CMD65	A4_BA2
FBK_CMD66	A5_BA1
FBK_CMD67	WE#
FBK_CMD68	A7_A8
FBK_CMD69	A6_A11
FBK_CMD70	AB1#
FBK_CMD71	A12_RFU
FBK_CMD72	A0_A10
FBK_CMD73	A1_A9
FBK_CMD74	RA5#
FBK_CMD75	RST#
FBK_CMD76	CKE#
FBK_CMD77	CAS#
FBK_CMD78	
FBK_CMD79	A3_BA3
FBK_CMD80	A2_BA0
FBK_CMD81	A4_BA2
FBK_CMD82	A5_BA1
FBK_CMD83	WE#
FBK_CMD84	A7_A8
FBK_CMD85	A6_A11
FBK_CMD86	AB1#
FBK_CMD87	A12_RFU
FBK_CMD88	A0_A10
FBK_CMD89	A1_A9
FBK_CMD90	RA5#
FBK_CMD91	RST#
FBK_CMD92	CKE#
FBK_CMD93	CAS#
FBK_CMD94	
FBK_CMD95	A3_BA3

[illegible]

Memory Partition C - Upper 32 bits

Diagram illustrating the memory partitioning and mapping for the upper 32 bits of Memory Partition C, showing connections to various memory banks (FBX_CMD0 to FBX_CMD31) and the GDDR5 Mode H - Mirror Mode Mapping.

The diagram is divided into two main sections: **FBX_CMD0 to FBX_CMD31** and **GDDR5 Mode H - Mirror Mode Mapping**.

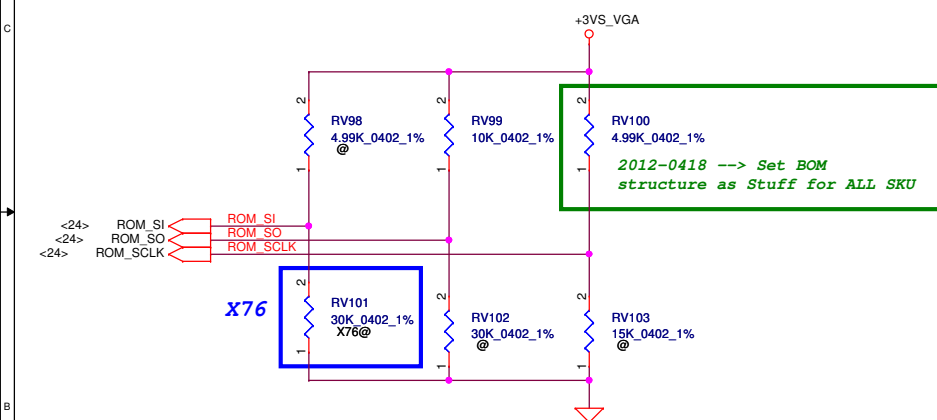
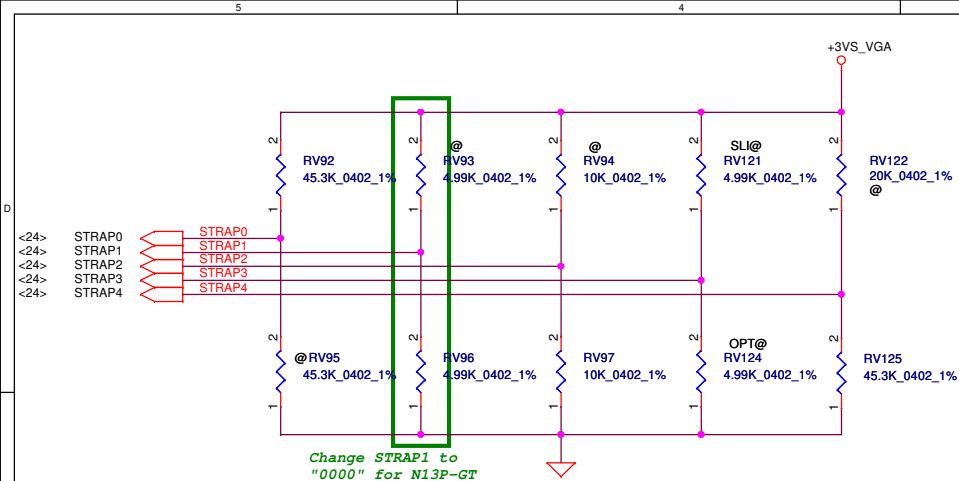
FBX_CMD0 to FBX_CMD31: This section shows the mapping of memory banks to the upper 32 bits of Memory Partition C. The banks are organized into four groups (FBX_CMD0 to FBX_CMD3, FBX_CMD4 to FBX_CMD7, FBX_CMD8 to FBX_CMD11, and FBX_CMD12 to FBX_CMD15) and are connected to the memory partition via various signals (e.g., FBX_CMD0, FBX_CMD1, FBX_CMD2, FBX_CMD3, FBX_CMD4, FBX_CMD5, FBX_CMD6, FBX_CMD7, FBX_CMD8, FBX_CMD9, FBX_CMD10, FBX_CMD11, FBX_CMD12, FBX_CMD13, FBX_CMD14, FBX_CMD15).

GDDR5 Mode H - Mirror Mode Mapping: This section shows the mapping of memory banks to the GDDR5 Mode H - Mirror Mode Mapping. The banks are organized into four groups (FBX_CMD0 to FBX_CMD3, FBX_CMD4 to FBX_CMD7, FBX_CMD8 to FBX_CMD11, and FBX_CMD12 to FBX_CMD15) and are connected to the memory partition via various signals (e.g., FBX_CMD0, FBX_CMD1, FBX_CMD2, FBX_CMD3, FBX_CMD4, FBX_CMD5, FBX_CMD6, FBX_CMD7, FBX_CMD8, FBX_CMD9, FBX_CMD10, FBX_CMD11, FBX_CMD12, FBX_CMD13, FBX_CMD14, FBX_CMD15).

The diagram also includes a table titled "GDDR5 Mode H - Mirror Mode Mapping" showing the mapping of memory banks to the upper 32 bits of Memory Partition C. The table has two columns: "Address" and "DATA Bus".

Address	DATA Bus
0..31	32..63
FBX_CMD0	CS#
FBX_CMD1	A3_BA3
FBX_CMD2	A2_BA0
FBX_CMD3	A4_BA2
FBX_CMD4	A5_BA1
FBX_CMD5	WE#
FBX_CMD6	A7_A8
FBX_CMD7	A6_A11
FBX_CMD8	ABI#
FBX_CMD9	A12_RFU
FBX_CMD10	A0_A10
FBX_CMD11	A1_A9
FBX_CMD12	RA5#
FBX_CMD13	RST#
FBX_CMD14	CKE#
FBX_CMD15	CAS#
FBX_CMD16	CS#
FBX_CMD17	A3_BA3
FBX_CMD18	A2_BA0
FBX_CMD19	A4_BA2
FBX_CMD20	A5_BA1
FBX_CMD21	WE#
FBX_CMD22	A7_A8
FBX_CMD23	A6_A11
FBX_CMD24	ABI#
FBX_CMD25	A12_RFU
FBX_CMD26	A0_A10
FBX_CMD27	A1_A9
FBX_CMD28	RA5#
FBX_CMD29	RST#
FBX_CMD30	CKE#
FBX_CMD31	CAS#

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				Size Custom	Document Number	Rev
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Physical Strapping pin	Power Rail	Logical Strapping Bit3	Logical Strapping Bit2	Logical Strapping Bit1	Logical Strapping Bit0
ROM_SCLK	+3VS_VGA	PCI_DEVID[4]	SUB_VENDOR	SLOT_CLK_CFG	PEX_PLL_EN_TERM
ROM_SI	+3VS_VGA	RAM_CFG[3]	RAM_CFG[2]	RAM_CFG[1]	RAM_CFG[0]
ROM_SO	+3VS_VGA	FB[1]	FB[0]	SMB_ALT_ADDR	VGA_DEVICE
STRAP0	+3VS_VGA	USER[3]	USER[2]	USER[1]	USER[0]
STRAP1	+3VS_VGA	3GIO_PAD_CFG_ADR[3]	3GIO_PAD_CFG_ADR[2]	3GIO_PAD_CFG_ADR[1]	3GIO_PAD_CFG_ADR[0]
STRAP2	+3VS_VGA	PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
STRAP3	+3VS_VGA	SOR3_EXPOSED	SOR2_EXPOSED	SOR1_EXPOSED	SOR0_EXPOSED
STRAP4	+3VS_VGA	RESERVED	PCIE_SPEED_CHANGE_GEN3	PCIE_MAX_SPEED	DP_PLL_VDD33V

Resistor Values	Pull-up to +3VS_VGA	Pull-down to Gnd
5K	1000	0000
10K	1001	0001
15K	1010	0010
20K	1011	0011
25K	1100	0100
30K	1101	0101
35K	1110	0110
45K	1111	0111

SLOT_CLK_CFG	
0	GPU and MCH don't share a common reference clock
1	GPU and MCH share a common reference clock (Default)

SUB_VENDOR	
0	No VBIOS ROM (Default)
1	BIOS ROM is present

3GIO_PADCFG	
3GIO_PADCFG[3:0]	
0000	Notebook Default

XCLK_417	
0	277MHz (Default)
1	Reserved

USER Straps	
User[3:0]	
1000-1100	Customer defined

PEX_PLL_EN_TERM	
0	Disable (Default)
1	Enable

PCIE_MAX_SPEED	
0	Limit to PCIE Gen1
1	PCIE Gen 2/3 Capable

FB_0_BAR_SIZE	
0	Reserved
1	Reserved
2	256MB (Default)
3	Reserved

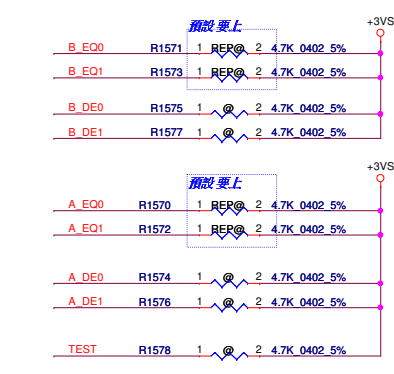
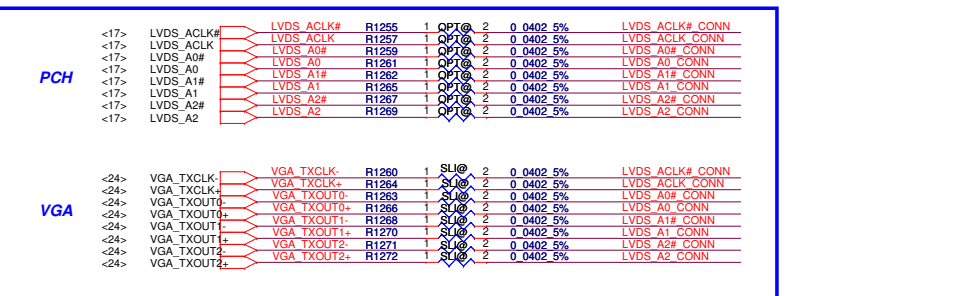
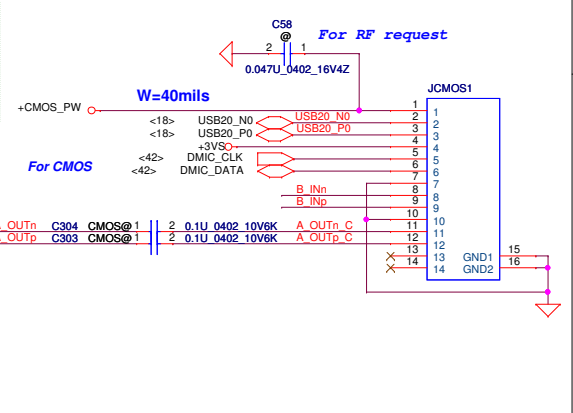
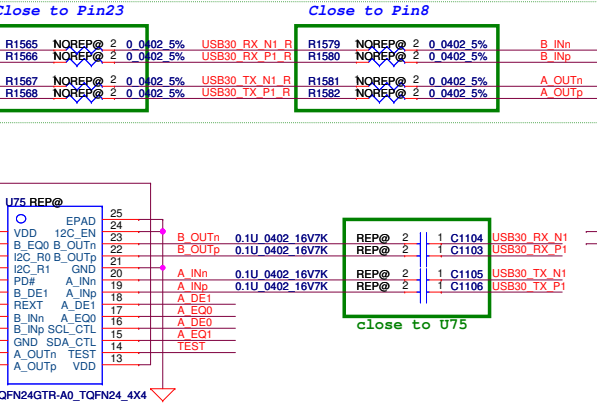
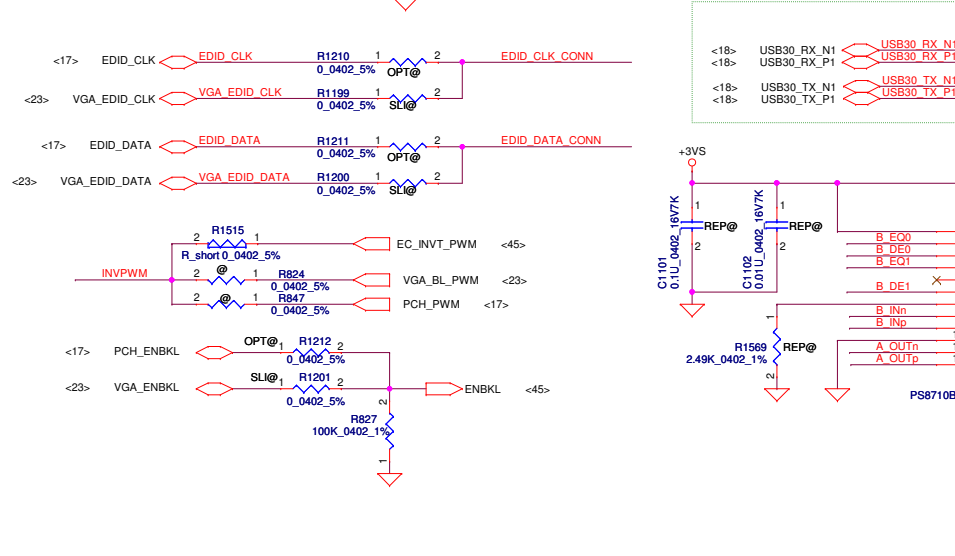
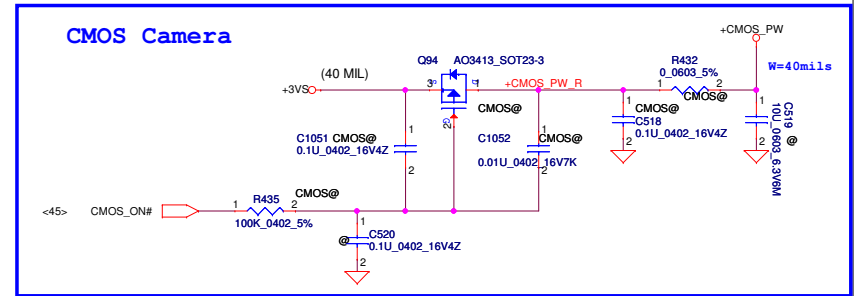
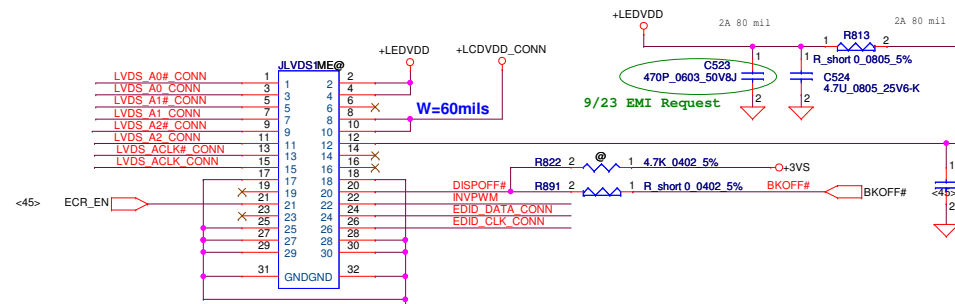
SMBUS_ALT_ADDR	
0	0x9E (Default)
1	0x9C (Multi-GPU usage)

VGA_DEVICE	
0	3D Device (Class Code 302h)
1	VGA Device (Default)

GPU		FB Memory (GDDR5)		ROM_SI	ROM_SO	ROM_SCLK	STRAP0	STRAP1	STRAP2	STRAP3	STRAP4
N13P-GT1 28nm	Samsung	K4G20325FD-FC04 2G 64Mx32		PD 30K	PU 10K	PU 5K (ALL SKU)	PU 45K	PD 5K	PD 10K	PU 5K	PD 10K
		K4G10325FG-HC04 1G 32Mx32		PD 45K							
	Hynix	H5GQ2H24MFR-T2C 2G 64Mx32	EOL	PD 25K							
		H5GQ1H24BFR-T2C 1G 32Mx32		PD 35K							
		H5GQ2H24AFR-T2C 2G 64Mx32		PD 5K							

VRAM	X76	VRAM P/N
Samsung	X76409JVL01 (2G 64Mx32)	SA00005B70J
	X76409JVL51 (1G 32Mx16)	SA00003RS0J
Hynix	X76409JVL02 (2G 64Mx32)	SA00004GD0J EOL
	X76409JVL02 (2G 64Mx32)	SA00004GD1J
	X76409JVL52 (1G 32Mx16)	SA00003WL1J

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				Date	Thursday, May 10, 2012
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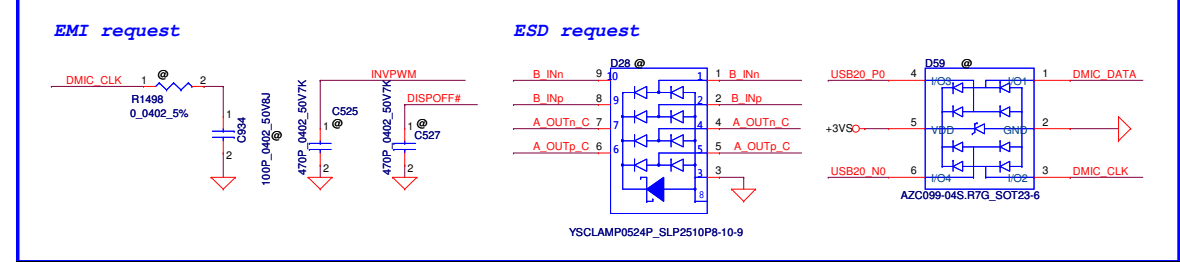
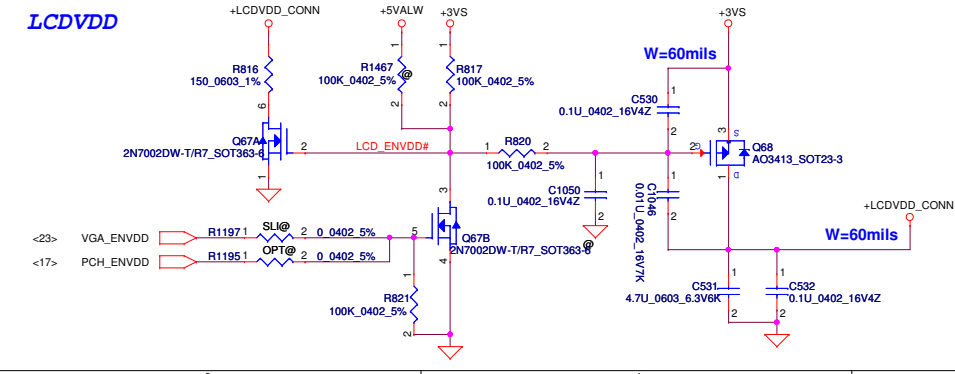
Equalizer control and program for channel B
3.3V tolerant. Internally pulled down at ~150KΩ
[B_EQ0, B_EQ0] ==
LL: adaptive EQ enable
LH: program EQ for channel loss up to 7dB
HL: program EQ for channel loss up to 14.5dB
HH: program EQ for channel loss up to 11.5dB

Programmable output pre-emphasis level setting for channel B
3.3V tolerant. Internally pulled down at ~150KΩ
[B_DE1, B_DE0] ==
**LL: 3.5dB de-emphasis
LH: No de-emphasis
HL: 7dB de-emphasis
HH: 5dB with boost output swing

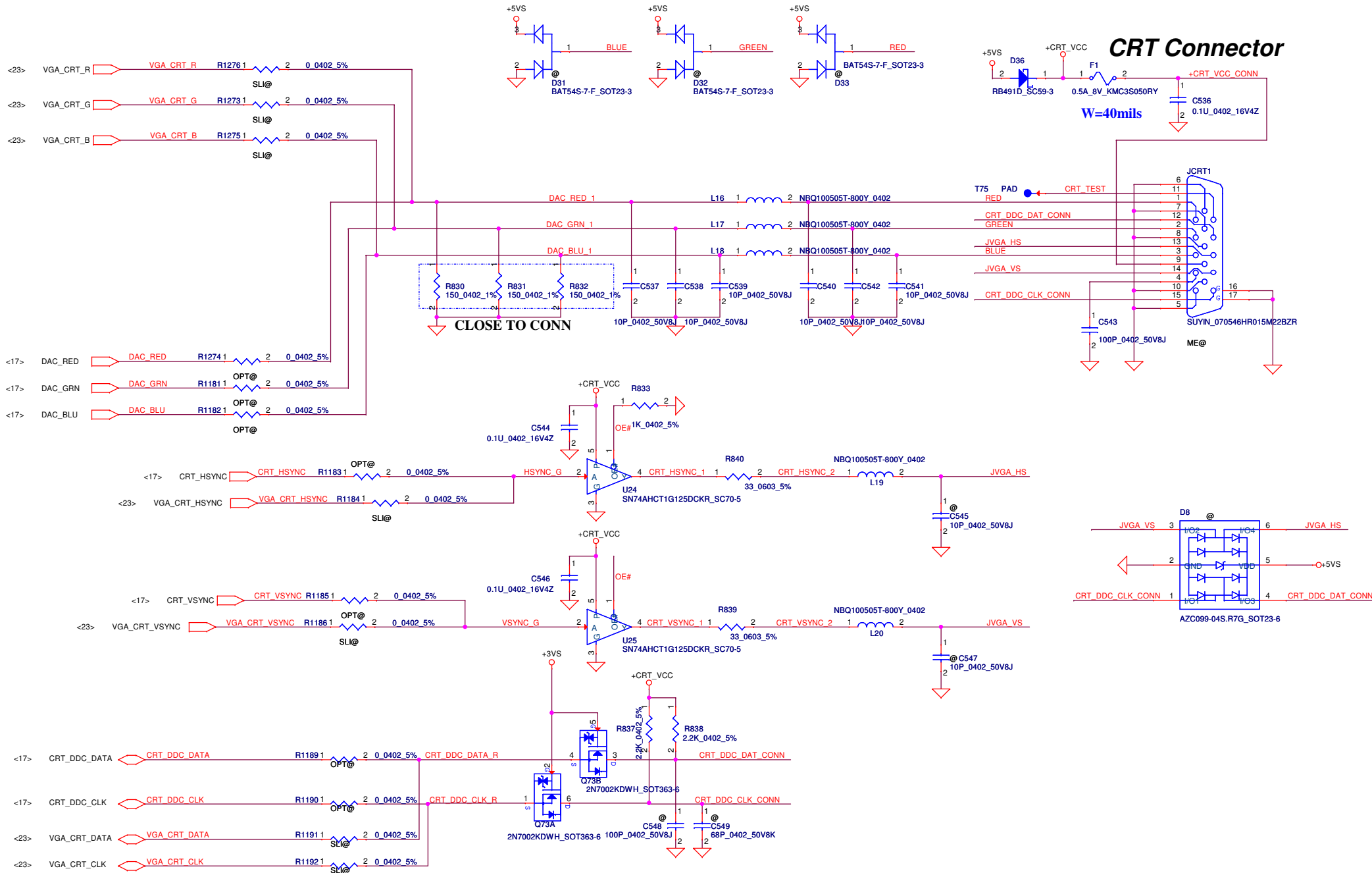
Equalizer control and program for channel A
3.3V tolerant. Internally pulled down at ~150KΩ
[A_EQ0, A_EQ0] ==
LL: adaptive EQ enable
LH: program EQ for channel loss up to 7dB
HL: program EQ for channel loss up to 14.5dB
HH: program EQ for channel loss up to 11.5dB

Programmable output pre-emphasis level setting for channel A
3.3V tolerant. Internally pulled down at ~150KΩ
[A_DE1, A_DE0] ==
**LL: 3.5dB de-emphasis
LH: No de-emphasis
HL: 7dB de-emphasis
HH: 5dB with boost output swing

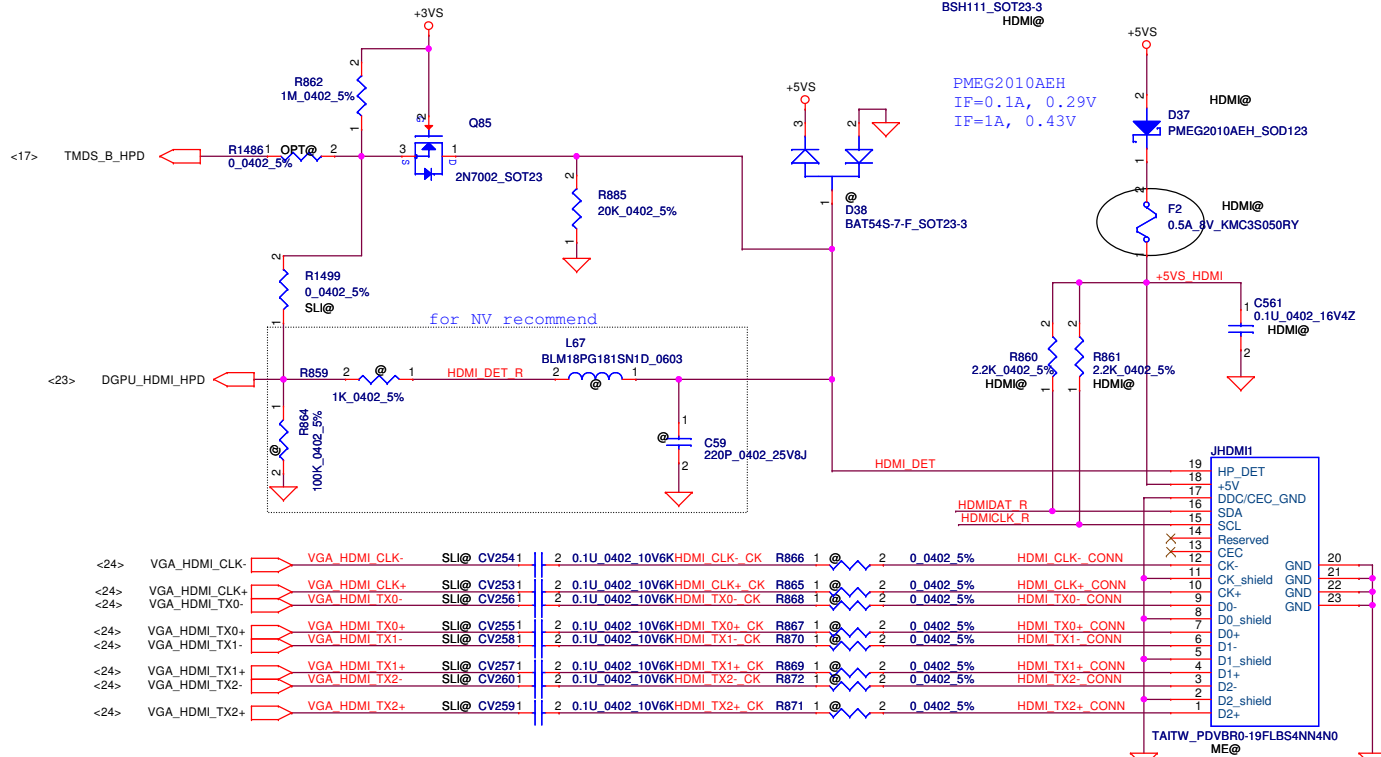
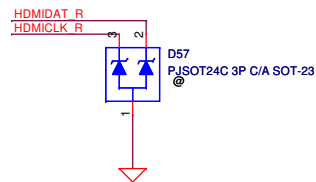
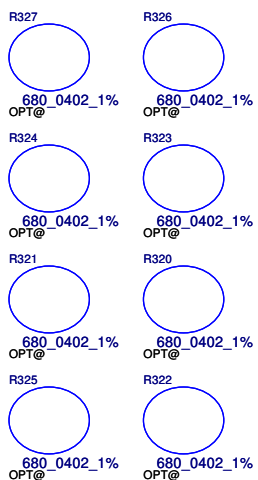
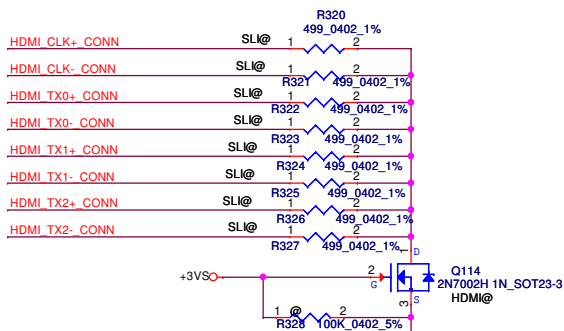
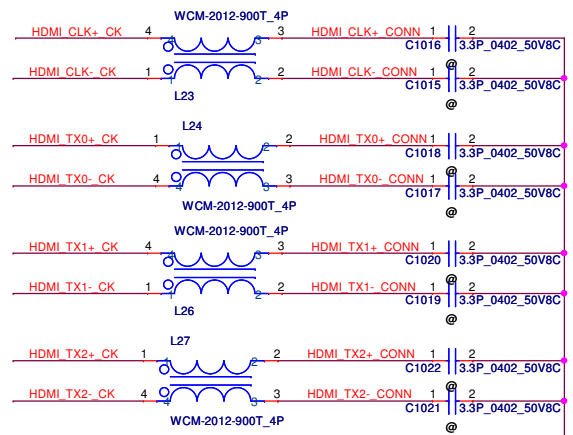
Chip test mode enable.
3.3V tolerant. Internally pulled down at ~150KΩ.
TEST ==
**L: Normal operation (default)
H: Test mode enable



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				Document Number						Rev	
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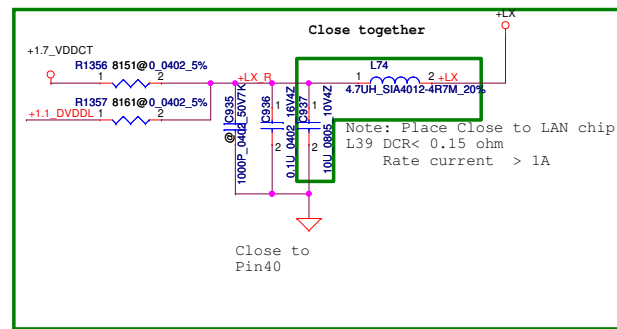
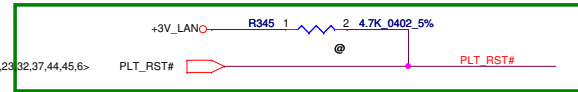
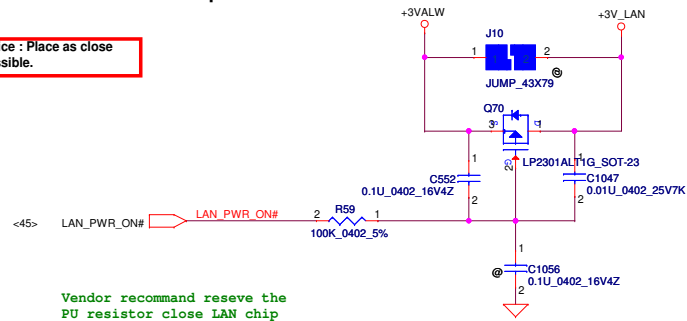
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<17>	TMDS_B_DATA1#_PCH	TMDS B DATA1#_PCH	OPT@	C203	1	2	0.1U	0402	10V6K	HDMI TX1- CK
<17>	TMDS_B_DATA1#_PCH	TMDS B DATA1#_PCH	OPT@	C204	1	2	0.1U	0402	10V6K	HDMI TX1+ CK
<17>	TMDS_B_DATA0#_PCH	TMDS B DATA0#_PCH	OPT@	C205	1	2	0.1U	0402	10V6K	HDMI TX0- CK
<17>	TMDS_B_DATA0#_PCH	TMDS B DATA0#_PCH	OPT@	C206	1	2	0.1U	0402	10V6K	HDMI TX0+ CK
<17>	TMDS_B_CLK#_PCH	TMDS B CLK#_PCH	OPT@	C207	1	2	0.1U	0402	10V6K	HDMI CLK- CK
<17>	TMDS_B_CLK#_PCH	TMDS B CLK#_PCH	OPT@	C208	1	2	0.1U	0402	10V6K	HDMI CLK+ CK



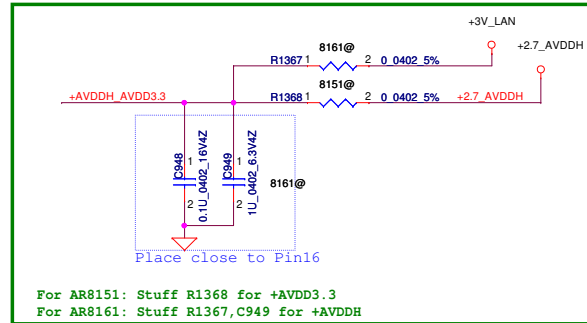
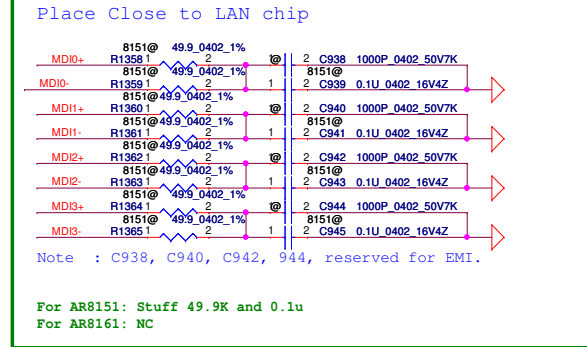
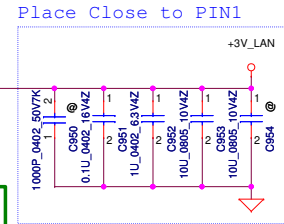
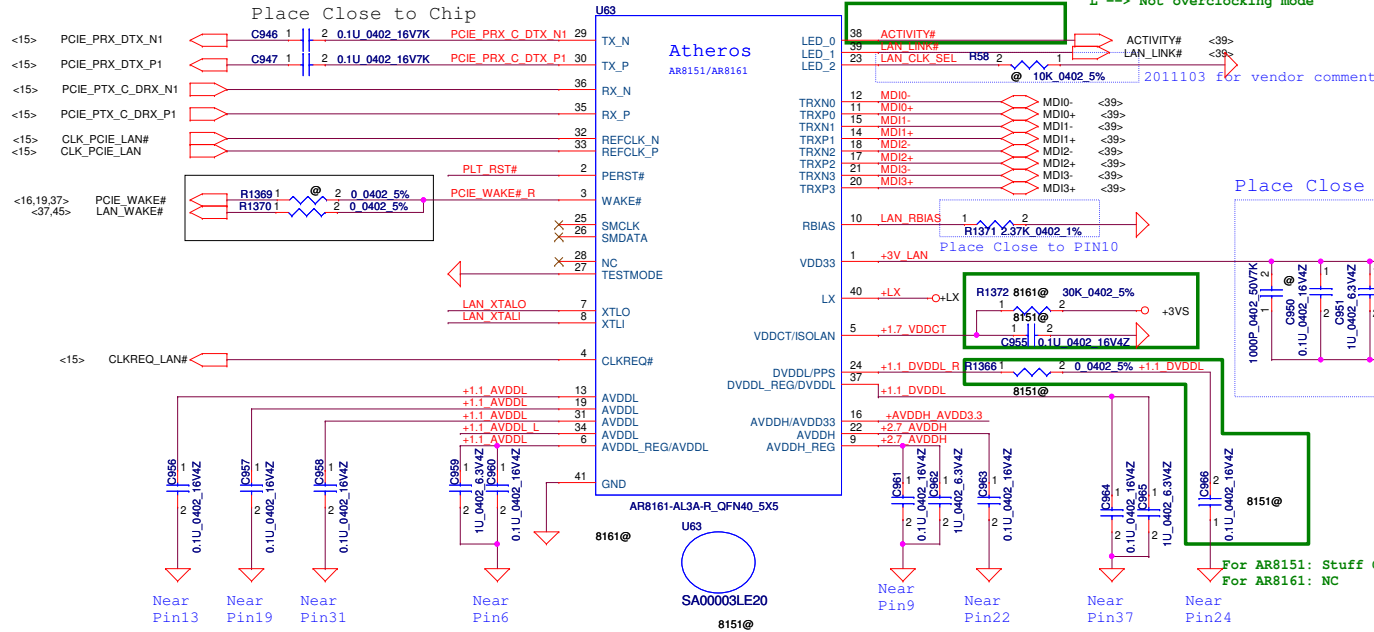
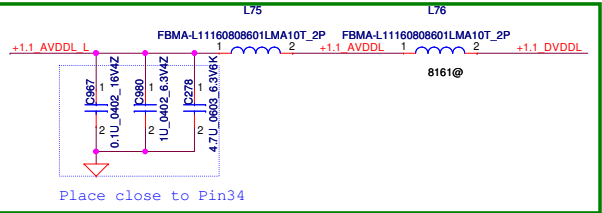
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Atheros request can't disable LAN power

**Layout Notice : Place as close
chip as possible.**



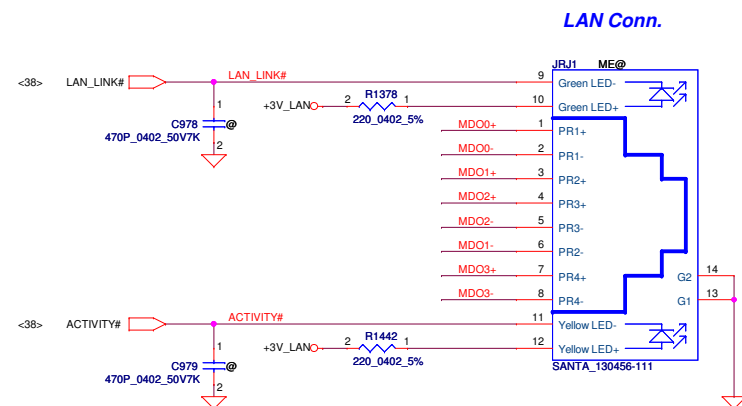
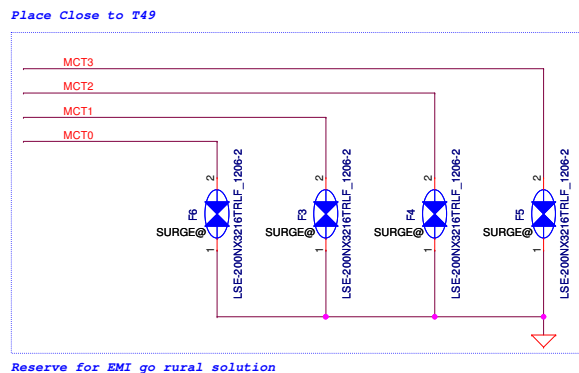
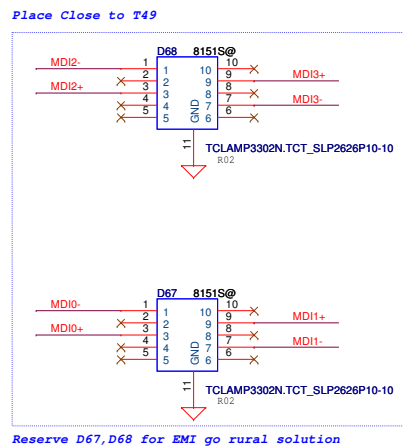
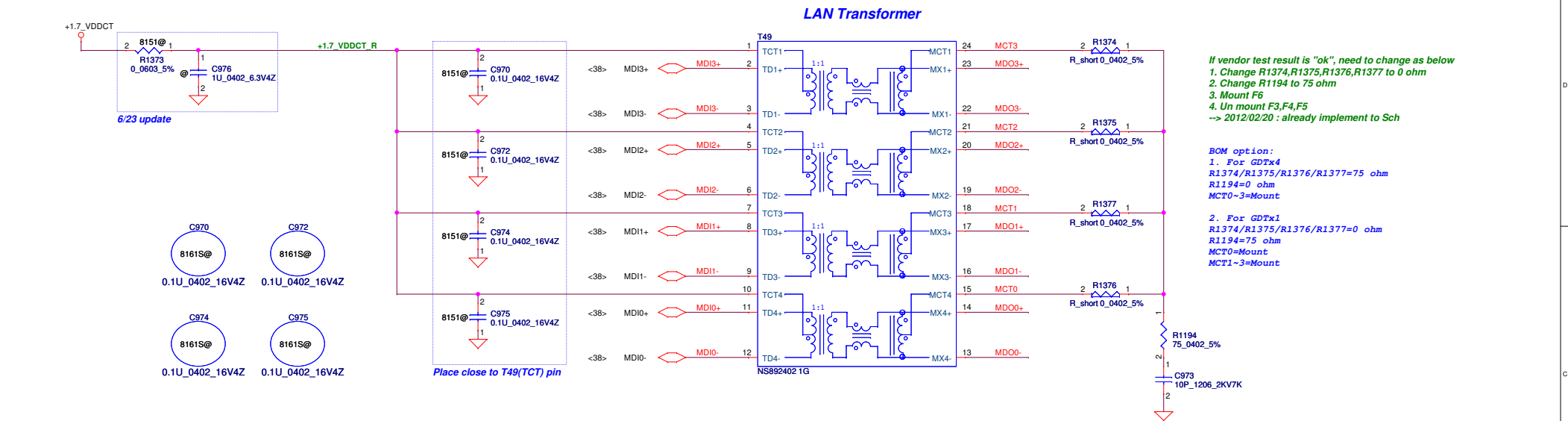
	LX Voltage <Pin 40>	Configure
AR8151	+1.7V <VDDCT>	R1356, C955
AR8161	+1.1V <DVDDL, AVDDL>	R1357, R1372, L76



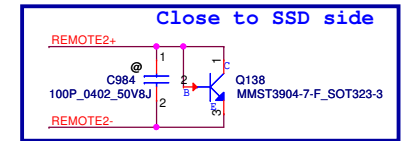
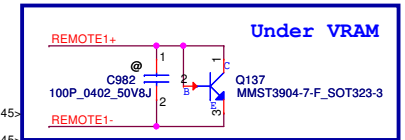
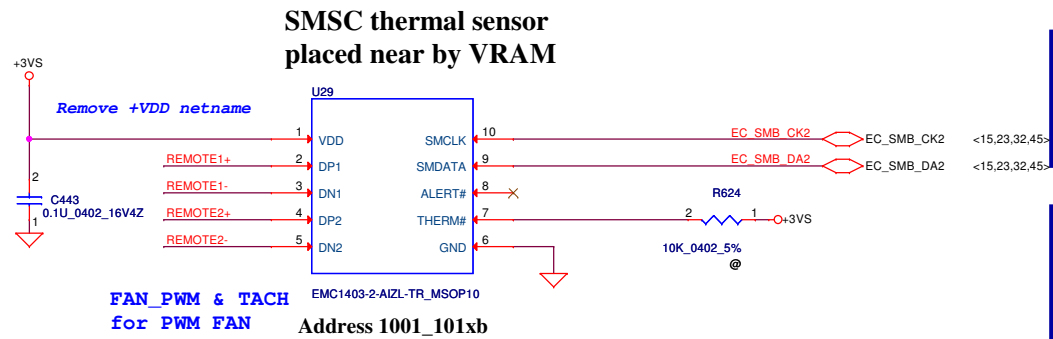
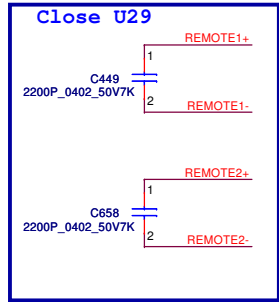
LAN IC	X76	VRAM P/N
8161	X76409JVL04	SA000050E1J
8151	X76409JVL03	SA00003LE2J

Change C968, C969 value of Cap from
33pF to 15pF
for TXC recommend

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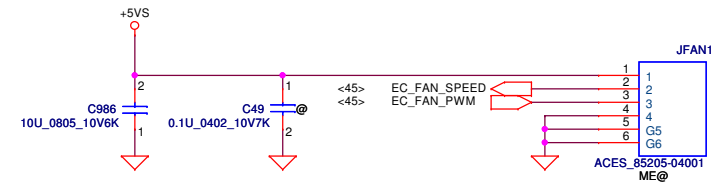


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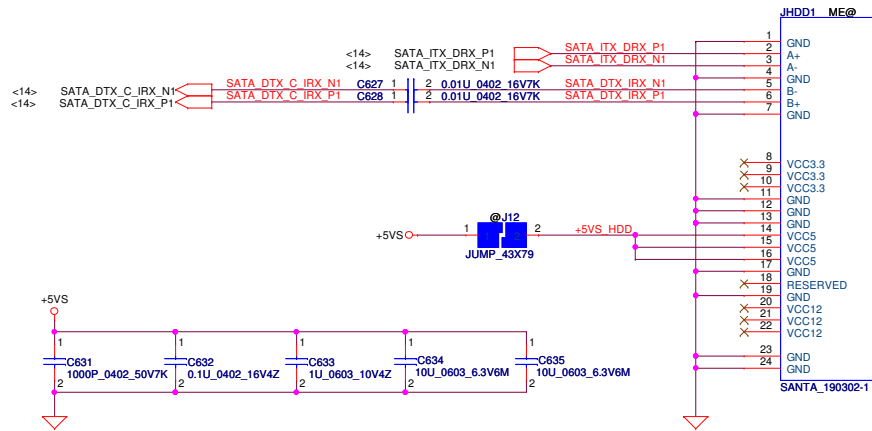
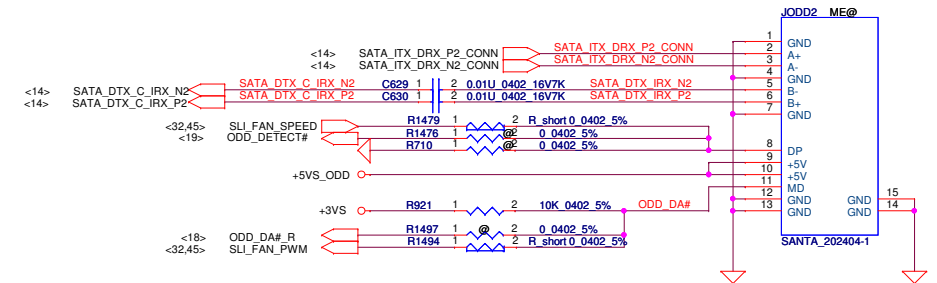


REMOTE2+/-:
Trace width/space:10/10 mil
Trace length:<8"

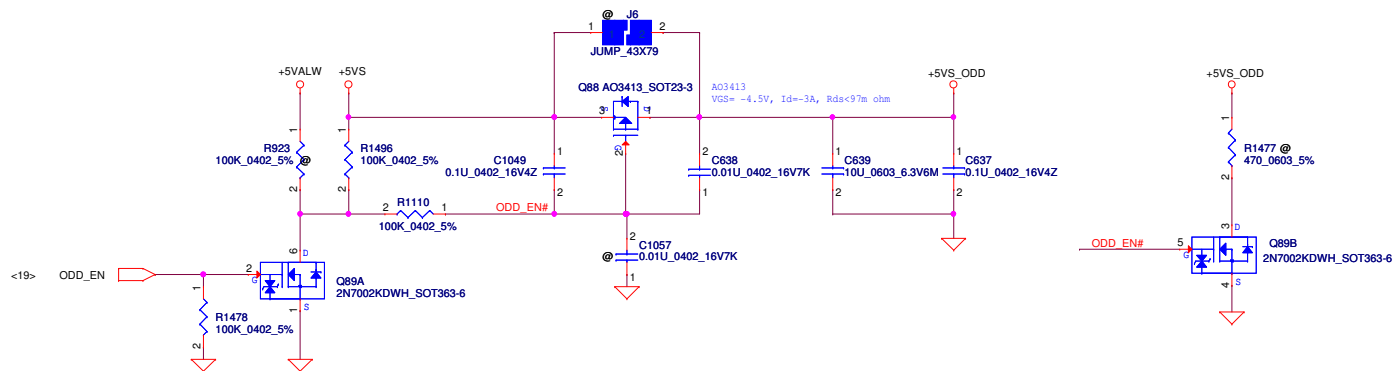
FAN1 Conn



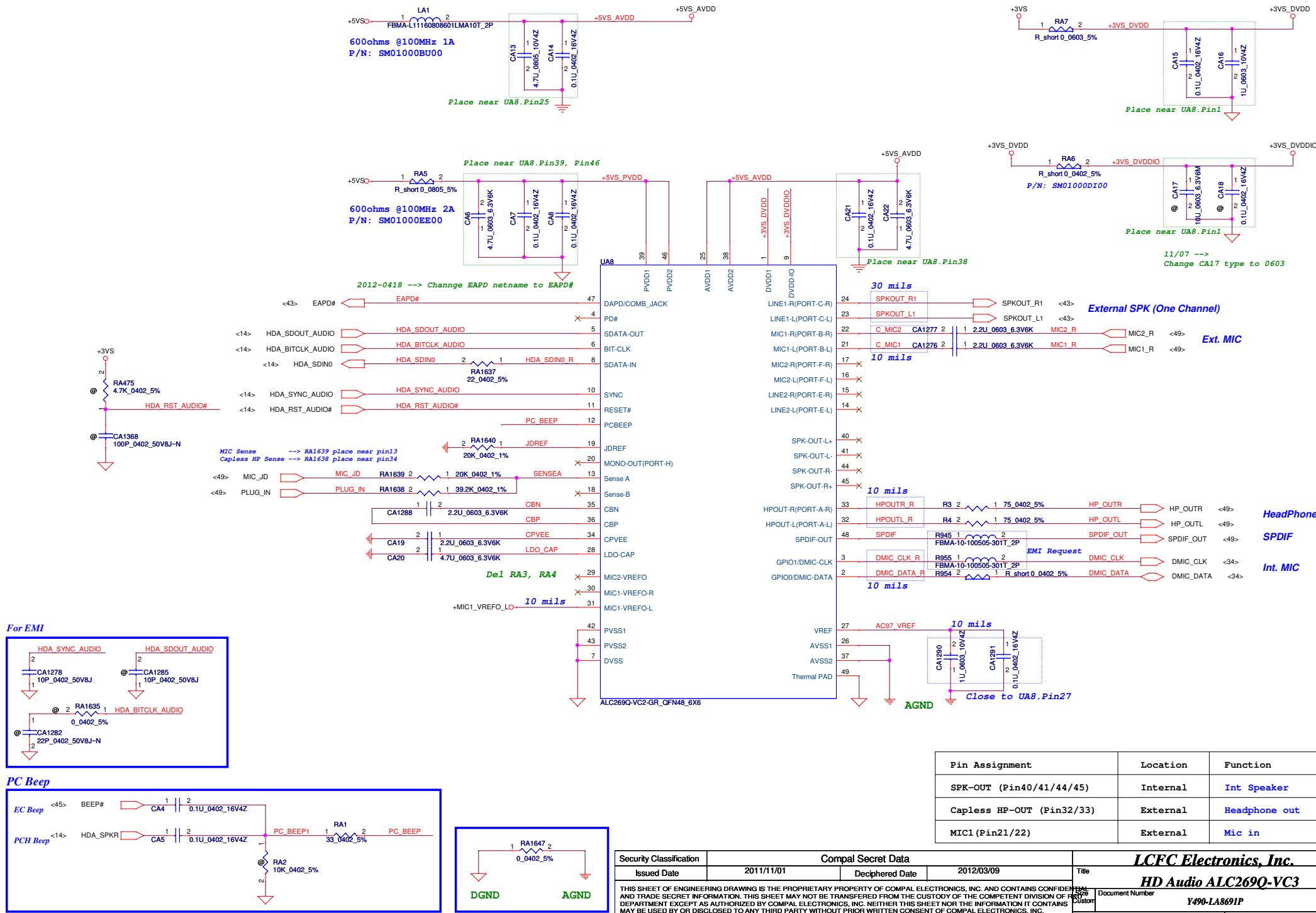
Security Classification		Compal Secret Data		LCFC Electronics, Inc.	
Issued Date	2011/11/01	Deciphered Date	2012/12/31	Title	EMC1403/2103 Thermal sensor/FAN
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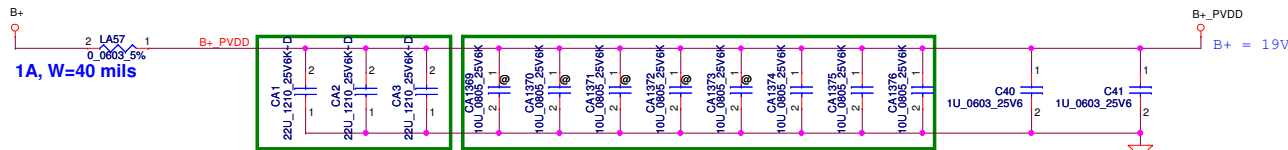
SATA HDD Conn.**SATA ODD Conn.**

ODD Power Control

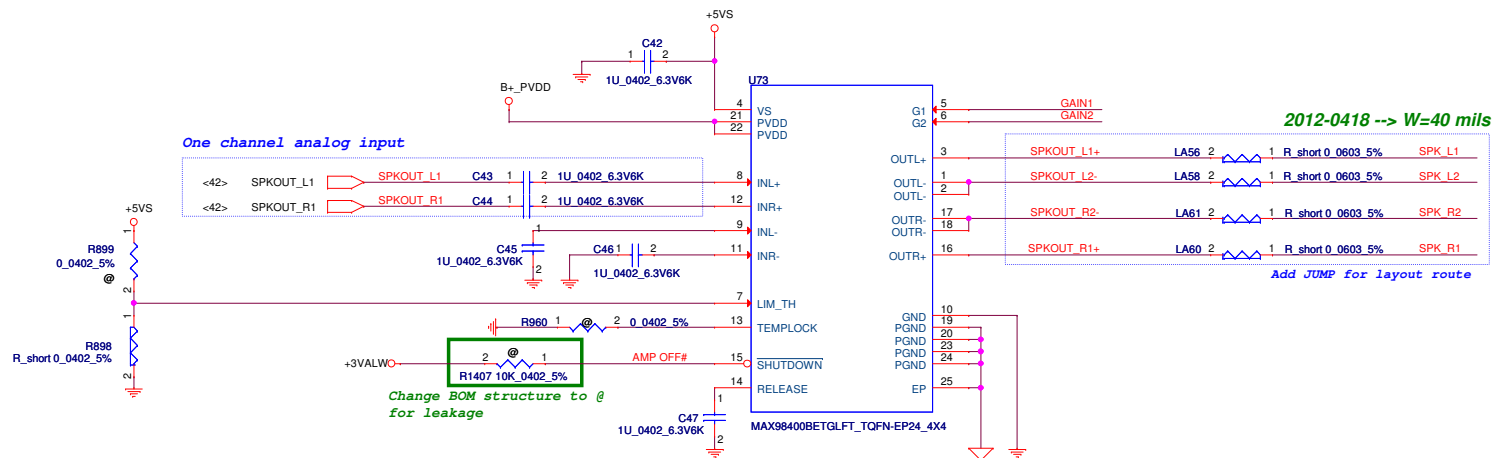


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Issued Date	2011/11/01	Deciphered Date	2012/12/31	Title	
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22uf*3, 10uf*8 for Damage issue
Stuff --> CA1, CA2, CA3, CA1374, CA1375 and CA1376



One channel analog input

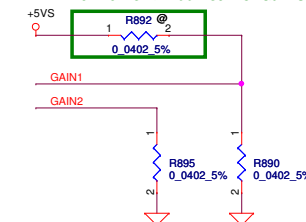
<42> SPKOUT_L1 SPKOUT_L1 C43 1 2 1U_0402_6.3V6K
<42> SPKOUT_R1 SPKOUT_R1 C44 1 2 1U_0402_6.3V6K

Change BOM structure to @ for leakage

2012-0429 --> Change C42-C47 Cap to X5R type for Vendor suggestion

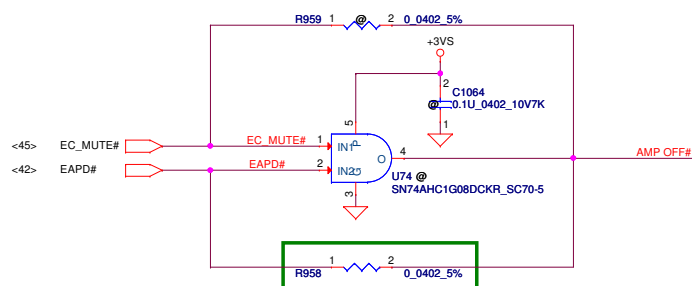
GAIN SETTING

2012-0418 --> Add R892 for Gain setting

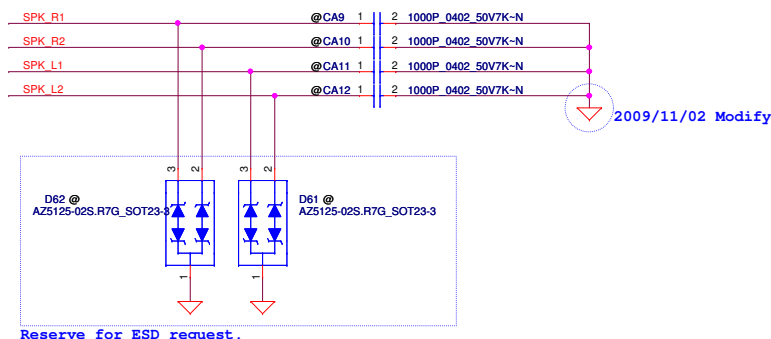


GAIN1	GAIN2	GAIN SETTING (dB)
* GND	GND	9 (Default)
NC	GND	13
+5VS	GND	16.7
GND	NC	20.1
NC	NC	23.3
+5VS	NC	26.4

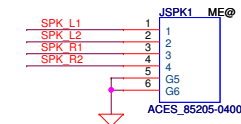
2012-0418 --> Set R890 BOM structure as Stuff

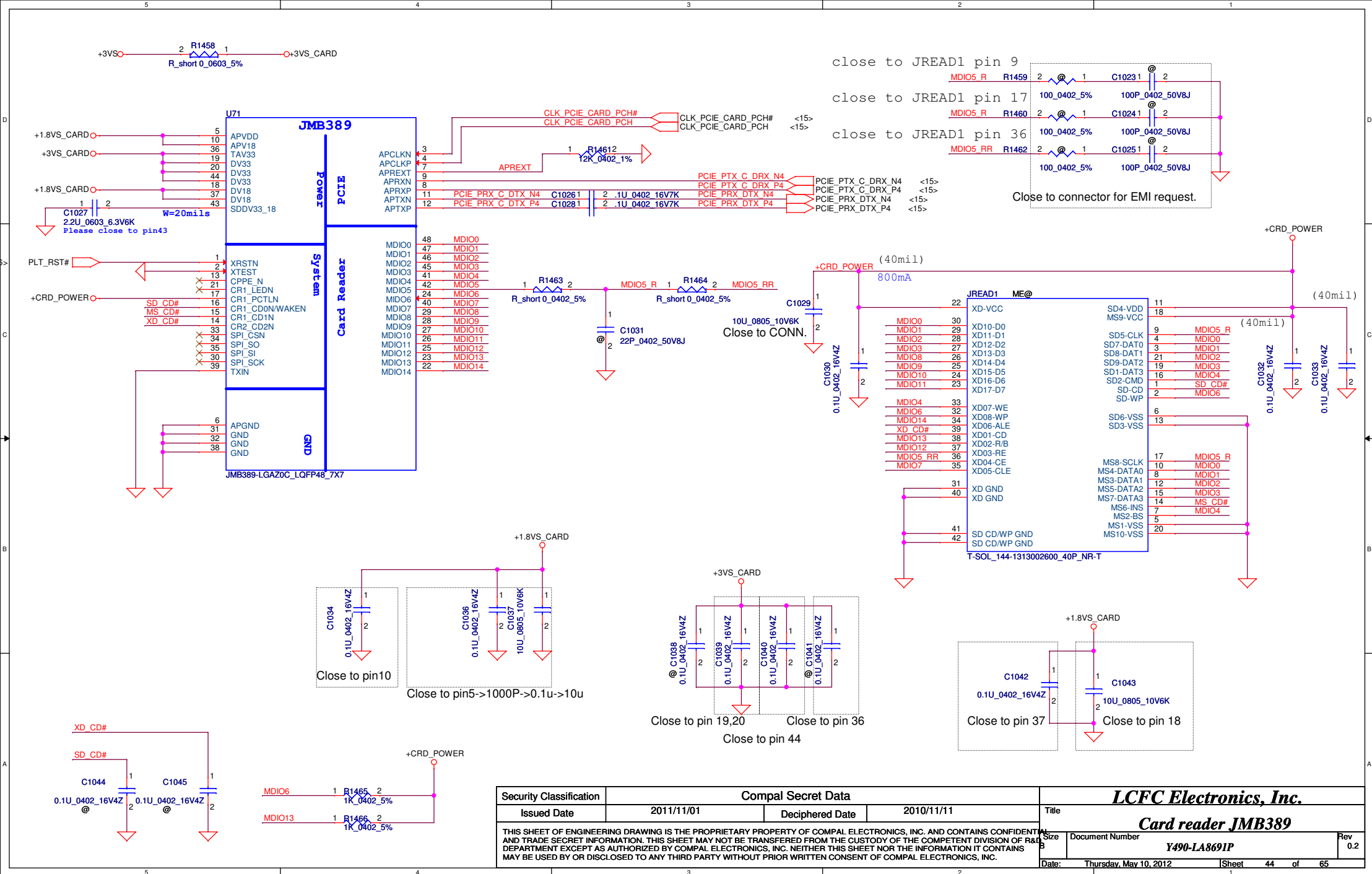


2012-0510B -->
1. R958, --> "Stuff"
2. U74, C1064, R959 --> "@"

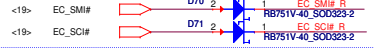


Speaker Conn.

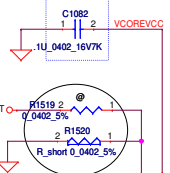




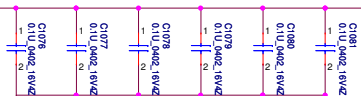
EC_SCI# / EC_SMI# pull up to PCH



close EC

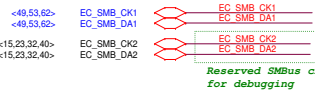


All capacitors close to EC

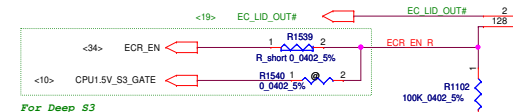
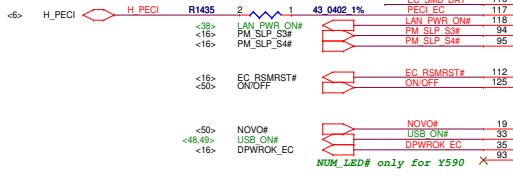


minimum trace width 12 mil

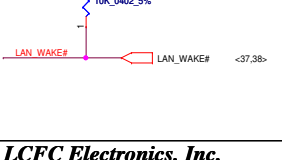
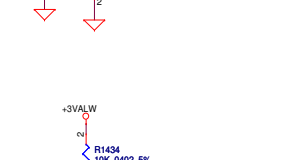
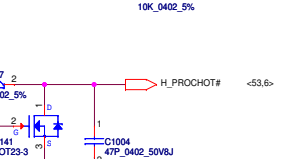
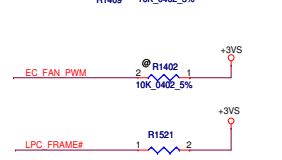
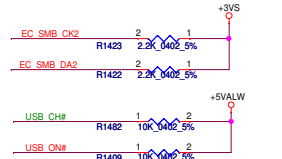
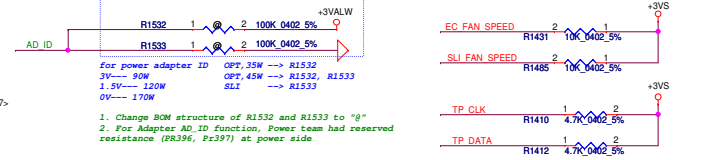
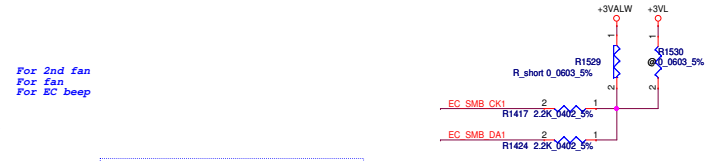
pull up to PCH



Please place R1435 close to EC within 790mil

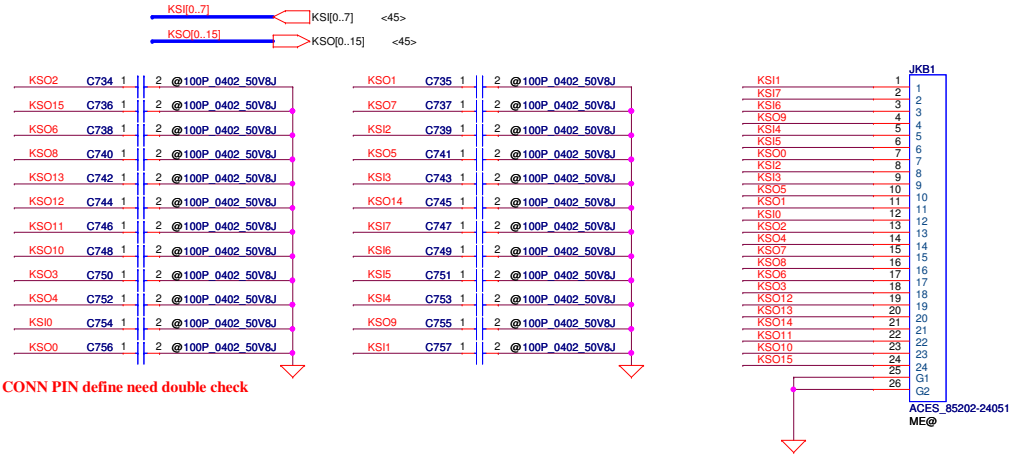


For factory EC flash

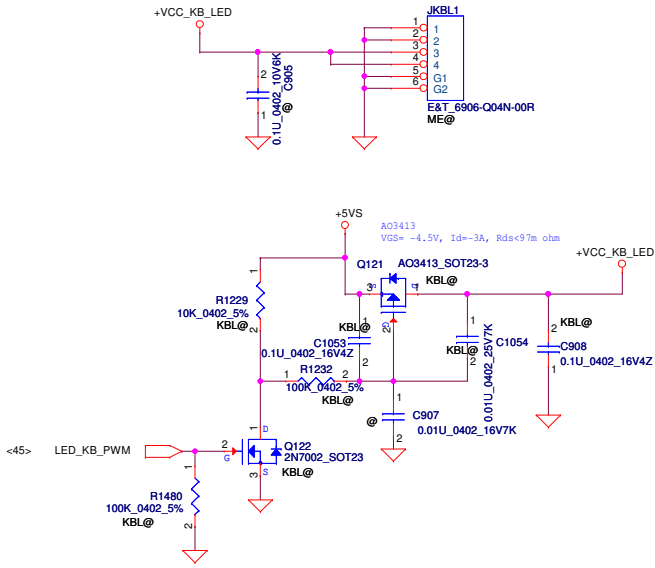


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								EC IT8580E	
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Size		Document Number				Rev			
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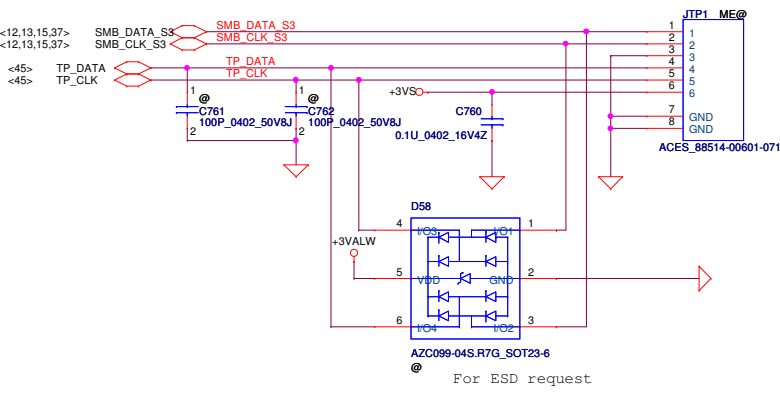
14" INT_KBD Conn.



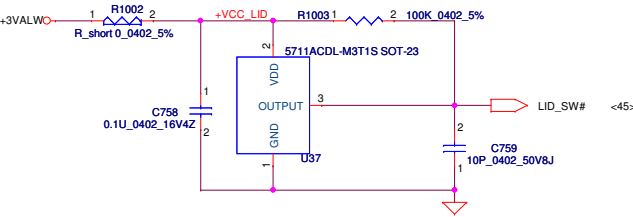
KB Lighting CONN.4pin

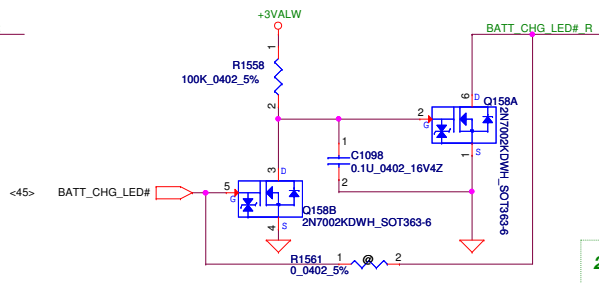
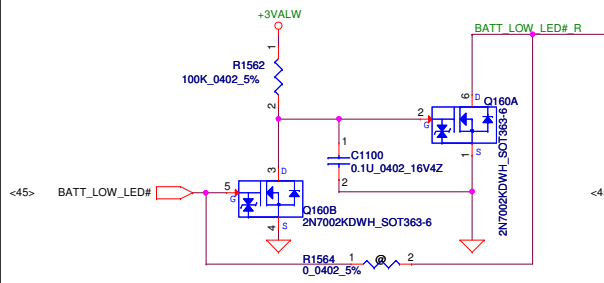


To TP/B Conn.

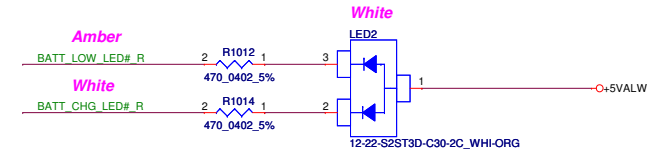


Lid Switch

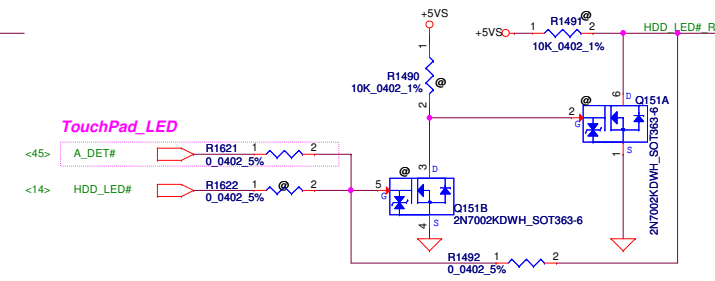
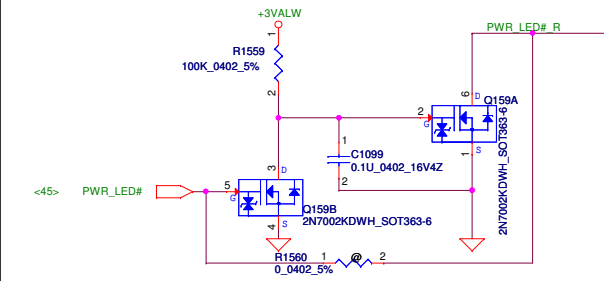




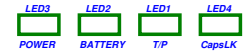
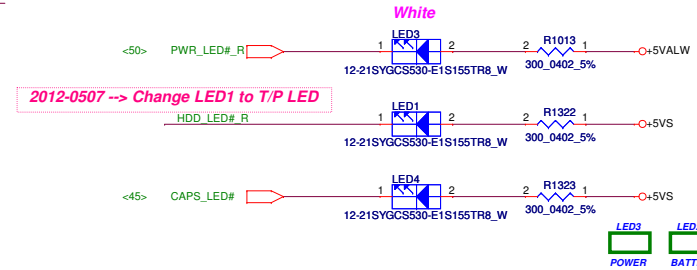
BATT CHARGE/LOW LED



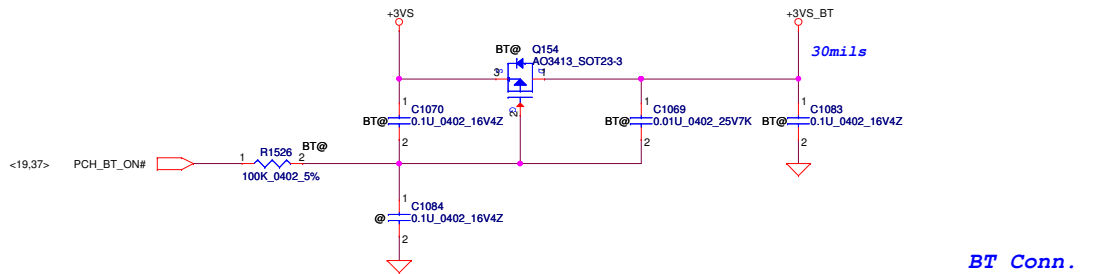
2012-0507 --> Add MOS solution onLED3, 2 to avoid the light blinked.



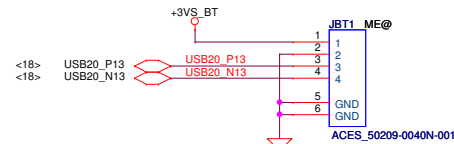
PWR LED HDD LED CapsLK LED



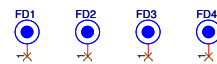
BlueTooth DC



BT Conn.



PCB Fedcal Mark PAD



Screw Hole

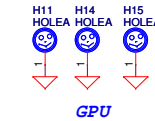
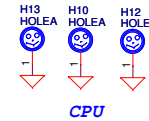
CPU and GPU: H_3P8X 6

C: H_3P8X 3

B: H_3P8X 3

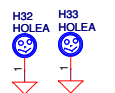
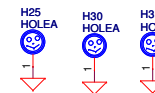
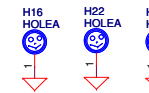
MIN PCIE: H_3P3 X 1

E: H_3P3X 1



ME: H_8P0 X 8; H_3P3X 1; H_4P0X3P0N X 2; H_2P0X 1

A: H_2P8X 8



E: H_3P3X 1

H_4P0X3P0NX 3

H_2P0X 2

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Low Active 2A

USB3 RX N3 2 1 USB3 RX R N3

USB3 RX P3 3 4 USB3 RX R P3

WCM-2012-900T_4P

L68

USB3 TX C N3 2 1 USB3 TX R N3

USB3 TX C P3 3 4 USB3 TX R P3

WCM-2012-900T_4P

L70

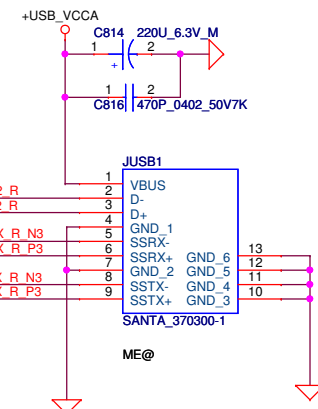
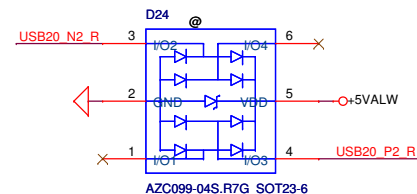
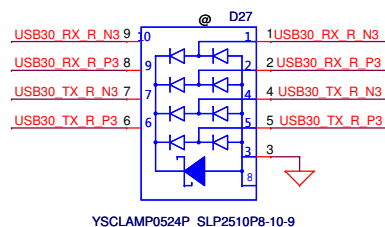
USB20 N2 2 1 USB20 N2 R

USB20 P2 3 4 USB20 P2 R

WCM-2012-900T_4P

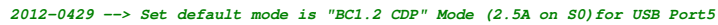
L72

For ESD request



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Right side USB Charger Port (USB_Port5, near JM1C1)

[illegible]

EXT. MIC

+MIC1_VREFO_L

Remove Diode (DA1, DA2)

RA1622 2.2K_0402_5%

RA1623 2.2K_0402_5%

RA1633 1K_0402_5%

<42> MIC2_R RA1634 2 1 1K_0402_5% EXT_MIC_R

<42> MIC1_R RA1633 2 1 1K_0402_5% EXT_MIC_L

	M1	M2	EM_EN	ACTIVE	MODE	
	0	0	1	Dedicated	Charger	Emulation Cycle
	0	1	1	Date Pass	through	
	0	0	0	BC1.2 CDP		
	1	0	1	Dedicated	Charger	Emulation Cycle
	1	1	0	Date Pass	through	
*	1	1	1	BC1.2 CDP		

Pull Low
OR-500mA
10K-900mA
12K-1000mA
15K-1200mA
18K-1500mA
22K-1800mA
27K-2000mA
★ 33K-2500mA

```

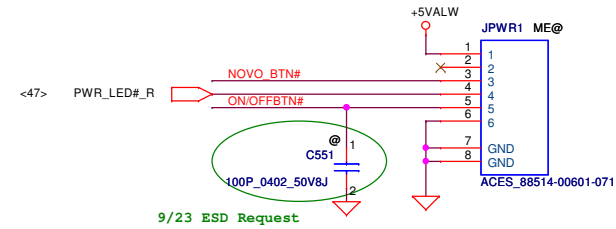
Pull LOW
0R -1010_000
★ 10K-1010_000
12K-1010_000
15K-1010_000
18K-0110_000
22K-0110_000
27K-0110_000
33K-0110_000

```

Pin connection diagram for the USB module. The diagram shows a 28-pin connector on the left and a 28-pin header on the right. The left side labels include: <18> USB20_P9, <18> USB20_N9, <42> MIC_JD, <42> HP_OUTR, <42> SPDIF_OUT, <42> PLUG_IN. The right side labels include: +5V5, +USB_VCCB, +5V_CHGUSB, EXT_MIC_L, EXT_MIC_R, MIC_JD, HP_OUTR, HP_OUTL, SPDIF_OUT, PLUG_IN, GND1, GND2. The pins are numbered 1 to 28 on the right side of the diagram.

need change to 無錫 material
COMPAL : SP010015W1J
Footprint : 88514-0240N-071

**Power Button/B link
to Function/B Conn. 10pin**

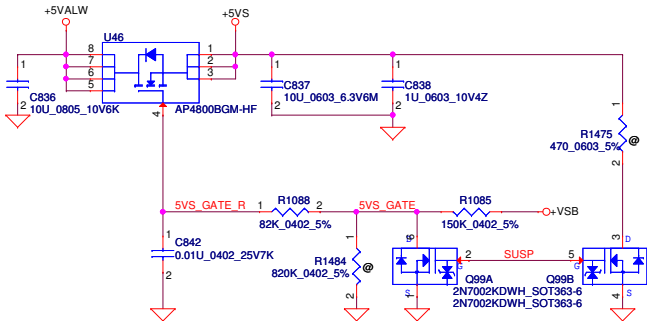


EMI REQUEST 1ST = SCA00000E00
2ST = SCA00000R00

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				Size Custom	Document Number	Rev
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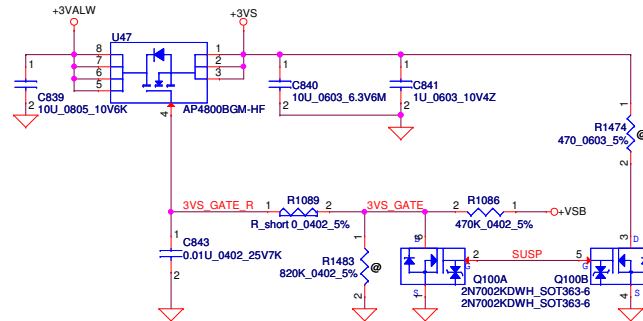
+5VALW to +5VS

AP4800BGM
VGS=10V, ID=9A, Rds=18m ohm
VGS=-25V

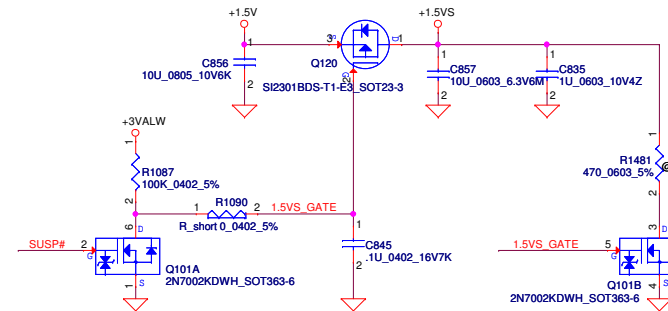


+3VALW to +3VS

AP4800BGM
VGS=10V, ID=9A, Rds=18m ohm
VGS=-25V

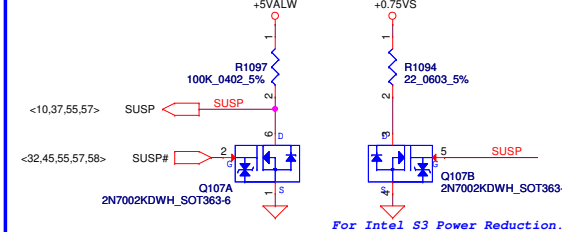
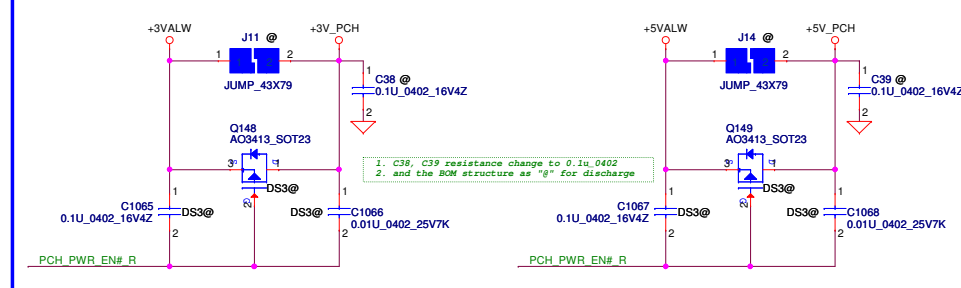


+1.5V to +1.5VS

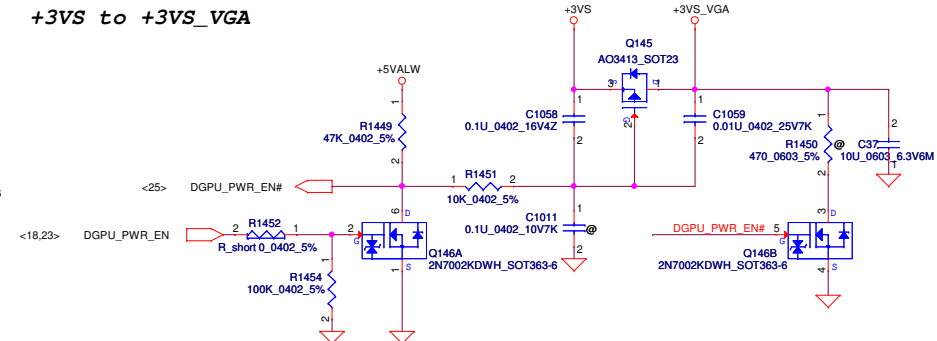


+3VALW to +3V_PCH

+5VALW to +5V_PCH

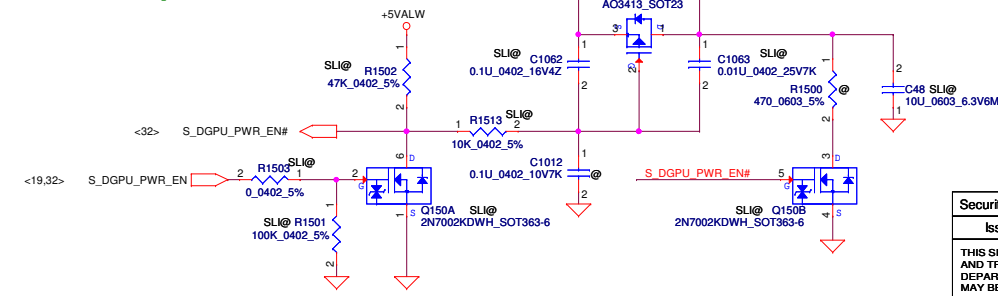


+3VS to +3VS_VGA



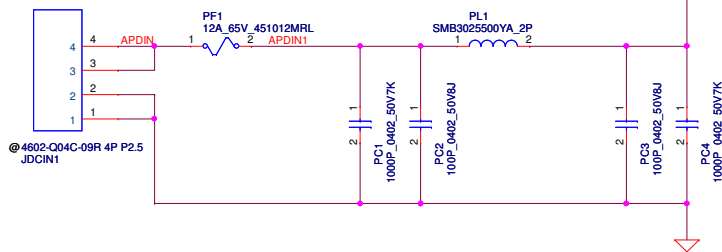
+3VS to +3VS_SLI

2012-0419 --> modify +3VS_SLI BOM structure to "SLI@"

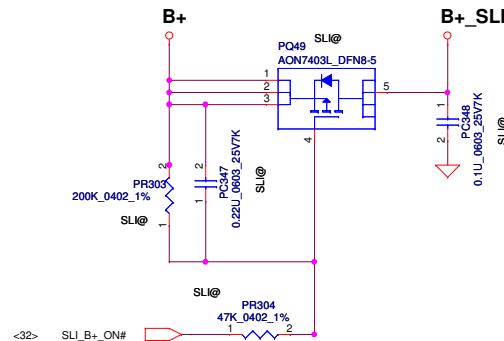


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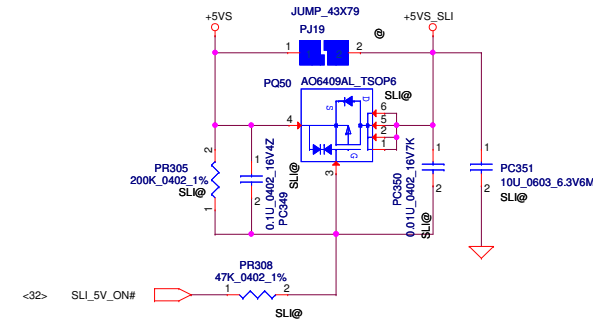
DC030006J00



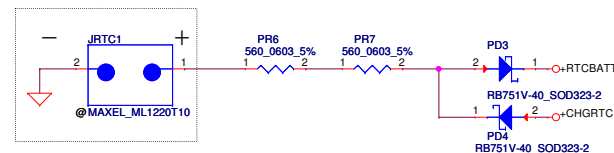
B+ to SLI_B+



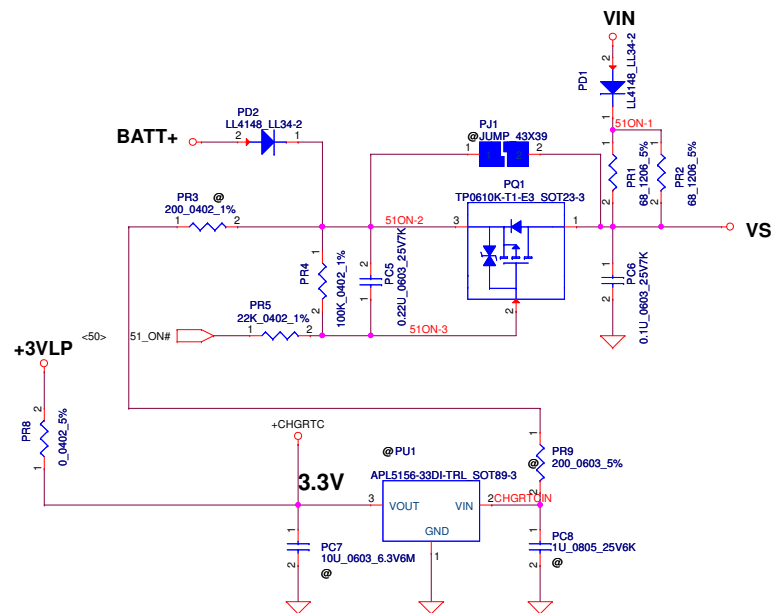
+5VS to +5VS_SLI



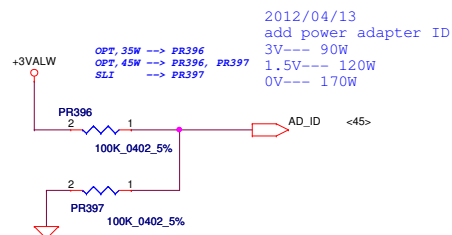
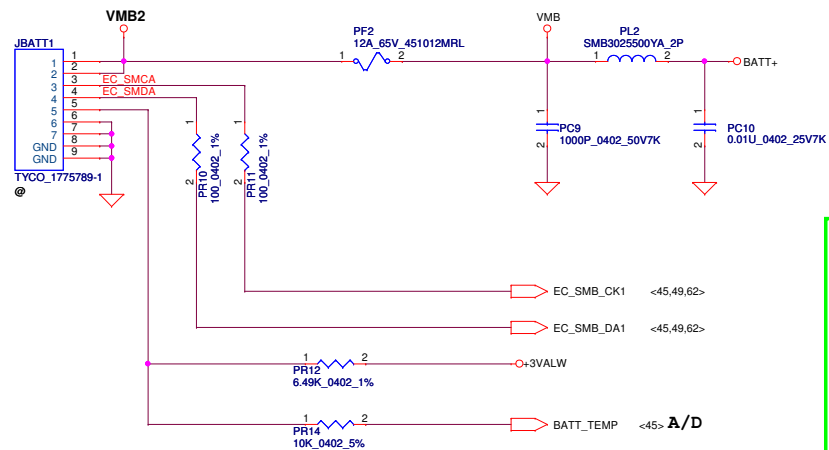
2012/04/13
add SLI Hot-plug Load-SW solution



RTC Battery



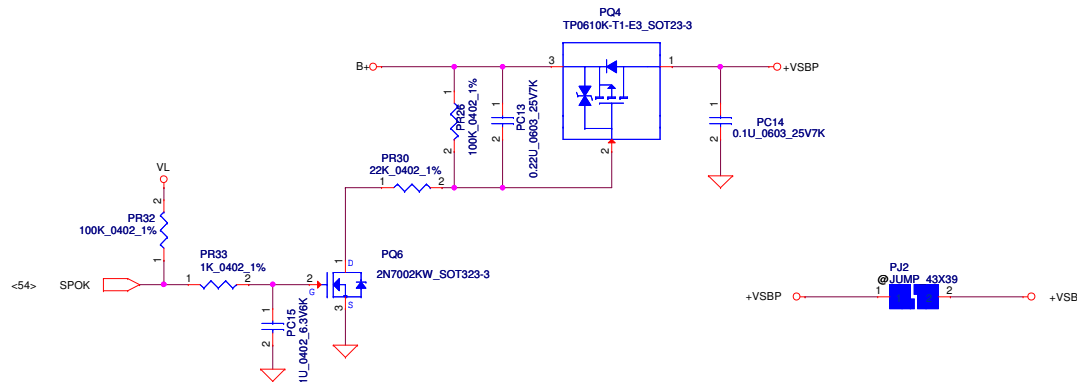
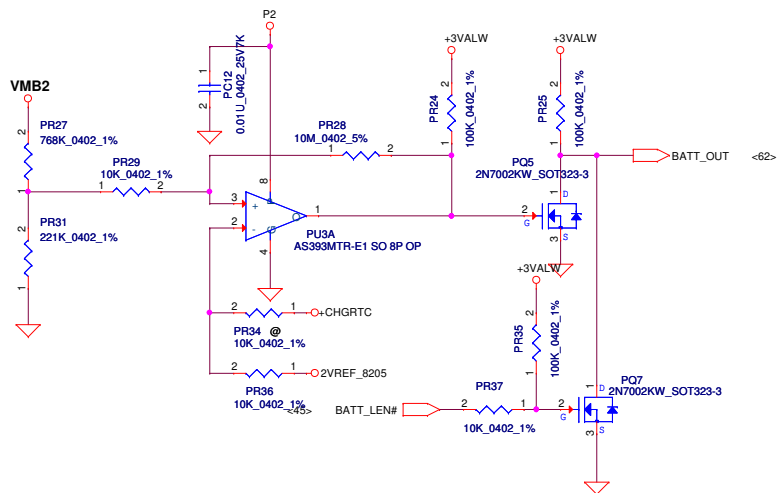
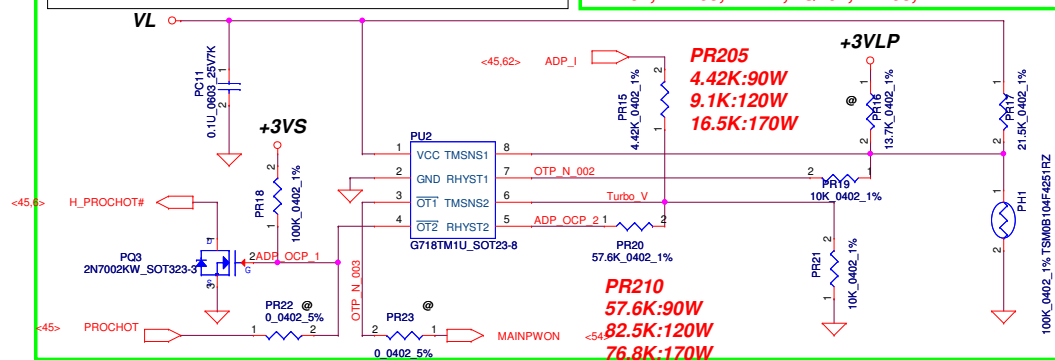
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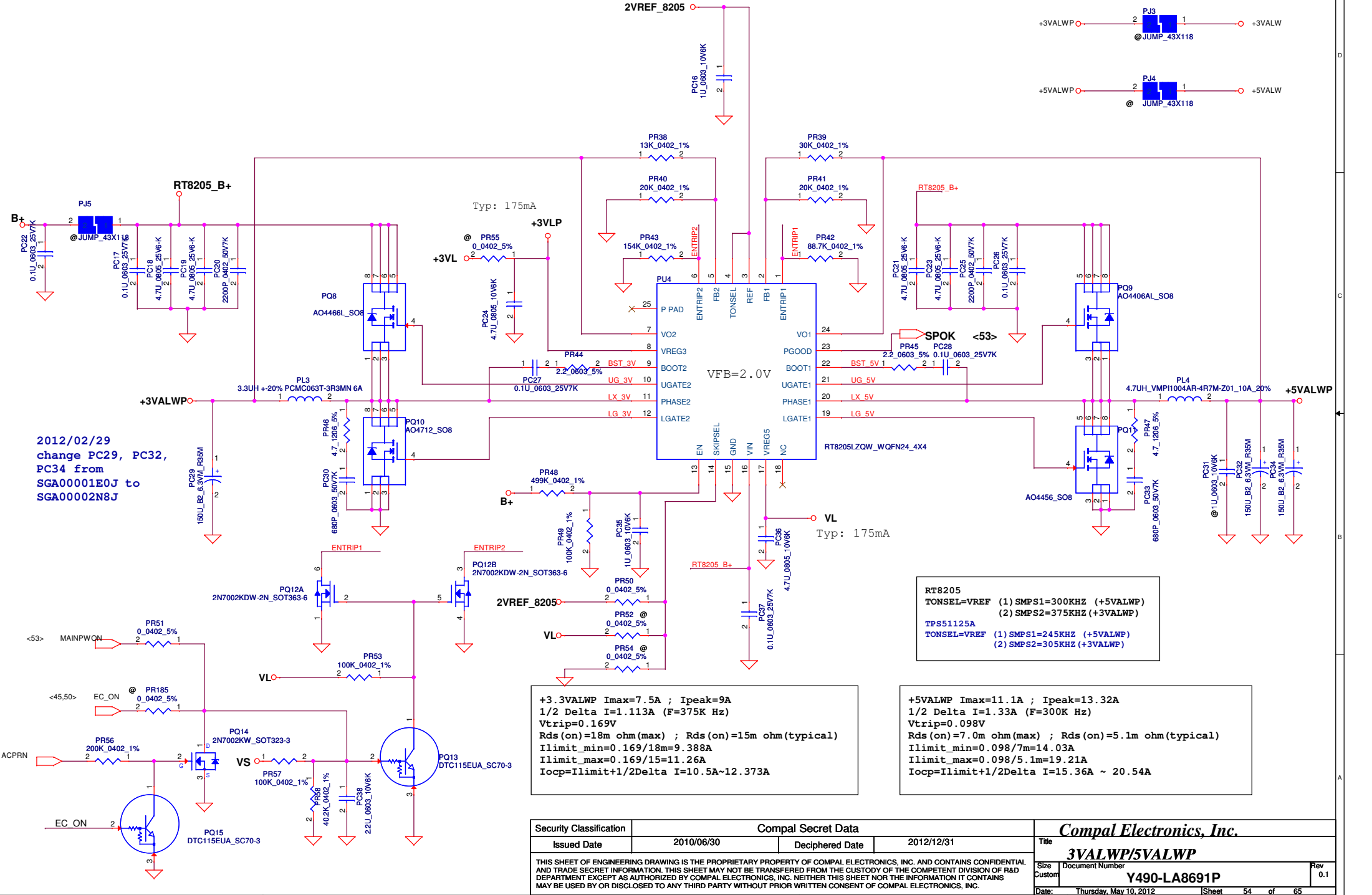
PH1 under CPU bottom side :
CPU thermal protection at 92+/-3 degree C
Recovery at 56 +/-3 degree C

For KB930 --> Keep PU1 circuit
(Vth = 0.825V)

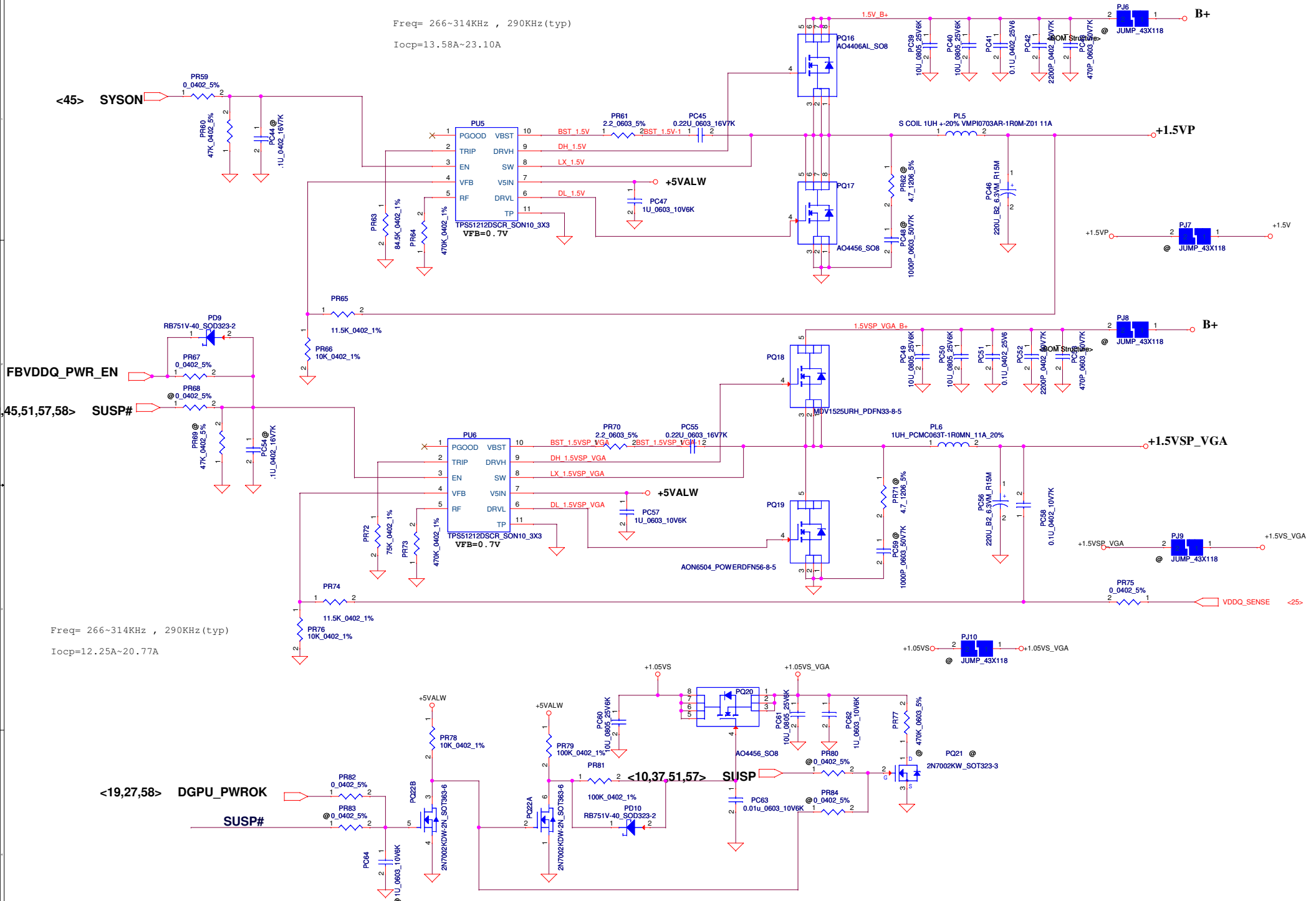
For KB9012 (Red square) --> Remove PU1 circuit, but keep PR206
PH201, PR205, PR211, PQ201, PR208, PR212



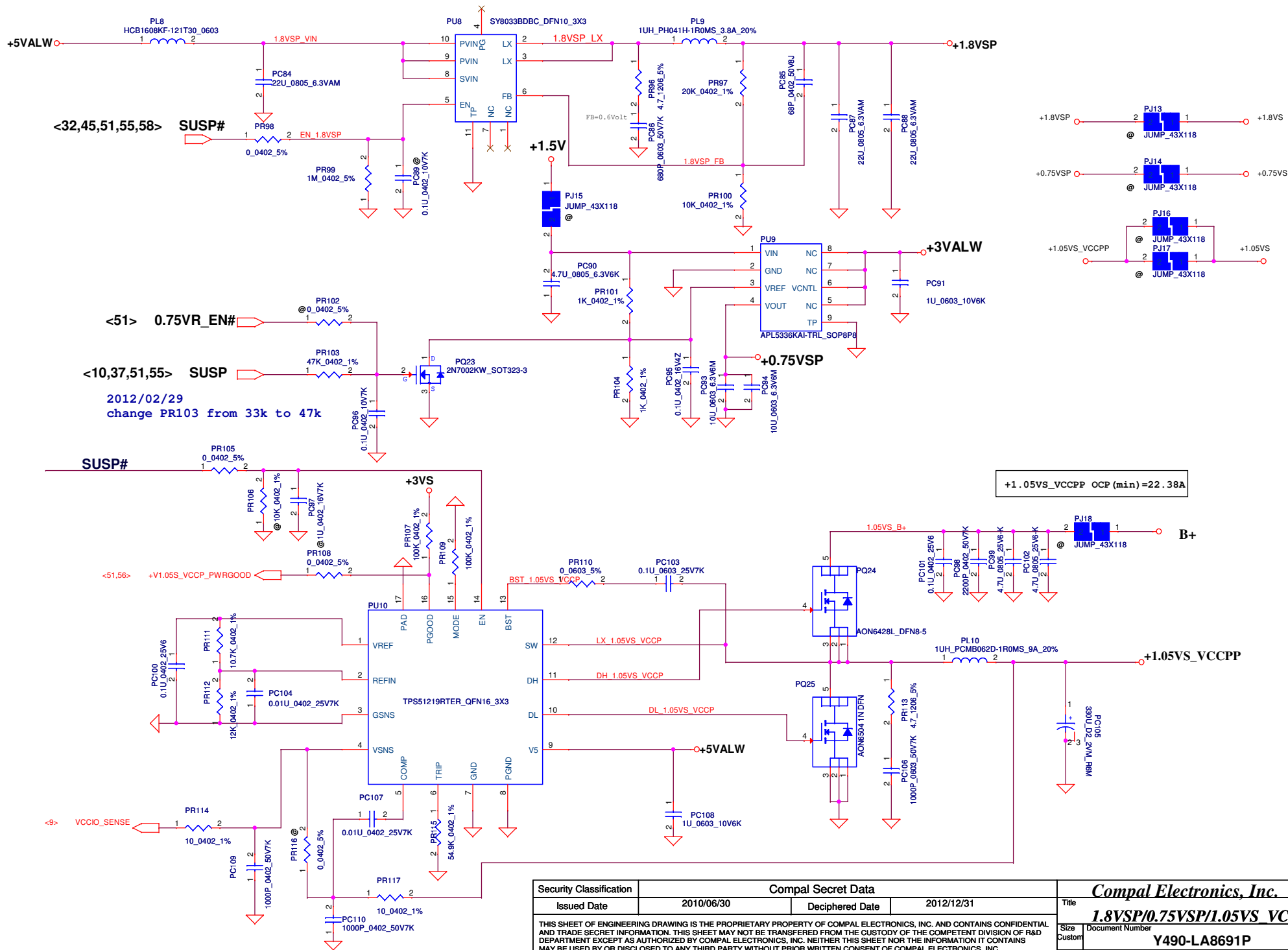
Note:
Use TPS51125 IC can remove RTC refernece LDO
Use TPS51427 IC must keep RTC refernece LDO

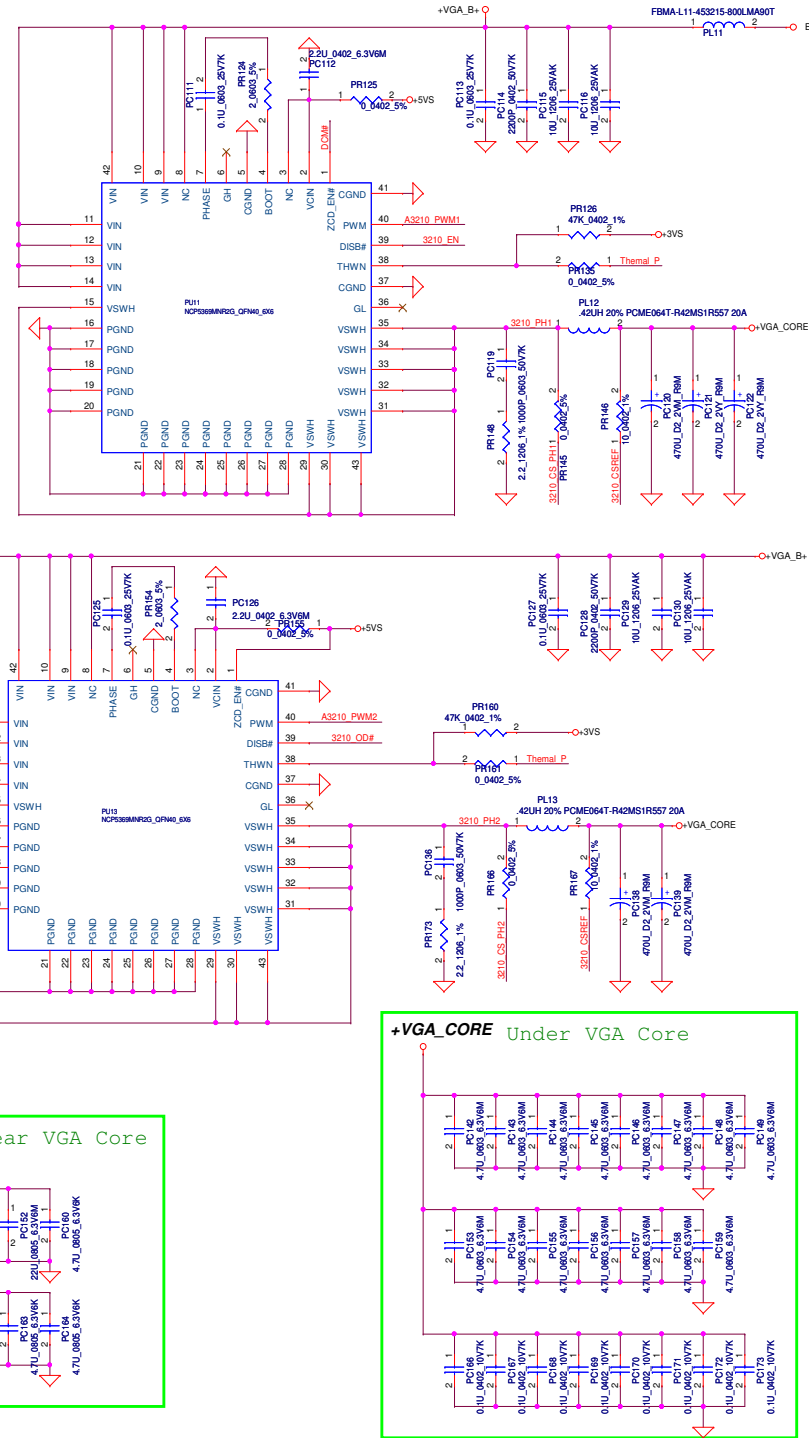


Freq= 266~314KHz , 290KHz(typ)
Iocp=13.58A~23.10A



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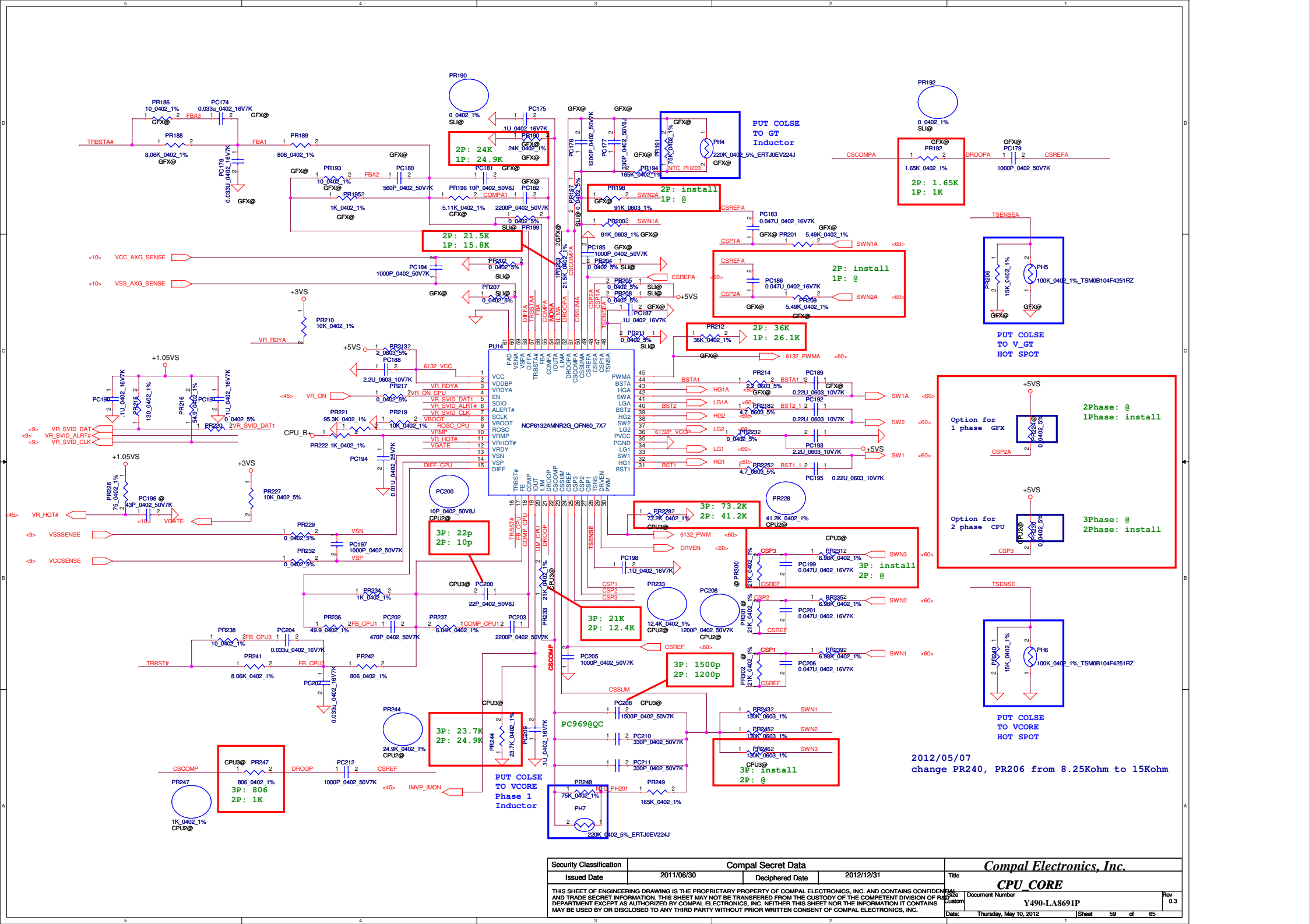
+VGA_CORE Near VGA Core

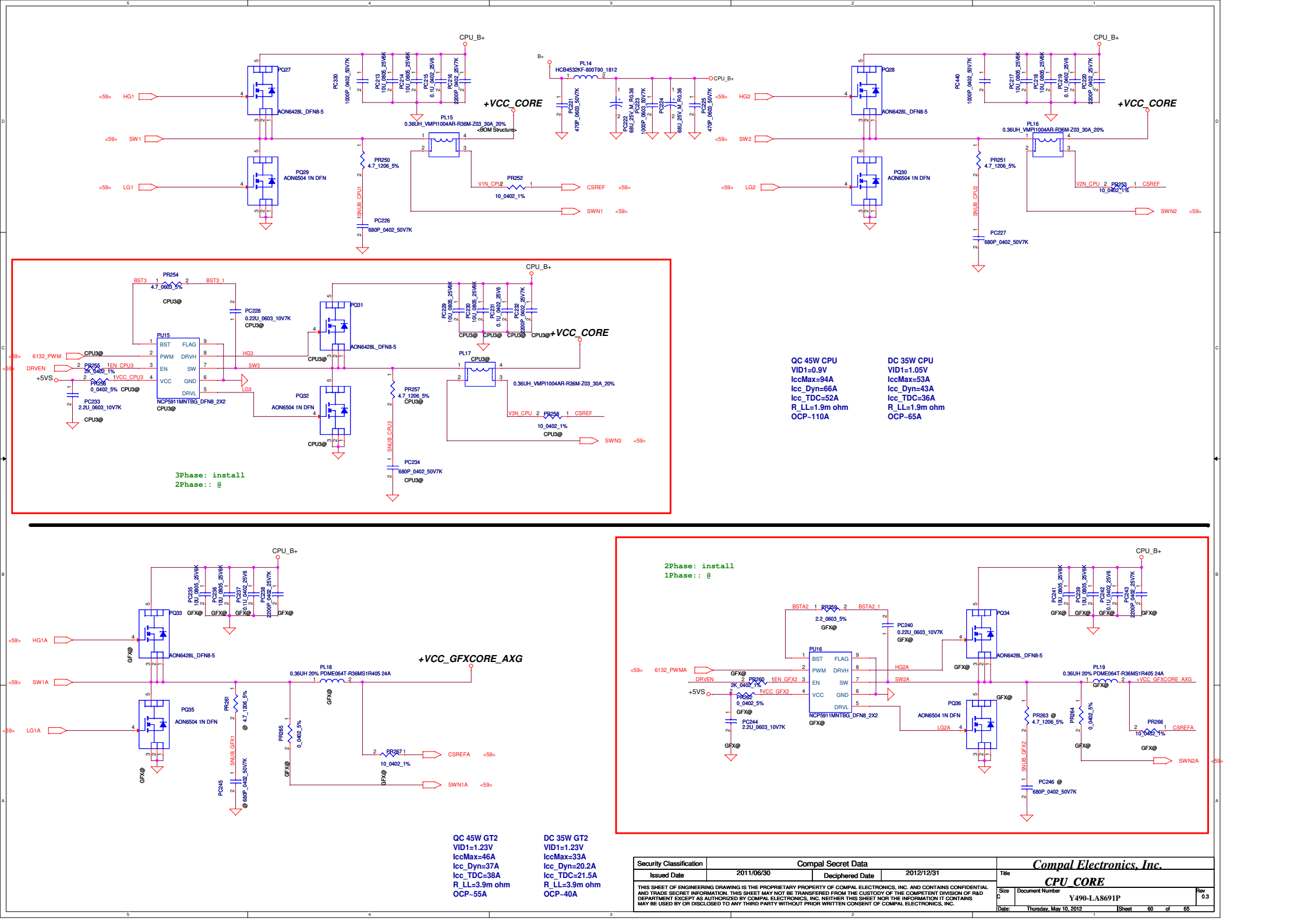
The diagram illustrates the power distribution network for the +VGA_CORE near the VGA Core. It features a central power rail connected to a common ground. The network includes several capacitors and resistors:

- PC150:** A 220uF capacitor connected to the power rail and ground.
- PC151:** A 470uF capacitor connected to the power rail and ground.
- PC160:** A 220uF capacitor connected to the power rail and ground.
- PC164:** A 470uF capacitor connected to the power rail and ground.

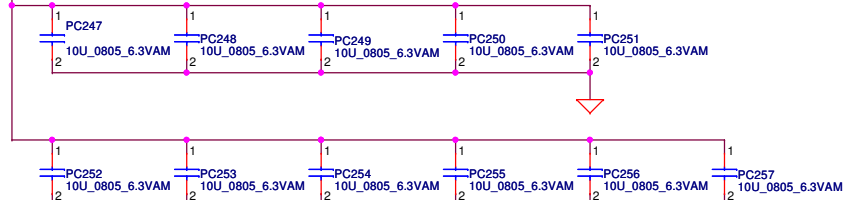
The capacitors are labeled with their values and the voltage they are connected to (6.3V6K or 6.3V6M). The resistors are labeled with their values (220u, 470u, 220u).

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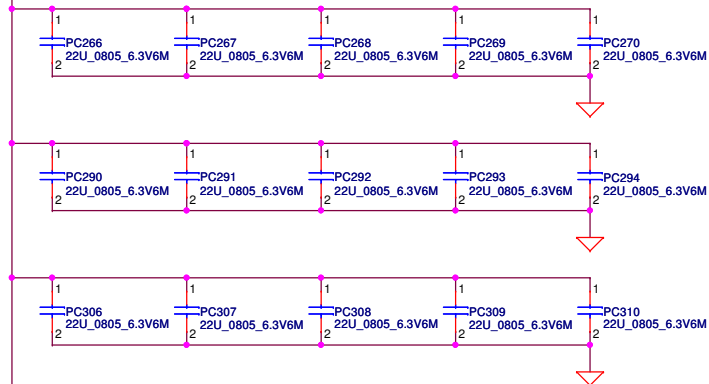




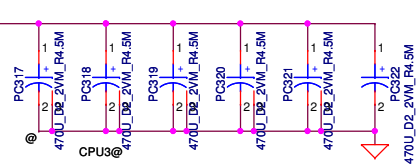
+VCC_CORE



+VCC_CORE



+VCC_CORE



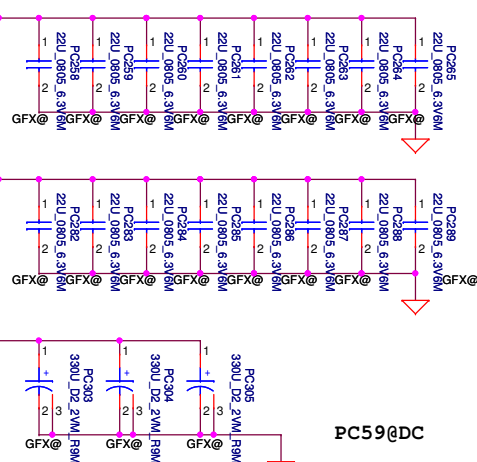
DC: PC73, PC74, PC75, PC76, PC77, PC78 (330uF/9m)
QC: PC76, PC78 (470uF/4.5m), PC73, PC74, PC75 (330uF/9m)

PC8, PC21, PC22, PC63

+CPU_CORE

+VCC GFXCORE_AXG

+VCC GFXCORE_AXG



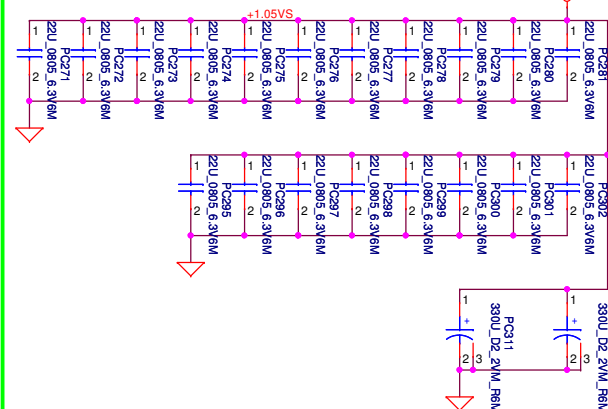
PC38, PC39, PC40, PC41

PC38, PC39, PC40, PC41

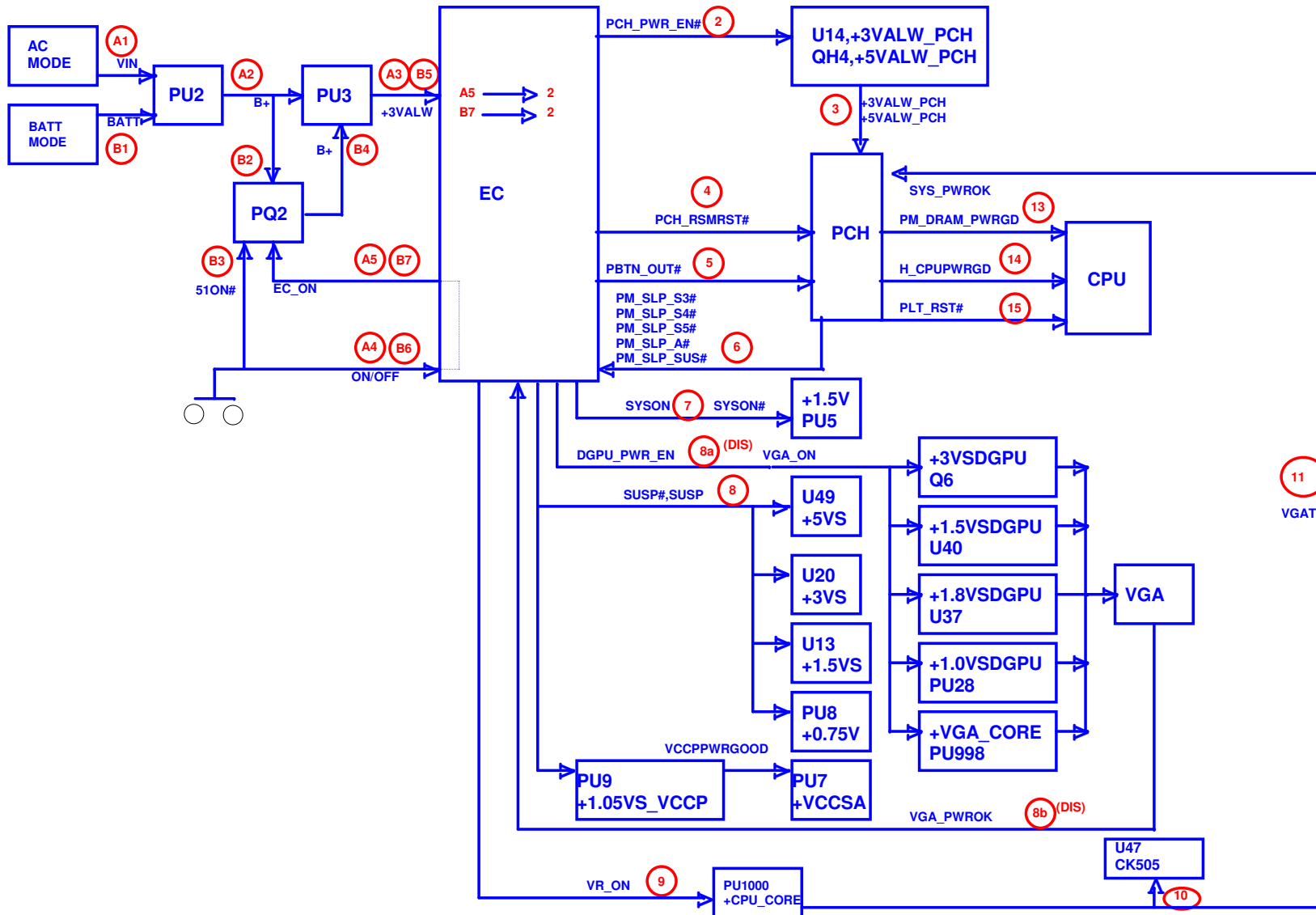
Below is 458544_CRV_PDDG_0.5 Table 5-8.

Socket Bottom	5 x 22 μ F (0805) 5 x (0805) no-stuff sites
Socket Top	7 x 22 μ F (0805) 2 x (0805) no-stuff sites

+1.05VS



PC32, PC49, PC54, PC55, PC56



Version change list (P.I.R. List)

Page 1 of 1
for PWR

Item	Reason for change	PG#	Modify List	Date	Phase
1	Reserve 0.1uF for Charger IC	51	Reserve PC321	201109/27	B test
2	EMI Request		change PR322,PR407,PR408,PR503,PR511,PR606,PR804,PR827 to 2.2 ohm add PC526,PC527,PC970,PC971(470uF)	201109/27	B test
3	Combine 1.05V	51	Remove one power rail +V1.05S_VCCPP Pop PR722,PR712,PR718	201109/27	B test
4	Discharge for +1.05VS_VGA by NV Request	53	Reserve PR528	201109/27	B test
5	Set VGA_CORE VBOOT voltage	56	unpop PR806 change PR813 to 147K ohm	201109/27	B test
6	For VGA_CORE power saving by NV Request	56	add PR838 0ohm	201109/27	B test
7	for CPU_CORE load line adjust	57	add PC969	201109/27	B test
8	to prevent MOS over temperature	55/58	change PQ702,PQ901,PQ902,PQ905 TPCA8065	201109/27	B test
9	for CPU_CORE test	59	Reserve PC77,PC78	201109/27	B test
10					
11					
12					
13					
14					
15					
16					
17					

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HW PIR (Product Improve Record)

QIQY5 LA-8691P SCHEMATIC CHANGE LIST
REVISION CHANGE: 0.2
GERBER-OUT DATE: 2012/03/09

NO	DATE	PAGE	MODIFICATION LIST	PURPOSE
01)	03/14	10	R64	Change R64 BOM structure from "a" to "DS3a"
				For Deep S3 Function

Security Classification		Compal Secret Data		Compal Electronics, Inc.	
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				LA-6882P PIR (HW)	
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				0.1	0.1
Date: Thursday, May 10, 2012				Sheet	65 of 65